

SONOMA:

a Small Open ecoNOmy for MAcrofinance

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ABSTRACT

We develop a new small open economy model (SONOMA) in which domestic corporate debt and equities coexist (Jermann and Quadrini, 2012) and are affected by shocks to both external credit and equity markets. In a novel empirical analysis of several small-but-developed

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economies, we show that both external debt and equity shocks are important determinants of domestic economic fluctuations, corporate leverage, and net foreign asset positions. SONOMA replicates our empirical facts about asset prices, financial flows, and economic activity.

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I. Introduction

Small open economy (SOE) models have been widely adopted to study emerging market economies (EMEs). In this study, we focus on small-but-*developed* open economies and ask the following question: How important are financial shocks to both internal and *external* equity and bond markets for the determination of domestic economic fluctuations and corporate financial dynamics?

We address this question by developing a new macrofinance SOE model whose analysis requires the collection and investigation of new data. We find that shocks to the cost of external debt and equities explain a prominent share of the medium- and long-run variance of investment, consumption, net external debt, and net external equities. In addition, we document that adverse realizations of these shocks are followed by sluggish long-run growth. Domestic credit shocks are important as well, mainly for output growth, domestic leverage, and external debt.

Importantly, several of our results are specific to small developed economies (SDEs), and these differ substantially from those documented focusing on EMEs. As in small open EMEs (see, among others, Smith and Valderrama, 2009; García-Cicco et al., 2010; Chang and Fernandez, 2013),

we find that financial frictions are first-order drivers of external positions in open SDEs. SDEs differ from EMEs in that shocks to the expected return of external equities are relatively more important for both domestic investment and corporate leverage decisions. In contrast, in EMEs, exchange rate shocks and domestic credit shocks feature prominently.

Other previous studies have focused on large-and-developed countries and have reached quite different conclusions. Focusing on a large and developed country such as the United States, the literature has found mixed results regarding the relevance of credit frictions (see, among others, Chari et al., 2007; Jermann and Quadrini, 2012). However, there is consensus about borrowing shocks being a key determinant of economic fluctuations (Blattner et al., 2023), financial flows (Maggiore, Neiman, and Schreger, 2020; Kalemli-Özcan and Varela, 2021), and sudden stops in EMEs (see, e.g., Chari, Kehoe, and McGratten, 2005). In addition, the literature has generally abstracted away from shocks to external equity markets. We propose a new perspective by (i) focusing on SDEs (as an exception, see Chodorow-Reich, Karabarbounis, and Kekre, 2023), and (ii) addressing the relevance of shocks to both external credit and equity markets.

SONOMA is a production-based general equilibrium model that successfully reproduces key properties of SDEs in terms of macroeconomic aggregates, external balances, and cost of capital. We start from a benchmark SOE model (see, among others, Mendoza, 1991; Schmitt-Grohé and Uribe, 2003; Mendoza, 2006) and modify it in three key dimensions. First, we introduce recursive preferences and productivity growth news shocks in the spirit of the long-run risk literature (among others, see Bansal and Yaron, 2004; Kaltenbrunner and Lochstoer, 2010; Croce, 2014). Growth news shocks are important sources of macroeconomic fluctuations, risk premia, and more broadly the cost of capital for private firms. Recursive preferences are important so that news shocks are directly priced by investors. We consider this step essential to properly address uncertainty about the long-term growth of economies in distress.

Second, we decompose net foreign asset positions into their debt and equity components. External debt is subject to exogenous external interest

rate shocks. External equities are subject to two distinct types of shocks: risk premium shocks and realized return shocks. This shock structure is consistent with our empirical evidence on the returns to external debt and equity positions. Furthermore, external equities and debt allow for different margins of adjustment in the domestic household’s balance sheet. While analyzing the role of external shocks is standard in the SOE literature, the decomposition of net foreign assets into debt and equity components is a unique aspect of our model.

Third, we introduce financial frictions to create a dynamic trade-off between equity and debt financing for domestic firms. In our model, a corporate tax shield on the cost of debt makes debt financing more appealing than equity. On the other hand, an enforcement constraint limits debt financing exactly as in Jermann and Quadrini (2012). The tightness of this constraint fluctuates with exogenous shocks that we interpret as domestic credit shocks. Even though there is no default at the equilibrium, these shocks distort labor demand and disrupt production.

Our new insights are produced from the interplay of three elements: (i) persistent credit shocks are priced because of recursive preferences; (ii) domestic financial frictions make corporate leverage important for economic dynamics; and (iii) domestic financial flows interact with external shocks to the cost of equity and debt.¹

Taking seriously a subset of the restrictions imposed by our model, we are able to measure both domestic and external borrowing conditions as well as shocks to the net external equity positions for a sizable cross section of

¹Alternatively, one could think of our SONOMA model as a Jermann and Quadrini (2012) model modified to (i) admit small open economy dynamics, and (ii) price news shocks. We do not focus on (real) exchange rate shocks in this study because (i) our empirical findings show that these shocks have a negligible effect on most macrofinancial variables in our study, and (ii) sudden stops, significant exchange rate shocks, or occasionally binding constraints are not a dominant or a common feature of the economies in our sample.

countries. As mentioned earlier, we find that adverse shocks to either external equities or external bonds are followed by lower expected productivity growth (negative growth news shocks). In addition, adverse shocks to the expected return of external equities are correlated with adverse domestic credit shocks. That is, a tightening of credit conditions is accompanied by notable increases in the cost of financing capital through the issuance of equities.

We also study the variance decomposition of several financial and macroeconomic variables of interest over long horizons. In the data, the costs of external debt and equities explain a prominent share of the medium- and long-run variance of investment, consumption, net external debt, and net external equities. Domestic credit shocks are important as well, mainly for output growth, domestic leverage, and net external debt.

We rely on these data and measurements to calibrate our model and show that SONOMA reproduces many empirical unconditional moments. The model inherits the success of a standard macroeconomic model in replicating the fluctuations of macroeconomic aggregates. At the equilibrium, the annualized equity premium is almost 6%, despite the ability of agents to hedge shocks through labor, domestic investment, and external debt and equity positions. Our model replicates the dynamics of corporate leverage and, most importantly, those of both the net foreign asset position and its subcomponents. Both net external debt positions and net external equity positions are countercyclical, as in the data. In addition, the contemporaneous correlation of these components is moderate. This implies that external debt and equity positions provide distinct hedging benefits.

SONOMA is also able to replicate conditional responses documented in the data. The model reproduces key patterns of the impulse responses of both domestic and external financial aggregates with respect to shocks to external debt and equity positions. Furthermore, in the model as in the data, adverse shocks to external equity and debt markets are as disruptive as adverse productivity shocks. We find similar successes in terms of variance decompositions. The main departure between the model and the data concerns the variance of the investment-output ratio. In the data, this variance is mainly driven by external financial shocks, whereas in the model,

productivity shocks are just as important.

We complete our analysis by inspecting the mechanism, specifically by removing features from our model one at a time. When we compare our open economy to an otherwise equivalent closed one, we find that shocks to external financing conditions can have a first-order impact on economic activity, and they are priced, that is, they are relevant for the determination of the equity risk premium. We also compare our results with and without financial frictions and financial shocks and find that both of these dimensions are essential for SONOMA to replicate the data.

Related literature. This paper relates to both the finance and macroeconomics literatures. The literature on SOEs (see, among many others, Mendoza, 1991; Schmitt-Grohé and Uribe, 2003; and Mendoza, 2006) focuses mostly on developing or emerging markets, and it devotes significant attention to sudden stops or exchange rate regimes (among many others, see Mendoza and Smith, 2006; Chari et al., 2005; Aghion, Bacchetta, and Banerjee, 2001; and Mendoza and Uribe, 2000). Other studies focus on sovereign default (see, for example, Uribe and Yue, 2006; Aguiar and Gopinath, 2007; and Arellano, 2008). In this study, we focus on SDEs. While they still qualify as SOEs, they typically do not face the same type of currency problems that EMEs face. In addition, we view the countries in our sample as being subject to frequent regular financing shocks instead of being subject to sudden stops or occasionally binding constraints.

Recently, more attention has been given to production-based asset pricing in SOEs. For example, Jahan-Parvar, Liu, and Rothman (2013) study an SOE production-based asset pricing model featuring Greenwood, Hercowitz, and Huffman (1988) preferences in a setting based on Jermann (1998). We differ from these studies in our adoption of Epstein and Zin (1989) preferences and growth news shocks, as in Bansal and Yaron (2004). In one of our model configurations, we also consider downside risk, that is, adverse co-skewness across financial shocks and growth news shocks. In this sense, this study is related to the disaster risk literature (see, among others, Gourio, 2012; Gourio, Siemer, and Verdelhan, 2013; and Wachter, 2013).

Our study is closely related to the growing literature on financially con-

strained firms and the implications of credit shocks in a dynamic stochastic general equilibrium macroeconomic framework (see, for example, Gertler and Karadi, 2011; Khan and Thomas, 2013; Guerrieri and Lorenzoni, 2017; and Guerrieri and Iacoviello, 2017). Unlike our setting, these models feature closed economies.

In contrast to the literature that considers frictionless environments (see, among others, Gourio et al., 2013, and Stathopoulos, 2016), we focus on the relevance of financing frictions, financial flows and cycles, and country risks (see, for example, Atkeson, Heathcote, and Perri, 2025; Hassan, Schreger, Schwedeler, and Tahoun, 2023; Avdjiev, Hardy, Kalemli-Özcan, and Servén, 2022; di Giovanni, Kalemli-Özcan, Ulu, and Baskaya, 2021; Miranda-Agrippino and Rey, 2020; and Borri and Verdelhan, 2012). For a broader review of the international literature with financial frictions and segmentation see Maggiori (2022). In our study, exchange rate fluctuations have a minor empirical role, and hence we abstract away from the observations in Itskhoki and Mukhin (2021, 2025). Coeurdacier, Guibaud, and Jin (2015) consider financial frictions in an SOE to assess three prominent global trends: a divergence in private saving rates between developed and emerging economies, large net capital outflows from the latter, and a sustained decline in the world interest rate. We focus on SDEs in which (i) sources of long-run growth risk are directly priced, and (ii) the interplay of corporate financing decisions and external balances has a first-order impact on macro aggregates.

The remainder of this article is organized as follows. We detail our model in section II. We describe our empirical investigation in section III. In section IV, we discuss our calibration and main results, and section V concludes.

II. The Economy

As discussed in the introduction, we view SONOMA as either a benchmark small open economy model modified in three dimensions, or as an augmented open economy version of Jermann and Quadrini (2012) (henceforth, JQ). The original JQ model is not intended for production-based asset pricing. We modify JQ to price both risky and risk-free assets in an SOE

environment by first embedding recursive preferences (Kreps and Porteus, 1978; Epstein and Zin, 1989) and long-run risks (Bansal and Yaron, 2004). We then open the economy and expose it to exogenous financing shocks.

As in the JQ economy, firms need to decide the optimal mix of corporate-debt and equity financing. Firms issue debt because of the existence of a tax shield on corporate interest payments. A borrowing constraint limits the amount of corporate debt and introduces a wedge into the labor market. Our economy differs from JQ's along two important dimensions. First, households can borrow from or lend to the rest of the world. Second, households can go long or short in foreign equities.

The model features six sources of risk. Similar to traditional real business cycle models, there are shocks to the level of the productivity growth rate. We follow Croce (2014) and introduce long-run productivity growth risk. As in JQ, there are domestic credit condition shocks, that is, shocks to the firm's borrowing collateral constraint. We also consider a shock to external credit conditions, namely, a shock to the cost of borrowing from abroad (see, among others, Jahan-Parvar et al., 2013; Aguiar and Gopinath, 2007; Uribe and Yue, 2006). Importantly, the agent in SONOMA also faces shocks to external equity returns. We consider shocks both to the unexpected component of the returns and to their expected value (Sharpe ratio shocks). To the best of our knowledge, this is the first study that considers all of these financing margins simultaneously. One of our goals is to explore the relevance of these shocks for the dynamics of the components of net foreign assets (NFA).

A. Households' Problem

Preferences. There is a continuum of homogeneous households endowed with Kreps and Porteus (1978) recursive preferences as specified in Epstein and Zin (1989):

$$U_t = \left[(1 - \beta) \tilde{C}(C_t, \ell_t)^{1 - \frac{1}{\psi}} + \beta (\mathbb{E}_t U_{t+1}^{1 - \gamma})^{\frac{1 - \frac{1}{\psi}}{1 - \gamma}} \right]^{\frac{1}{1 - \frac{1}{\psi}}}, \quad (1)$$

where $\tilde{C}(C_t, \ell_t)$ is a consumption bundle defined over consumption (C_t) and leisure (ℓ_t), γ is a determinant of the coefficient of relative risk aversion, and ψ drives the elasticity of intertemporal substitution (IES). We define leisure as the portion of time not worked, $\ell_t \leq 1 - H_t^s$, where H_t^s is the household supply of labor. We define the consumption bundle as

$$\tilde{C}_t = \left(\varphi C_t^{1-\frac{1}{f}} + (1-\varphi) (SL_t \ell_t)^{1-\frac{1}{f}} \right)^{\frac{1}{1-\frac{1}{f}}}.$$

We set the weight φ so that steady-state labor supply is equal to 0.3. To ensure balanced growth with these preferences, we introduce an exogenous preference shock process, SL_t , cointegrated with measured productivity, A_t . Specifically, we define $e^{sla_t} := \frac{SL_t}{A_t}$ and assume that

$$sla_t = (1-\theta)(sla_{t-1} - (\Delta \ln A_t - \mu)).$$

We set θ to a moderate value so that SL_t mimics an exogenous linear trend and hence productivity shocks do not act as volatile demand shocks (Colacito, Croce, Ho, and Howard, 2018). The parameter μ determines the average growth of the economy.

The stochastic discount factor (SDF) of the household is

$$M_{t+1} = \beta \left(\frac{\tilde{C}_{t+1}}{\tilde{C}_t} \right)^{-\frac{1}{\psi}} \left(\frac{U_{t+1}}{E_t \left[U_{t+1}^{1-\gamma} \right]^{\frac{1}{1-\gamma}}} \right)^{\frac{1}{\psi}-\gamma} \frac{\partial \tilde{C}_{t+1} / \partial C_{t+1}}{\partial \tilde{C}_t / \partial C_t}. \quad (2)$$

Since the consumption bundle is a composite of both leisure and consumption, the last term in equation (2) reflects marginal variations due to the consumption good. The second term captures aversion to long-run risks.

Budget constraint. The household maximizes lifetime utility by choosing the levels of consumption and leisure and a portfolio of financial assets, subject to the following constraints:

$$\begin{aligned} C_t &\leq \overbrace{w_t H_t + (S_t - S_{t+1}) V_t^{ex} + S_t \cdot d_t + (1 + r_t) D_t - D_{t+1} - T_t^H}^{\text{Domestic variables}} \\ &\quad + \underbrace{(NED_{t+1} - (1 + r_t^{ED}) NED_t) - (NEE_{t+1} - (1 + r_t^{EE}) NEE_t)}_{\text{International Flows}} \\ 1 &\geq H_t^s + \ell_t, \end{aligned}$$

where w_t represents wages earned; S_t (D_t) is a predetermined variable measuring domestic equities (corporate bonds) owned by the household at the beginning of the period; V_t^{ex} is the ex-dividend share price; and d_t and r_t are the net equity payout and domestic corporate bond interest rate, respectively. The government rebates corporate taxes, T_t^H , to the households. Given this notation, the net domestic corporate debt payout (NDP) is $NDP_t = (1 + r_t)D_t - D_{t+1}$.

Net foreign assets. Turning our attention to the international side of the budget constraint, we have net external debt (NED_t) and net external equities (NEE_t) at time t . We define net foreign assets (NFA_t) in the economy as

$$NFA_t = NEE_t - NED_t.$$

We do not impose any restriction on the sign of these variables, meaning that a country can be long or short in both external debt and equity.

Let r_t^{ED} be the interest rate paid to external (i.e., foreign) lenders. Since our model does not admit sovereign default, we will have $r_t = r_t^{ED}$ at equilibrium. The return on the household's net external equity position is denoted by r_t^{EE} , and the current account (CA) can be written as the sum of the changes in net foreign assets and net factor incomes from abroad:

$$CA_{t+1} \equiv \Delta NFA_{t+1} - r_{t+1}^{EE} NEE_t + r_t^{ED} NED_t.$$

In this model, only households can borrow from abroad (lend to the rest of the world) by increasing (decreasing) their foreign debt position. Similarly, households can adjust their net external equity position. Proceeds from these activities can then be used for domestic investment. This assumption is irrelevant for our main results; however, we find it useful when comparing our open economy model with its closed-economy counterpart, as the domestic asset holdings are concentrated in the domestic households' portfolios in both configurations.

Cost of external financing. Following Schmitt-Grohé and Uribe (2003), Jahan-Parvar et al. (2013), and numerous other studies in international finance, we posit that the interest paid to external lenders, r_t^{ED} , is a func-

tion of the predetermined and exogenous world financing rate, r_{t-1}^w , and a predetermined and endogenous country-specific spread, P_t^{ED} . The country spread increases when the external debt position of the country goes above its steady-state value, $\overline{NEDY}$:

$$r_t^{ED} = r_{t-1}^w + P_t^{ED} \quad (3)$$

$$P_t^{ED} = p_{ED,2} e^{p_{ED,1}(NED_t/Y_t - \overline{NEDY})}. \quad (4)$$

Neumeyer and Perri (2005) argue that country spreads have both endogenous and exogenous components. In this study, the spread (P^{ED}) is defined to be endogenous, but exogenous random factors can affect the external interest rate (r_t^w) as follows:

$$r_t^w = (1 - \rho_{rw}) \mu_{rw} + \rho_{rw} r_{t-1}^w + \sigma_{rw} \epsilon_{rw,t}. \quad (5)$$

These shocks must be interpreted broadly, as they capture both changes in the world risk-free rate and changes in the sentiment of external lenders toward a specific SOE. Hence we refer to them as external credit condition shocks.

The returns on the net external equity position require more attention, as they represent a new dimension in our study. Since this position equals the difference between global equities held by country i 's residents and country i 's equities held by the rest of the world, the returns of the NEE portfolio are identical to those of an investment strategy long in global equities and short in domestic equities (for a detailed discussion, see section A). We model the log returns on the NEE portfolio, $\log R^{EE}$, as having three components:

$$r_t^{EE} := \log R_t^{EE} = \eta_{t-1} + P_t^{EE} + \sigma_{ee} \cdot \epsilon_{ee,t} + \rho_{rw,ee} \sigma_{rw} \epsilon_{rw,t} \quad (6)$$

$$P_t^{EE} = p_{EE} \cdot (\overline{NEEY} - NEE_t/Y_t), \quad (7)$$

where η_{t-1} is an exogenous and persistent process, P_t^{EE} is an endogenous wedge that depends on the predetermined NEE_t position, $\rho_{rw,ee}$ allows for correlation with interest rate shocks, and $\epsilon_{ee,t}$ is a pure unexpected *i.i.d.* shock which generates changes in the market value of the NEE portfolio (also known as the valuation channel). Because of $\epsilon_{ee,t}$ and $\epsilon_{rw,t}$, the return

to external equities is ex-ante uncertain, in contrast to the cost of external debt financing.²

We impose $p_{EE} > 0$ so that the expected return from investing in foreign equity declines as NEE_t increases and hence there is no incentive to take infinitely long (or short) positions in external equities. Equivalently, equations (4) and (7) ensure a unique steady state and stationarity. We model the exogenous part of the expected log returns of the NEE portfolio as

$$\eta_t = (1 - \rho_\eta) (\mu_{rw} + s_{rw}) + \rho_\eta \eta_{t-1} + \sigma_\eta \epsilon_{\eta,t}. \quad (8)$$

Given our formulation, when the NEE position is equal to its long-run value $\overline{NEEY}$, the expected foreign cost of equity equals exactly η_t . The parameter s_{rw} is computed as a residual adjustment such that the return on the NEE is equal to the cost of external debt at the deterministic steady state. Equivalently, absent risk, all assets pay the same risk-free return and no arbitrage exists. Since in our model we abstract away from time-varying volatility shocks, $\epsilon_{\eta,t}$ must be interpreted as generating small but long-lived fluctuations in the foreign equity Sharpe ratio. In our empirical analysis, we recover these shocks by working with returns adjusted by their ex-ante time-varying volatility. We show that Sharpe ratio shocks, $\epsilon_{\eta,t}$, are more relevant than unexpected realized shocks, $\epsilon_{ee,t}$.

Optimality and exchange rate. All first-order conditions are standard and are detailed in [Appendix A](#). In the empirical study, we focus on European countries and benchmark them against Germany. As a result, real exchange rate fluctuations have a marginal role in explaining the variability of our variables of interest.³

²This shock captures unexpected variation in both exchange rates and foreign equity returns.

³In [Appendix D.1](#) we show that for the SDEs in our sample, exchange rate fluctuations are far less important as drivers of macrofinancial quantities compared to EMEs. Given this observation, we work with a parsimonious model in which there is only one good and there is no need to account for shocks to the terms of trade.

B. Firms' Problem

Technology. There is a continuum of identical firms with production function

$$F_t := F(A_t, K_t, H_t) = K_t^\alpha (A_t H_t)^{1-\alpha},$$

where K_t , H_t , and A_t represent capital, labor, and the stochastic level of productivity, respectively. K_t is predetermined at time $t - 1$, whereas labor is determined flexibly at time t . Capital evolves according to

$$K_{t+1} \leq (1 - \delta)K_t + I_t - \Phi\left(\frac{I_t}{K_t}\right) K_t, \quad \Phi\left(\frac{I_t}{K_t}\right) = \frac{\phi_1}{1 - \frac{1}{\phi_2}} \left(\frac{I_t}{K_t}\right)^{1 - \frac{1}{\phi_2}} + \phi_3,$$

where δ is the depreciation rate for capital, I_t is investment at time t , and $\Phi\left(\frac{I_t}{K_t}\right)$ is the capital adjustment cost as defined in Jermann (1998). The elasticity of adjustment costs is determined by ϕ_2 .

In this study, we assume that productivity growth evolves as follows:

$$\Delta a_{t+1} = \mu_a + x_t + \sigma_a \epsilon_{a,t+1}, \tag{9}$$

$$x_{t+1} = \rho_x x_t + \sigma_x \epsilon_{x,t+1} + \rho_{rw,x} \sigma_{rw} \epsilon_{rw,t+1} + \rho_{\eta,x} \sigma_\eta \epsilon_{\eta,t+1}, \tag{10}$$

where x_t is the long-run productivity risk (LRR) component (Croce, 2014), and the volatilities of short- and long-run risks are represented by σ_a and σ_x , respectively. The coefficients $\rho_{rw,x}$ and $\rho_{\eta,x}$ capture potential correlations of the long-run risk process with the external interest rates (equation (5)) and the Sharpe ratio shocks (equation (8)), respectively.

Financing problem. The firm finances its operations by issuing corporate debt or equities that are sold to the domestic household. The household can use foreign funds to buy local equities or debt. This convention enables us to replicate the closed-economy setting of JQ when we impose $NEE \equiv 0$ and $NED \equiv 0$. As in JQ and Hennessy and Whited (2005), corporate debt carries a tax advantage over equity and thus it is the preferred source of funding for the firm. The corporate capital structure is well defined because of the existence of a working capital constraint.

More specifically, the firm chooses labor, investment, and debt and equity

issuance to maximize its cum-dividend value V_t ,

$$V_t = d_t + \mathbb{E}_t [M_{t+1} V_{t+1}],$$

where d_t is the net equity payout, and M_t is the domestic SDF.⁴ This optimization is subject to the following constraints:

$$d_t \leq F(A_t, K_t, H_t) - w_t H_t - I_t - \chi(d_t) + D_{t+1} - D_t(1 + r_t) - T_t^c \quad (11)$$

$$F_t \leq \xi_t (K_{t+1} - D_{t+1}). \quad (12)$$

Equation (11) represents a standard firm's budget constraint in which D denotes corporate debt. Equation (12) is a working-capital borrowing constraint. As in JQ, expanding production requires intraperiod borrowing that must be backed by collateral measured as book equity, $K - D$. Following JQ, we impose a quadratic cost on the equity payout to make equities and bonds imperfect substitutes,

$$\chi(d_t) = A_{t-1} \cdot \kappa \left(\frac{d_t}{SL_{t-1}} - \bar{d} \right)^2,$$

where $\kappa \geq 0$, $\bar{d}$ is the steady-state payout target, and SL_t mimics a time trend that enables us to have balanced growth. The government collects a corporate revenue tax, T_t^c ,

$$T_t^c = \tau_F (F_t - w_t H_t - D_t r_t), \quad (13)$$

where τ_F is a constant tax rate, and the tax base equals corporate revenues $F_t - w_t H_t$ net of the cost of predetermined corporate debt, $D_t r_t$.

In equation (12), we impose a credit constraint subject to time-varying leverage shocks, as in JQ. The process ξ_t captures exogenous changes in

⁴Given our assumptions, the domestic household is the marginal investor, and only its SDF matters.

domestic credit conditions, and it evolves as follows:

$$\xi_{t+1} = (1 - \rho_\xi) \mu_\xi + \rho_\xi \xi_t + \epsilon_{\xi,t+1} + \rho_{rw,\xi} \sigma_{rw} \epsilon_{rw,t} + \rho_{\eta,\xi} \sigma_\eta \epsilon_{\eta,t}. \quad (14)$$

The parameters $\rho_{rw,\xi}$ and $\rho_{\eta,\xi}$ determine the contemporaneous impact of shocks to external financing conditions on domestic credit conditions.

Optimality. Our optimality conditions are identical to those in JQ and are reported in [Appendix A](#). Our financial constraint causes a wedge in the labor market such that a tighter credit constraint reduces labor demand and economic activity. In this class of models, there is a possibility that the credit constraint may not be binding in some states. In our implementation, we observe negligible instances of such an event. For robustness, we also solve the companion convexified version of the problem by removing the credit constraint and introducing convex distress costs on working capital. Our results remain unchanged (see [Appendix B](#)).

In what follows, the return on domestic equities is denoted as R_E , whereas the return on assets is denoted as R_K . At the equilibrium, $R_{E,t+1} = \frac{V_{t+1}}{V_t - d_t}$. The expression for the (unlevered) return on assets can be found in [Appendix A](#), equation (A3).

C. Tax Policy and Market Clearing

Government. The government runs a balanced budget policy; specifically, it collects corporate taxes and rebates them in a lump sum to the household. The main purpose of this margin is to generate an incentive to issue corporate debt.

Market clearing conditions. The recursive competitive equilibrium in this economy requires that (i) the labor market clears, (ii) all equities and corporate bonds are held domestically by the household, and (iii) the goods market clears:

$$F_t - \chi(d_t) = C_t + I_t + CA_t.$$

III. Empirical Evidence

The goal of our empirical investigation is twofold. First, we establish important and novel empirical macrofinancial regularities in SDEs. Second, we use our empirical findings to provide new empirical moments that can guide both our calibration and future quantitative studies.

We start our empirical analysis by investigating the properties of the fundamental shocks that drive the economic dynamics in the SONOMA model. We then look at the contribution of our shocks to the long-run variance of relevant macrofinancial variables. We focus on a cross section of 10 Western European SDEs: Austria, Belgium, Denmark, Finland, Italy, the Netherlands, Portugal, Spain, Sweden, and Switzerland. The main data sources that we use are reported in [Appendix C](#). We visualize our results on this webpage: <https://sites.google.com/view/sonoma-macrofin/home>.

A. Country-Level Measurements

For each country in our sample, we compute essential aggregate variables needed for our analysis. In what follows, we focus on the series that are necessary to compute our fundamental shocks.

Capital stock. In each country j , we measure the capital stock as in JQ by capitalizing quarterly investment net of depreciation,

$$K_{t+1}^j = K_t^j(1 - \delta_t^j) + I_t^j.$$

Quarterly investment (I^j) data are from the OECD dataset, whereas the depreciation rate (δ^j) is from the Penn World Table (PWT).⁵

Productivity. For each country j , we compute total factor productivity A_t^j by postulating the Cobb-Douglas production function $Y_t^j =$

⁵For each country, we initialize the recursion above by ensuring that (i) the initial and the final values of capital-to-output have the same value (as in JQ); and (ii) the average capital-to-output ratio is equal to that computed from the annual data from PWT. Our empirical approach abstracts away from adjustment costs.

$A_t^j K_t^{j\alpha_{j,t}} H_t^{j1-\alpha_{j,t}}$, in which K_t^j measures beginning-of-quarter aggregate capital, H_t^j refers to total labor hours, and $\alpha_{j,t}$ is a moving average of the labor income share of GDP as reported in the PWT. JQ recommend using business value added to measure Y_t^j . However, we believe that GDP is more appropriate in our case, since the capital stock and labor series are total measures that include all sectors and economic activities. In some of the countries in our sample, nonprofit- and government-sector activities are sizeable and should be accounted for in our empirical exercise.

Long-run productivity shocks. As in Croce (2014), we estimate the long-run component of productivity growth by forecasting the demeaned productivity growth rate ($\Delta a_{t+1}^j - \mu_a^j$) using a set of lagged variables as predictors,

$$\Delta a_{t+1}^j - \mu_a^j = \overbrace{b_x^j X_t^{j,dm}}^{x_t^j} + \epsilon_{a,t+1}^j \quad (15)$$

$$x_t^j = \rho^j x_{t-1}^j + \epsilon_{x,t}^j. \quad (16)$$

Specifically, the vector $X_t^{j,dm}$ includes the demeaned lagged values of the log price-dividend ratio, inflation, and risk-free rate. This approach is common in the long-run growth news shock literature. See [Appendix D](#) for more details.

Domestic credit conditions. We take seriously the JQ approach and measure credit tightness, ξ_t^j , directly from the following ratio derived from equation (12),

$$\xi_t^j = \frac{Y_t^j}{K_{t+1}^j - D_{t+1}^{j,end}},$$

where K_{t+1}^j ($D_{t+1}^{j,end}$) measures the end-of-quarter capital stock (corporate debt, provided by the BIS). In contrast to JQ, we do not focus on detrended variables, that is, on deviations from a long-run trend. Given our attention to long-run dynamics and expectations, we extract our measures of domestic credit conditions, ξ_t^j , using raw aggregate variables in levels.

External credit conditions. We model the external interest rates as specified in equations (3) and (4). In the data, this rate is measured using country-specific long-term sovereign yields, $r_t^{ED,j}$, which we decompose as follows:

$$(1 + r_t^{ED,j}) = (1 + r_t^{Ger}) \cdot \underbrace{(1 + P_t^{ED,j})}_{\text{driven by NED}} \cdot \underbrace{\exp(\epsilon_t^{ED,j})}_{\text{residual country spread}},$$

where r_t^{Ger} is the real German rate, our measure for the world risk-free interest rate. To identify the country-specific endogenous spread, $P_t^{ED,j}$, we proceed as follows. We compute the log difference between the real interest rate of country j , $1 + r_t^{ED,j}$, and $1 + r_t^{Ger}$. These data are obtained from the OECD and refer to long-term debt. We measure the demeaned country-level net external debt ratio, $(NED/Y^j - \overline{NEDY}^j)$, using data from the IMF IIP/BOP database (see the methods in Lane and Milesi-Ferretti, 2007). We assume that the parameters $p_{ED,1}$ and $p_{ED,2}$ are common across countries and estimate them through a panel regression that enables us to isolate the country-specific spread:

$$\ln(1 + \hat{P}_t^{ED,j}) = \underbrace{\alpha_{IV}}_{\substack{0.0008 \\ (0.0004)}} + \underbrace{\beta_{IV}}_{\substack{0.0026 \\ (0.0007)}} \left(NED_t^j / Y_t^j - \overline{NEDY}^j \right) \quad j = 1, 2, \dots, 10, (17)$$

where α_{IV} and β_{IV} refer to the fact that we instrument NED_t^j / Y_t^j with its lagged value and report corrected standard errors. We derive the exogenous component of the external interest rate for each country, $\hat{r}_t^{j,w}$, in the following residual way:

$$\hat{r}_t^{j,w} = \frac{1 + r_t^{ED,j}}{1 + \hat{P}_t^{ED,j}} - 1 \approx r_t^{Ger} + \epsilon_t^{ED,j}.$$

Consistent with our model description, we refer to $\hat{r}_t^{j,w}$ as a measure of external credit conditions throughout the remainder of our paper. This process embodies both changes to the world risk-free rate and changes to the external credit sentiment toward country j . Additional details and results are available in [Appendix D](#).

External cost of equity. In our model, we do not have frictions that determine gross positions, that is, the foreign holdings of domestic assets and the domestic holdings of foreign assets. As a result, we focus on $R_t^{EE,j}$, the gross return on the NEE position of country j , which is measured as in Lane and Milesi-Ferretti (2007). We construct this return using gross positions. Specifically, in each country j , the time- t value of the local NEE portfolio net of new purchases, $NPEE_t^j$, is

$$NEE_t^j - NPEE_t^j = A_{t-1}^{EE,j} R_t^{E,w} - L_{t-1}^{EE,j} R_t^{E,j},$$

where $R_t^{E,w}$ is the return on the rest-of-the-world equity asset; $R_t^{E,j}$ is the return on equity in country j ; and $A_{t-1}^{EE,j}$ and $L_{t-1}^{EE,j}$ are the time $t-1$ balances of external equity assets and liabilities, respectively. We divide and then multiply by $NEE_{t-1}^j \equiv A_{t-1}^{EE,j} - L_{t-1}^{EE,j}$ to find

$$NEE_t^j - NPEE_t^j = \left(\underbrace{\frac{A_{t-1}^{EE,j}}{A_{t-1}^{EE,j} - L_{t-1}^{EE,j}} R_t^{E,w} - \frac{L_{t-1}^{EE,j}}{A_{t-1}^{EE,j} - L_{t-1}^{EE,j}} R_t^{E,j}}_{R_t^{EE,j}} \right) NEE_{t-1}^j. \quad ^6$$

In our analysis, we use the Major Markets International Index Portfolio provided by Kenneth French to measure the world returns. Kenneth French's Data Library also provided the country-specific equity market returns used to measure the cost of external equity liabilities. Both sets of raw returns are measured in U.S. dollars, which are then converted to real returns using country-specific inflation. For world returns, we use U.S. inflation.

⁶If the country holds foreign equities but has no international equity liability (i.e., $L_{t-1}^{EE} = 0$), then $R_t^{EE} = 1 + r_t^{E,w}$ and $NEE_t = R_t^{EE} A_{t-1}^{EE}$. If the country holds no foreign equity asset (i.e., $A_{t-1}^{EE} = 0$), then $R_t^{EE} = 1 + r_t^{E,j}$ and the external position would be $NEE_t = -R_t^{EE} L_{t-1}^{EE}$. If $A^{EE} = L^{EE}$, R_t^{EE} is undefined, but the NEE evolution can still be computed as $NEE_t = (R_t^{E,w} - R_t^{E,j}) A_{t-1}^{EE}$.

Composition of R^{EE} . To estimate the components in equation (6), we must be able to both identify η_{t-1}^j and estimate the key parameters of the endogenous spread between internal-equity ($R^{E,j}$) and external-equity ($R^{E,w}$) expected returns. In our model, volatility is assumed to be constant. Hence when η_{t-1}^j increases, foreign stocks are relatively more appealing, as they exhibit a better Sharpe ratio. In our data, by no-arbitrage, expected equity returns are driven by both volatility shocks and Sharpe ratio shocks. Empirically, we must disentangle pure volatility shocks from Sharpe ratio shocks to properly calibrate our model. We work with a volatility-adjusted spread (VAS_t^j) defined as follows:

$$VAS_t^j := \frac{E_t[R_{t+1}^{E,w} - R_{t+1}^{E,j}]}{\sigma_t^j},$$

where $\sigma_t^j := \text{vol}_t(R_{t+1}^{E,w} - R_{t+1}^{E,j})$. We recover $\hat{\sigma}_t^j$ by estimating a GARCH(1,1) model over the full sample. In the spirit of the finance literature, we use the world price-dividend ratio, pd^w , to capture variations in the global investment frontier by assuming that

$$VAS_t^j = \alpha_{1,j} + \beta_{pd} pd_t^w \quad j = 1, \dots, 10.$$

Equivalently, we allow for a country-specific constant, but we assume a common β_{pd} to achieve a tighter identification of this parameter. We then impose that

$$\eta_t^j := VAS_t^j \times \bar{\sigma},$$

where $\bar{\sigma} = \text{Avg}(\sigma_t^j)$ so that our empirical measure of the external equity market conditions is driven only by country-level Sharpe ratio shocks.

For empirical investigation purposes, no additional assumptions are required. Given (i) the linear dependence of η_t^j on pd_t^w , and (ii) the restriction that all countries share the same β_{pd} , in our VAR analysis we can directly use data on pd_t^w in order to capture the role of shocks to η_t^j . In order to calibrate our model, however, additional work is required to estimate the coefficient $\hat{p}_{EE}$. We discuss this task in section A. Here we anticipate that $\hat{\beta}_{pd} = -0.445$, with a HAC-adjusted standard error of 0.048. The negative sign of this estimate implies that when global sentiment is high (high pd^w), foreign equities offer lower Sharpe ratios relative to domestic equities. In what follows, we refer to our fitted η_t^j as the NEE expected equity return

(NEE-ER) process.

Visual inspection. Figure D1 (Appendix D) displays the time series of our main exogenous variables. Our long-run component captures the positive revision of expected growth in the late 1990s, as well as that during the global financial crisis (GFC). Our process for domestic credit conditions starts to decline in 2008, and it drops even further during the European debt crisis. It starts to pick up again post-2015, in the recent period of improved credit. The external cost of debt shows a downward trend consistent with the well-known secular decline in interest rates across the globe. Our VAS picks up the increases in external Sharpe ratios that we observed in the post-IT bubble (post-2000), and post-GFC (post-2008).

B. Aggregation and VAR Specification

In what follows, we report the results that we obtain by estimating VAR models using the time series of data averaged across countries. We do so for two reasons. First, at the country level we have an unbalanced panel (see Appendix C) with a small cross section when we focus on a long time series. The use of cross-country averaged data enables us to work with long time series. Second, our results could be viewed as applying to a representative SDE, and hence they can be used to calibrate our model. Alternatively, our approach captures common components which affect all the countries in our sample. By focusing on common shocks, as opposed to idiosyncratic shocks, we mitigate concerns about the ‘third-country effect’ due to the strong integration of our countries.

In addition, we observe that the country-level results are qualitatively similar to those which we report here for the averaged variables (as shown in Appendix D.2). Our results are confirmed both using GDP-weighted data and using equal-weighted data. We find these results reassuring, as they confirm the robustness of our main findings.

We begin our empirical analysis by studying the relevance of all of our exogenous shocks for key endogenous variables of interest. Specifically, we

document novel salient empirical facts by estimating the following VAR:

$$Y_t = \Phi Y_{t-1} + \Sigma u_t, \quad (18)$$

in which

$$Y_t = \begin{bmatrix} r_t^w & pd_t^w & r_t^{EE} & \xi_t & x_t & Z_t \end{bmatrix}', \quad (19)$$

and where r_t^w , pd_t^w , r_t^{EE} , ξ_t , and x_t denote the exogenous component of the external interest rate, the world price-dividend ratio, the returns on net external equity position, the domestic credit tightness, and the long-run productivity component, respectively. As explained in the previous section, the variable pd_t^w identifies time series variation in expected returns, NEE-ER, whereas r_t^{EE} captures unexpected shocks to realized NEE returns.

In what follows, we use a lower-triangular Cholesky decomposition to orthogonalize our shocks. As a result, external shocks are treated as ‘more exogenous,’ in the spirit of the SOE literature. The last variable, Z_t , refers to an endogenous variable of interest. In the next section, we focus on the net external debt-to-output ratio; net external equity-to-output ratio; investment-to-output ratio; and corporate leverage, that is, the debt-to-assets ratio.

In section D, since we study only the dynamic properties of our exogenous shocks, we replace equation (19) with

$$Y_t = \begin{bmatrix} r_t^w & pd_t^w & r_t^{EE} & \xi_t & x_t \end{bmatrix}'. \quad (19b)$$

C. SOE Accounting: Variance Decomposition

We depict the forecast error variance decomposition (FEVD) values recovered from our VAR in Figure 1. We note several interesting facts. First, credit shocks are first-order in explaining the medium- and long-run adjustments of both net external debt and equity. A substantial share of external debt variance over a 10-year horizon is explained by internal credit shocks. Through the lens of our equilibrium model, this is consistent with the fact

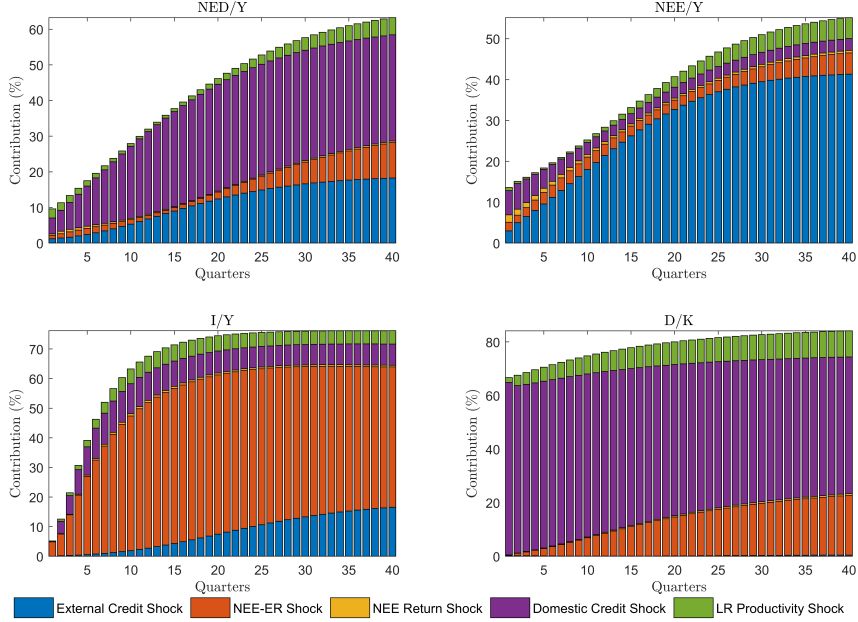


Figure 1. Relevance of Exogenous Shocks. This figure shows the forecast error variance decomposition over a 40-quarter horizon for the following variables of interest: net external debt-to-output ratio (NED/Y); net external equity-to-output ratio (NEE/Y); investment-to-output ratio (I/Y); and corporate leverage, i.e., debt-to-assets ratio (D/K). All results are based on the VAR specified in equations (18) and (19) using data constructed from cross-country averages. Our sources are detailed in [Appendix C](#), and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

that internal credit shocks can be hedged by adjusting external leverage. In contrast, internal credit shocks play a minor role in explaining net external equity fluctuations. In the next section, we show that this is consistent with the observation that international credit conditions are correlated with adverse external equity shocks, and hence internal shocks are not hedged by adjusting the external equity positions. Interestingly, external credit shocks explain a large share of the fluctuations of the net external equities, implying that external credit shocks prompt a substantial reallocation across external debt and equity. This dominance of financial shocks is also consistent with the evidence for EMEs emphasized by García-Cicco et al. (2010) and Chang and Fernandez (2013), who show that financial frictions play a central role in shaping business-cycle dynamics.

When we turn our attention to investment, we see that financial shocks play a dominant role in explaining the medium- and long-run fluctuations. This is a novel and relevant empirical fact that goes beyond what JQ have established about the United States. More specifically, shocks to external equity markets are a key determinant of the investment-to-output ratio.

The role of long-run shocks deserves several clarifications. First of all, it is well documented that there is a strong global component in long-run expected productivity (Colacito et al. (2018)). In our VAR, this common component is picked up mostly by the world price-dividend ratio and the external interest rate. According to our FEVD analysis, these two variables explain about 50% of the variance of x . For this reason, in what follows, we think of external financing shocks as important determinants of long-run productivity dynamics. Not surprisingly, the residual (i.e., Cholesky-orthogonalized) part of our x process explains a moderate share of the variance of the macro-financial aggregates that we study. In the model, these residual shocks contribute to deliver a high equity premium.

Finally, we note that domestic corporate leverage is primarily driven by internal credit shocks. Given how we measure credit shocks, this is not surprising. However, we find it important to note that external equity shocks explain about 20% of the fluctuations of the corporate capital structure over the long-run, which is a relevant portion. As for investment, long-run productivity shocks explain a significant portion of variation over long horizons. In the next section, we reassess these empirical facts through the lens of our equilibrium model.

SDEs versus EMEs. The SOE empirical literature has focused primarily on EMEs (among others, see García-Cicco et al., 2010; Chang and Fernandez, 2013). We find it important to complement this literature by comparing our results across SDEs. Specifically, we repeat our VAR analysis using a sample of five Latin American EMEs: Brazil, Chile, Colombia, Mexico, and Peru. We focus on a single geographic region in order to obtain a relatively homogeneous comparison group exposed to similar regional financial conditions, mirroring the European focus of our SDE sample. These countries also

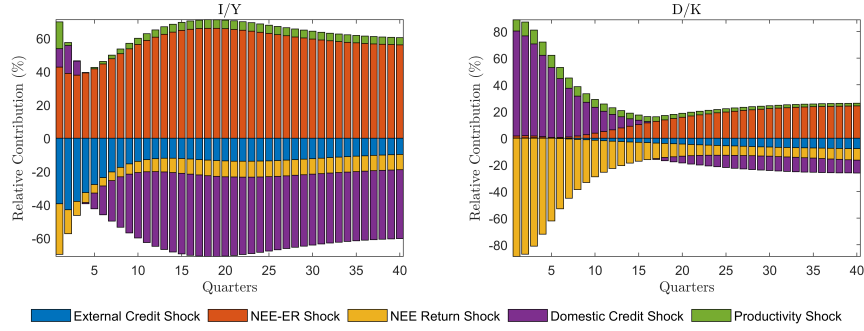
overlap substantially with the benchmark EME samples studied in the related SOE literature.⁷ Although additional Latin American EMEs have been examined in the literature, the joint availability of the variables required for our VAR analysis remains an important constraint.⁸

In Figure 2, we report differences in the FEVD shares observed for SDEs less those for EMEs. An exogenous shock is assigned a positive (negative) bar when its relevance is relatively more prominent in SDEs (EMEs). For the sake of focus, we study only domestic variables in the main text and report additional results and details in Appendix D.1. Here we point out that in EMEs, estimating the long-run component x with statistical power is challenging. Hence, when comparing EMEs and SDEs in our VAR we use Δa , as opposed to x . Correspondingly, in our Figure 2 we refer to productivity shocks, not to long-run news shocks.

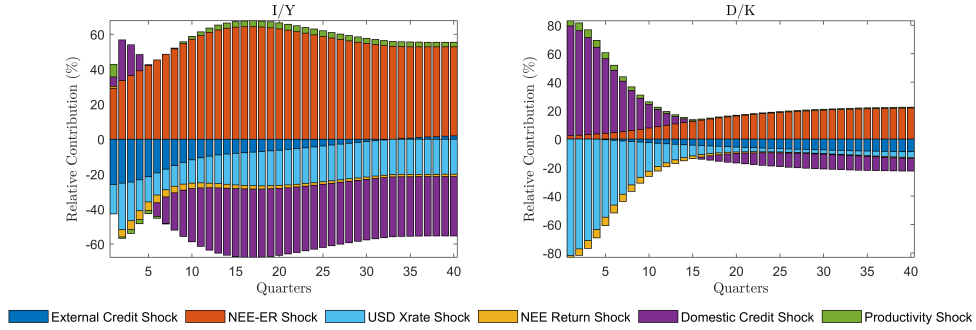
We note three key differences across SDEs and EMEs. First, NEE-ER shocks are a prominent source of variation in SDEs. Given that international equity markets are both more internationally integrated and developed in SDEs than in EMEs, external equity markets represent a key source of long-term financing in SDEs that should not be neglected. This result is both

⁷All five countries appear in Uribe and Yue (2006), although Chile and Colombia are considered only in their robustness exercise. Brazil, Mexico, and Peru are also included in Aguiar and Gopinath (2007). Brazil and Chile are studied in Jahan-Parvar et al. (2013), while Brazil and Mexico appear in Neumeyer and Perri (2005). Mexico is the primary focus of Mendoza (2010) and features prominently in Chang and Fernandez (2013) and García-Cicco et al. (2010).

⁸As a prominent example, Argentina is commonly studied in the emerging-market literature (e.g., García-Cicco et al., 2010) but is not included in our EME sample because the long-term sovereign yield series required to construct our external credit measure is unavailable for most of our quarterly sample. Argentina is not covered by our benchmark OECD long-term government bond yield database, and the corresponding Bloomberg 10-year government bond yield series begins only in 2016. Additional details regarding country-level data sources and availability are provided in Appendix C.



(a) VAR Specification Without Exchange Rate Shocks



(b) VAR Specification Including Exchange Rate Shocks

Figure 2. Difference in Relevance of Exogenous Shocks: SDEs Minus EMEs.

The top panel reports the differences in the values reported in the bottom panels of Figure D4 and Figure D5. The bottom panel, which incorporates exchange rate shocks relative to the U.S. dollar, reports the differences in the values reported in top and bottom panels of Figure D6. In all panels, the underlying values from each set of countries are the relative shares of variance explained by each variable, expressed as a proportion of the total variance explained among the given variables. A positive (negative) bar denotes a shock whose share of variance is more prominent in SDEs (EMEs).

novel and distinct from what we know about EMEs.

Second, in EMEs, domestic investment is more sensitive to credit shocks than to NEE-ER shocks. This is consistent with the existing literature on EMEs linking corporate investment to external borrowing shocks.

Third, unexpected external equity shocks are relatively more important in EMEs; that is, sudden adjustments in corporate investment are associated with sudden changes in key drivers of equity valuation.⁹ Since the NEE re-

⁹Figure D7 documents that domestic investment is more sensitive to unexpected ex-

turns are expressed in U.S. dollars, we consider it important to check whether this result is mostly driven by changes in exchange rates or by returns in local units. We add the exchange rate variation relative to the U.S. dollar averaged across countries in our VAR right after the world price-dividend ratio, so that unexpected returns capture shocks unrelated to currency adjustments. We report our results in the bottom portion of Figure 2.¹⁰

Consistent with prior literature, our results confirm that exchange rate shocks are a key driver in EMEs. We show that this is true for both investment and leverage. In contrast, SDEs feature more stable currencies, and hence unexpected changes in the exchange rate are less relevant than in EMEs. This is another important dimension in which we differ from earlier literature.

In Appendix D.1, we apply a similar analysis to the NED- and NEE-to-GDP ratios (Figure D8). We confirm that unexpected equity return shocks are relatively more important in EMEs than in SDEs. In addition, external credit shocks explain a relatively larger portion of short-run NED variance in EMEs.

Additional robustness. In what follows, we explore the sensitivity of our benchmark results for our SDEs when we account for (i) exchange rate risk, and (ii) monetary policy shocks. In principle, both of these omitted channels might change our results regarding financial shocks. In what follows, however, we show that our main results are robust. In the spirit of what was done for Figure 2, we compute shares of variance for our endogenous variables and focus on their variation across our benchmark VAR and a modified version of it. We focus on 10-year horizon shares and report our results in Table I.

We augment equation (19) by placing the real exchange rate as the second-

 ternal equity shocks in EMEs than in SDEs.

¹⁰Real exchange rate fluctuations are the percent change in the country’s local currency nominal exchange rate relative to the U.S. dollar minus the difference in the local and U.S. inflation rates. The cross-country average is simply the equal-weighted average within each set of countries.

to-last variable, and find that our results are practically unchanged whether we benchmark against Germany (i.e., EUR) or the United States (i.e., USD). Exchange rate fluctuations have some additional explanatory power in terms of variance decomposition, but their inclusion does not fundamentally alter our prior findings.

To capture monetary policy shocks, we follow the market-based procedure of Cieslak and Pang (2021). For robustness, we consider three different measures of monetary policy surprises extracted from German short-term debt, German long-term debt, and euro-area equities. Similar to the case of exchange rate fluctuations, we find no substantial change in our results.

Panel VAR. As an additional robustness exercise, we re-estimate our benchmark VAR using a first-order fixed-effects panel VAR on the underlying country-level data rather than cross-country averages. The resulting FEVDs are qualitatively very similar to those obtained from the benchmark specification. In particular, the panel VAR reproduces both the relative importance of exogenous shocks for SDEs (Figure 1) and the differences between SDEs and EMEs (Figure 2). Additional discussion and corresponding figures are reported in Appendix D.3.

D. Dynamic Properties of Exogenous Shocks

In this section, we document novel salient empirical facts about the dynamics and comovements of macrofinancial shocks in SDEs. We do so by studying the impulse responses obtained by estimating the VAR specified in (18) and (19b).

Our VAR analysis highlights several relevant results about our fundamental processes. First, both adverse NEE-ER shocks and external credit condition shocks are correlated with negative long-run growth news, meaning that tight external markets tend to be associated with periods of lower expected long-run growth (Figure 3, left panel). As mentioned in the previous section, these two shocks alone explain about 50% of the variance of the long-run component. As a robustness exercise, in Appendix D.4 we reconstruct $\hat{x}$

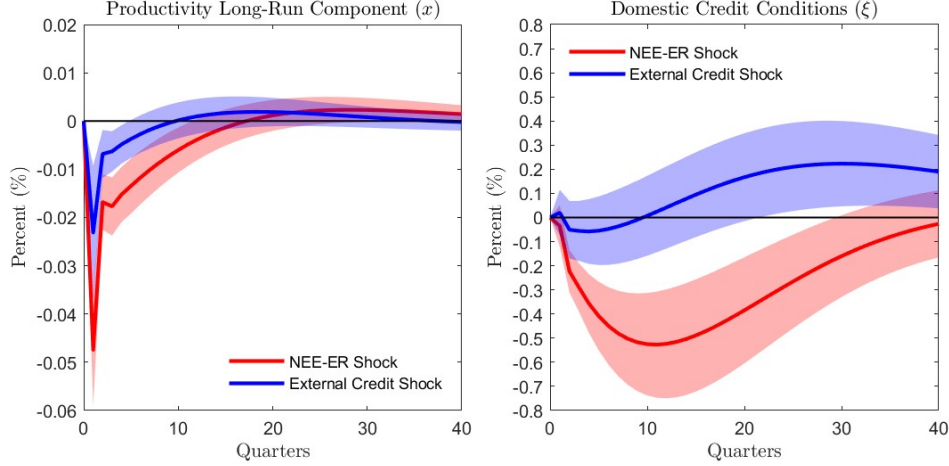


Figure 3. Empirical Impulse Responses to External Financial Shocks. This figure shows the responses of the long-run component of productivity (x) and domestic credit conditions (ξ) to adverse shocks from external credit (r^w) and equity (NEE-ER, η) markets. All results are based on the VAR specified in equations (18) and (19b) using data constructed from cross-country averages. Confidence intervals are HAC-adjusted. Our sources are detailed in [Appendix C](#), and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

without using the domestic price-dividend ratio. Under this alternative construction, the estimated η -to- x coefficient declines in magnitude from -0.12 to -0.04 , but remains negative and statistically significant. We also show that this attenuation does not materially alter our conclusions regarding the relative importance of international shocks.

Second, adverse NEE-ER shocks (η_t increases) are a leading indicator of tighter internal credit markets, that is, the ξ_t process declines (Figure 3, right panel). Consistent with the FEVD results in the previous subsection, these findings are also robust to estimating the VAR using a first-order fixed-effects panel VAR on the underlying country-level data (see [Appendix D.3](#)). The effects of external financial shocks are long-lived and robust to many other Cholesky orderings (see [Appendix D](#)).

Interestingly, in this VAR, external credit shocks do not have a significant correlation with internal credit conditions. In order to better interpret this result, we must recall that the external financing rate of a country, r_t^w , can be decomposed into a common risk-free component and a country-specific exter-

nal sentiment component. In our case, the sentiment component is captured by the country-specific spread ($P_t^{ED,j}$) relative to Germany (see equation (17)). When international investors suddenly become more concerned about a country’s default risk, the spread increases. In what follows, we link our sentiment measure to the sovereign credit default swap spread ($SovCDS$) of our sample countries.

In addition, we note that our model features one representative corporation. In reality, there are many corporations, and an internal credit shock can result in tighter credit conditions for the riskier firms, i.e., it manifests as a higher credit spread. In what follows, we also consider the spread between corporate yields for BBB-rated and AAA-rated bonds in Europe, which we refer to as the European corporate spread ($r^{CorpSpd}$). Our sources are described in [Appendix C](#).

Given this premise, we estimate the same VAR specified in (18) with the following expanded set of variables:

$$Y_t = \begin{bmatrix} r_t^{Germany} & r_t^{CorpSpd} & SovCDS_t & P_t^{ED} & pd_t^{world} & \xi_t & x_t \end{bmatrix}'. \quad (20)$$

As in the previous specification, the external financial variables are treated as ‘more exogenous,’ in the spirit of the SOE literature. Our main results are reported in [Figure 4](#). First, when the CDS spread increases (i.e., the sentiment deteriorates), the external credit cost increases and, simultaneously, domestic credit conditions deteriorate. Hence, confidence in government solvency is a common component across both external and internal credit conditions, consistent with [Bocola \(2016\)](#). Second, when confidence about corporate BBB bonds declines (i.e., the corporate spread increases), domestic credit conditions decline, but the external country spread does not react. This observation has two important consequences. The response of the domestic component in [Figure 3](#) is mostly capturing the lack of correlation between a pure internal corporate credit shocks and external sentiment. Second, external bonds represent a hedge against pure domestic credit crunches.

Lastly, we note that in our 1995–2018 sample, CDS-related shocks explain at most 30% (15%) of the external credit cost (domestic credit conditions)

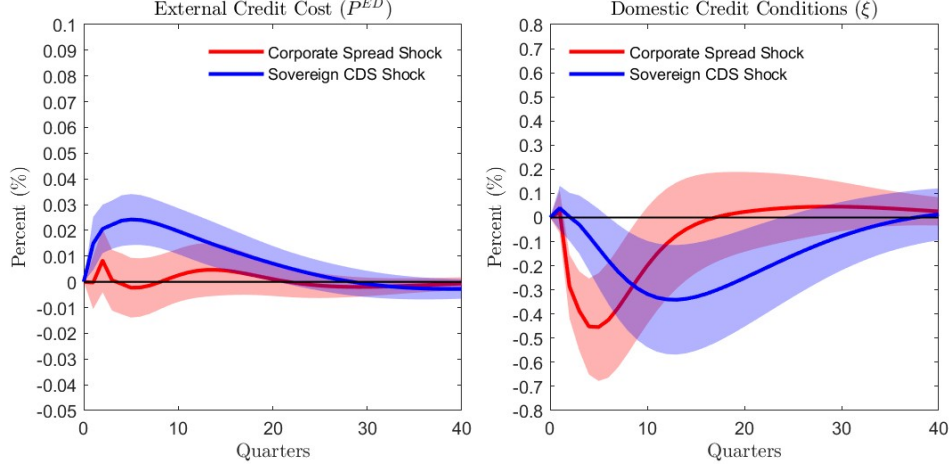


Figure 4. Empirical Impulse Responses to Corporate Spread and Sovereign CDS Shocks. This figure shows the responses of the external cost of credit (P^{ED}) and domestic credit conditions (ξ) to adverse shocks from corporate spreads ($r^{CorpSpd}$) and sovereign CDS ($SovCDS$). All results are based on the VAR specified in equations (18) and (20) using data constructed from cross-country averages. Confidence intervals are HAC-adjusted. Our sources are detailed in [Appendix C](#), and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

variance over a 10-year horizon. Equivalently, sovereign default concerns have not been a dominant source of external credit costs variation. As a result, we calibrate our model according to the benchmark VAR defined in equations (18)–(19).

IV. Calibration and Results

In this section, we discuss our calibration strategy and address our model results. We solve our model using a second-order perturbation method (`dynare++4.2.1` package). Non-negative variables are expressed in log units and approximated to the second order. Given our functional forms, our model is close to being log linear in a neighborhood of the steady state. See [Appendix A](#) for additional details. Our simulated results are obtained after pruning.¹¹

¹¹ In our simulations, the maximum cumulative absolute error for our Euler equations is $.29e^{-10}$. The concerns of de Groot et al. (2025) are mitigated by the following features

A. Calibration

Our medium-scale model features a total of 34 parameters. In what follows, we discuss the calibration of the parameters that are specific to SONOMA (Table II). The remaining parameters are calibrated similarly to those in both the production-based asset pricing literature and the international finance literature. Our calibrated values are broadly consistent with those reported in empirical studies of the euro area (among others, see Smets and Wouters, 2003; Gerali et al., 2010; Brinca et al., 2016). We discuss in detail the calibration of our canonical parameters in Appendix E.

In Table II, our parameters can be split into two groups. In the first group, parameters are calibrated in order to match relevant targeted moments.¹² The second group features parameters for which (i) we have direct estimates from our data, and (ii) we select values consistent with our confidence intervals.

Domestic variables. The first block of parameters pertains to the domestic financial conditions. We choose the average domestic credit conditions parameter, μ_ξ , to target the average domestic corporate leverage in our sample, $\mathbb{E}[D/K]$. The volatility of this shock, σ_ξ , is set to a conservative level that enables us to keep the volatility of corporate leverage, $\sigma[D/K]$, consistent with the data. Our estimated results point to notable persistence in domestic financial conditions. Accordingly, we set the persistence parameter,

of our setting: (i) by allowing agents to trade both international bonds and equities, we avoid extreme market incompleteness; (ii) our NFA has a moderate persistence of 0.85, not 0.99 as in many cases assessed by de Groot et al. (2025); (iii) our calibration for SDEs features less-volatile shocks; and (iv) we do not have occasionally binding constraints.

¹²As in all medium-scale dynamic stochastic general equilibrium (DSGE) models, our calibration is a multidimensional object in which each moment, $m_1, m_2, \dots, m_p$, is affected by many parameters simultaneously, $\theta_1, \theta_2, \dots, \theta_q$. For each parameter θ_j , we report the targeted moment m_{i*} that is more sensitive to it, that is, $m_{i*} : |\partial m_{i*} / \partial \theta_j| \geq |\partial m_i / \partial \theta_j| \quad \forall i$.

ρ_ξ , to 0.97. The loadings of the domestic financial conditions process on two of the shocks to the external conditions, $\rho_{rw,\xi}$ and $\rho_{\eta,\xi}$ (see equation (14)), are recovered from our VAR estimation. Our choice for the equity adjustment cost coefficient, κ , is based on Jermann and Quadrini (2012).

External credit conditions. The external exogenous interest rate process (equation (5)) features three parameters, μ_{rw} , σ_{rw} , and ρ_{rw} . We set σ_{rw} and ρ_{rw} so that are within our empirical confidence intervals. We calibrate μ_{rw} to match the average domestic interest rate in the data. The endogenous component of the external interest rate spread, P^{ED} , requires us to calibrate three parameters (equation (4)). We choose $\overline{NEDY}$ so that the endogenous average level of the external debt-to-output ratio is consistent with our data. Starting from the regression in equation (17), we compute a confidence interval for $p_{ED,1}$ and set this parameter to 6.¹³ We set $p_{ED,2}$ to match the volatility of the estimated country spread in our data. The loading of the external credit conditions process on the long-run component, $\rho_{rw,x}$, is recovered from our VAR estimation.

External equity conditions. The discussion of the net external equity position is a novel feature of our study. We choose $\overline{NEEY}$ so that the endogenous average level of the net external equity-to-output ratio is consistent with our data. The elasticity in equation (7), p_{EE} , is consistent with the empirical estimates described below.

Specifically, we augment equation (7) with a country-specific constant and use an overidentified GMM to jointly estimate our key parameters.¹⁴ More precisely, we use forty moments (four moments per country, for a total of ten countries) to estimate a total of twenty parameters. Country-specific coef-

¹³In our empirical specification, the gross spread is defined as $\exp(\alpha + \beta z)$, where $z = NEDY - \overline{NEDY}$. In our model, the gross spread is $1 + p_{2,ED} \exp(p_{1,ED} z)$. When $z = 0$, the two functions have the same slope if $p_{1,ED} = \frac{\exp(\alpha)\beta}{\exp(\alpha)-1}$. Standard errors are derived by the delta method.

¹⁴We recover the parameters of interest using a joint GMM estimation strategy featuring

ficients help us in capturing cross-sectional heterogeneity in external equity positions ($A_t^{EE,j}$ and $L_t^{EE,j}$), that is, the key source of difference between the return on NEE, $R_t^{EE,j}$, and the external spread, $R^{E,w} - R^{E,j}$. The estimated J -statistic is 5.375, which corresponds to a p -value of 0.99, indicating that our restrictions cannot be rejected.

For the sole sake of calibrating our model, we are interested in the following parameter estimate (HAC-adjusted standard error in parentheses):

$$\hat{p}_{EE} = \frac{0.117}{(0.019)}. \quad (21)$$

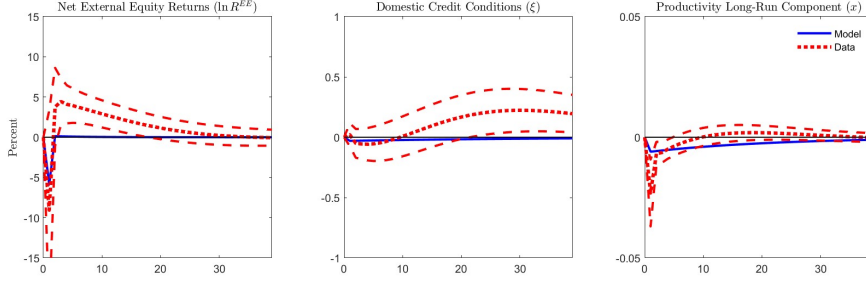
This coefficient confirms that countries that take longer positions in external equities accept lower expected returns.

We use our empirical results on NEE returns to calibrate the components reported in equations (6)–(8). Our VAR specified in equation (19b) gives us confidence intervals for (i) the persistence and volatility parameters for the exogenous component in external expected returns (ρ_η and σ_η); (ii) the volatility of unexpected external equity shocks (σ_{ee}); (iii) the loading of the external credit conditions process on the equity return ($\rho_{rw,ee}$); and (iv) the loading of the expected return process on the long-run component ($\rho_{\eta,x}$). As

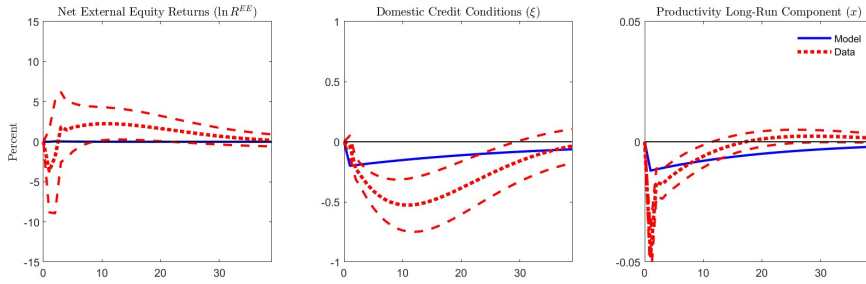
the following moment conditions for each country $j = 1, \dots, 10$:

$$\begin{aligned} 0 &= \mathbb{E} \left[\frac{R_t^{E,w} - R_t^{E,j}}{\hat{\sigma}_{t-1}^j} - \alpha_{1,j} - \beta_{pd} p d_{t-1}^w \right] \\ 0 &= \mathbb{E} \left[\left(\frac{R_t^{E,w} - R_t^{E,j}}{\hat{\sigma}_{t-1}^j} - \alpha_{1,j} - \beta_{pd} p d_{t-1}^w \right) p d_{t-1}^w \right] \\ 0 &= \mathbb{E} \left[\left(\ln(R_t^{EE})^j - \hat{\sigma}_t^j \times (\alpha_{1,j} + \beta_{pd} \times p d_{t-1}^w) - \alpha_{2,j} - p_{EE} \left(\overline{NEEY}^j - \frac{NEE_{t-1}^j}{Y_{t-1}^j} \right) \right) \right] \\ 0 &= \mathbb{E} \left[\left(\ln(R_t^{EE})^j - \hat{\sigma}_t^j \times (\alpha_{1,j} + \beta_{pd} \times p d_{t-1}^w) - \alpha_{2,j} - p_{EE} \left(\overline{NEEY}^j - \frac{NEE_{t-1}^j}{Y_{t-1}^j} \right) \right) \cdot \frac{NEE_{t-1}^j}{Y_{t-1}^j} \right]. \end{aligned}$$

The first two conditions pin down the country fixed-effects $\alpha_{1,j}$ and the common slope β_{pd} , respectively; the last two conditions pin down the constant $\alpha_{2,j}$ and the cost function slope p_{EE} .



(a) External Credit Conditions Shock (r^w)



(b) NEE-ER Shock (η)

Figure 5. Model versus VAR: Fundamental Shocks. This figure shows the responses of net external equity returns ($\ln R^{EE}$), domestic credit conditions (ξ), and the long-run component of productivity (x) to shocks from external credit conditions (r^w) and the NEE-ER (η). The model responses are based on the calibration in Table II. The data responses are based on the VAR specified in equations (18)–(19) using data constructed from cross-country averages. Confidence intervals are HAC-adjusted. Our sources are detailed in Appendix C, and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

discussed in Appendix D.4, excluding the domestic price-dividend ratio from the construction of $\hat{x}$ materially attenuates the estimated η -to- x relation, implying an alternative calibration of $\rho_{\eta,x} = -0.04$, compared with -0.06 in our benchmark calibration. Recalibrating the model accordingly produces only modest changes in the model’s quantitative implications.

Visualization. To better assess our calibration strategy, in Figure 5 we depict the implied dynamics of key exogenous processes in our model and compare them with their VAR counterparts. Specifically, we show the responses of net external equity returns ($\ln R^{EE}$), domestic credit conditions (ξ), and the long-run component of productivity (x) to external credit con-

dition shocks (r^w) and NEE-ER shocks (η). Our parsimonious model specification captures the main features highlighted in the data: (i) an adverse shock to external credit conditions is associated with both lower expected productivity and negative external equity returns; and (ii) an adverse NEE-ER shock is associated with adverse adjustments in both domestic credit conditions and expected productivity growth.

In principle, we could perfectly replicate our empirical responses by specifying a richer VAR structure for our exogenous processes in our model. However, we prefer our approach, as it reproduces the main patterns in the data parsimoniously. In addition, our approach is sophisticated enough to highlight the transmission of internal and external financial conditions to long-term growth. This is a novel and relevant dimension of our study.¹⁵

B. Results

We simulate our SONOMA model and show that it captures several key dimensions of both macroeconomic and financial data across SDEs. Furthermore, we show that the model is successful in matching a notable number of moments that are not directly targeted in our calibration.

Simulated moments. In Table III, we present several moments obtained from simulations and compare them to their empirical counterparts. Specifically, in the data we compute each moment at the country level and report its median as well as its cross-sectional dispersion, that is, the range implied by the second-smallest and the second-largest country-level estimates.

In the first panel of Table III, we focus on standard macroeconomic moments. SONOMA inherits the ability of many standard SOE models to match the behavior of macroeconomic aggregates in the data. In our model, investment growth (Δi) is more volatile than both output (Δy) and consumption (Δc), as in the data. In the model, consumption growth tends to be as volatile as output. This is due to the fact that the representative agents

¹⁵All of the other contemporaneous correlations among our shocks are set to zero, as they are empirically small and irrelevant for our simulation results.

use the current account to hedge domestic shocks and keep investment and consumption slightly more correlated than in a closed economy.

Most importantly, the model is very successful in replicating the moderate correlation between output and labor growth (Δh), and it produces a slightly positive correlation between investment growth and corporate debt issuance ($\Delta \log D$). The latter dimension is very relevant, given our interest in the role played by both financial frictions and shocks.

The second panel in Table III refers to our cross section of returns. SONOMA produces a low risk-free rate (r) while delivering a very sizable equity risk premium of 5.60% ($\mathbb{E}[r_E - r]$). To the best of our knowledge, this is the first model that produces such a large difference in the cost of debt and equity in an open economy setting. This model's risk premium is slightly lower than the median estimated in the data, but it is both empirically plausible and close to the lower bound of our cross-sectional range. A similar result applies to the average unlevered excess returns ($\mathbb{E}[r_K - r]$). In addition, our calibration produces (i) a mild positive correlation between output growth and the risk-free rate, and (ii) a country-level spread between the cost of debt and the world risk-free rate ($r - r^w$) consistent with the data.¹⁶

Our success in explaining the behavior of equity returns is related to our ability to reproduce many empirical features of corporate leverage. In the third panel of Table III, we show that our corporate debt-to-assets ratio (D/K) is as high and volatile as in the data. In addition, book-leverage variations ($\Delta(D/K)$) are slightly procyclical, and the equity payout-to-asset ratio (d/K) is 1.5 times more volatile than output, consistent with the data.

More broadly, our equilibrium model simultaneously tracks many features of financial leverage both domestically and internationally. In the last two panels of Table III, we focus on moments that are specific to our open economy setting. Given our calibration strategy, the average levels of both the

¹⁶The equity volatility puzzle is unresolved in our model. It is well known that adding time-varying volatility could resolve the problem, but this would come at the cost of making the model less parsimonious. We leave this task to future research.

NEE-to-output (NEE/Y) and NED-to-output (NED/Y) ratios are consistent with our empirical range. Our model produces an excessively smooth NED-to-output ratio, but it replicates the volatility of the NEE position. In addition, our model reproduces the countercyclicality of these components, that is, their negative correlation with output growth. Both in the model and in the data, the variation of the NED-to-output ratio ($\Delta(NED/Y)$) and that of the NEE-to-output ratio ($\Delta(NEE/Y)$) feature a very moderate contemporaneous correlation with each other.

According to our data, the median contemporaneous correlation between book leverage and the subcomponents of the NFA is slightly positive. Our empirical range, however, is very wide and spans both negative and positive values for both NEE and NED. In general, we consider our model satisfactory, as it features very small correlations. For example, our model produces a negative correlation between book leverage and NED in the order of -0.52, while our empirical lower bound is about -0.16. To better understand this result, we depict the responses of several variables of interest to our adverse financial shocks in Figure 6.

The representative firm responds to all of these shocks by reducing the equity payout in order to repurchase debt and reduce leverage. The response of NED, in contrast, changes across shocks. Specifically, when there is an external shock to the cost of debt, NED declines together with domestic corporate leverage, implying a positive comovement. In contrast, an adverse domestic credit shock produces a negative comovement, as the households find it optimal to partially hedge the internal credit crunch by both borrowing more from abroad and reducing their NED position.

When external equities become more convenient, we observe a reallocation toward foreign equities financed by increasing foreign leverage, that is, by increasing NED. This reallocation increases the country spread (see equation (4)) and hence the cost of debt. As a result, domestic leverage declines and there is a negative comovement with NED. At the equilibrium, domestic corporate leverage has a mild negative correlation with external debt.

Finally, we note that the current account-to-output ratio ($\Delta NFA/Y$) in our model is as volatile and persistent as in the data. To the best of our

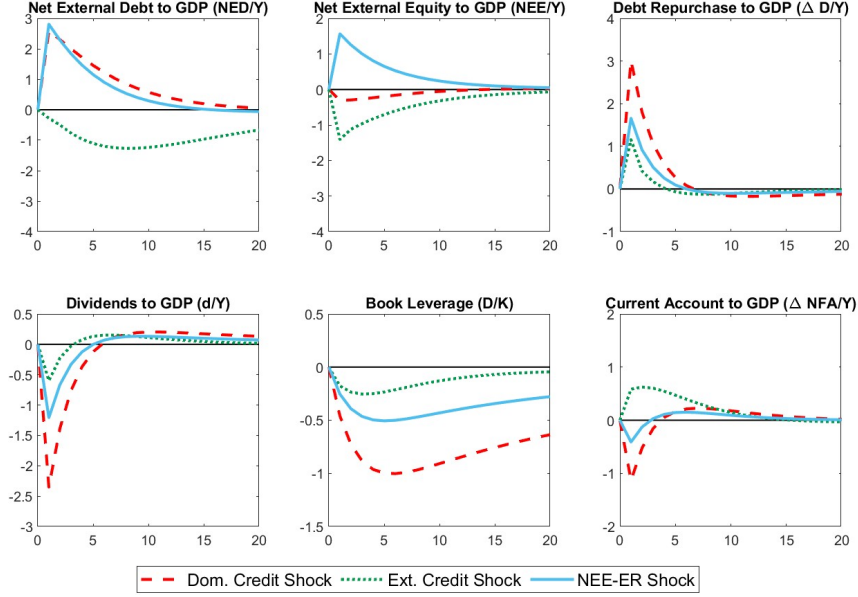


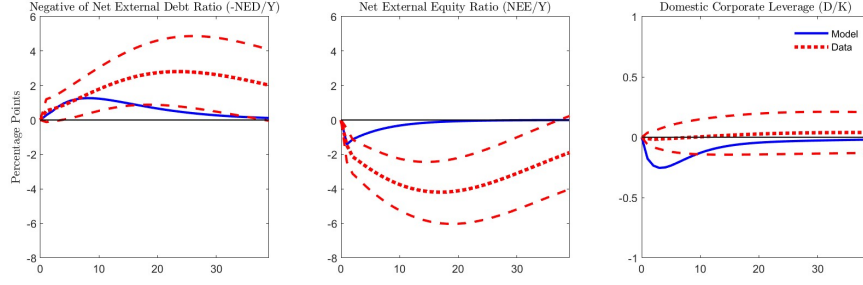
Figure 6. Impulse Responses in SONOMA. This figure shows percentage deviations from steady state for variables expressed in logs (percentage point deviations from steady state for variables expressed in levels) in response to a one-standard-deviation adverse shock to domestic credit conditions (ξ), external credit conditions (r^w), or the NEE-ER (η). Our benchmark calibration is reported in Table II.

knowledge, we are the first to match key properties of both the NFA and its subcomponents.

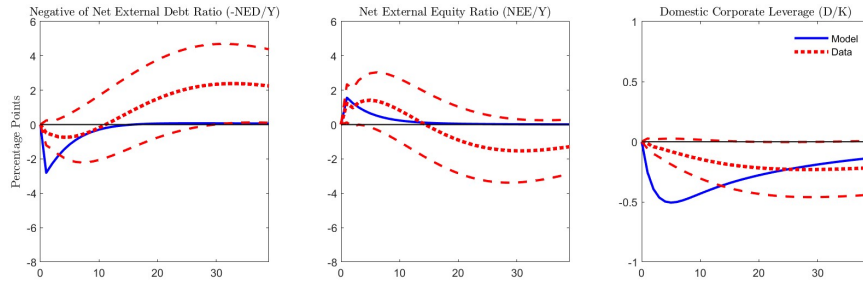
We also examine the Feldstein–Horioka correlation, $\text{corr}(S/Y, I/Y)$. In our sample of SDEs, the data imply a median correlation of 0.12, with a range of $[-0.34, 0.73]$. These values are smaller than those typically reported in the earlier SOE literature. Most importantly, the slightly negative correlation produced by our model falls within our empirical range.

Impulse responses after external shocks. We produce a complete analysis of the impulse responses generated by our model in section C. For the sake of focus, in Figure 7 we depict only the model-implied responses studied so far and compare them with their empirical counterparts. Overall, the model’s predictions align well with our data-based responses, meaning that the model’s responses are similar to those in the data.

Specifically, in the aftermath of an adverse external credit shock, the com-



(a) External Credit Conditions Shock



(b) NEE-ER Shock

Figure 7. Model versus VAR: External Positions and Leverage. This figure shows the responses of the negative of the net external debt ratio ($-NED/Y$); the net external equity ratio (NEE/Y); and the domestic book leverage ratio (D/K) to shocks from external credit conditions (r^w) and the NEE-ER (η). The model responses are based on the calibration in Table II. The data responses are based on the VAR specified in equations (18) and (19) using data constructed from cross-country averages. Confidence intervals are HAC-adjusted. Our sources are detailed in Appendix C, and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

position of the NFA changes, and we observe a reallocation of resources from external equities to external credits. For a prolonged number of quarters, net external equities decline in order to free up resources so as to decrease net external debt. In the data, this process builds up slowly and picks up after 20 quarters. In the model, this reallocation takes place mainly in the first 10 quarters. Domestic corporate leverage is nearly unchanged in the data at all horizons; in the model, there is a slight reduction of about 0.25% over the first ten quarters.

In the aftermath of an adverse NEE-ER shock, domestic corporate leverage declines in the data. This is a novel empirical fact that our model repli-

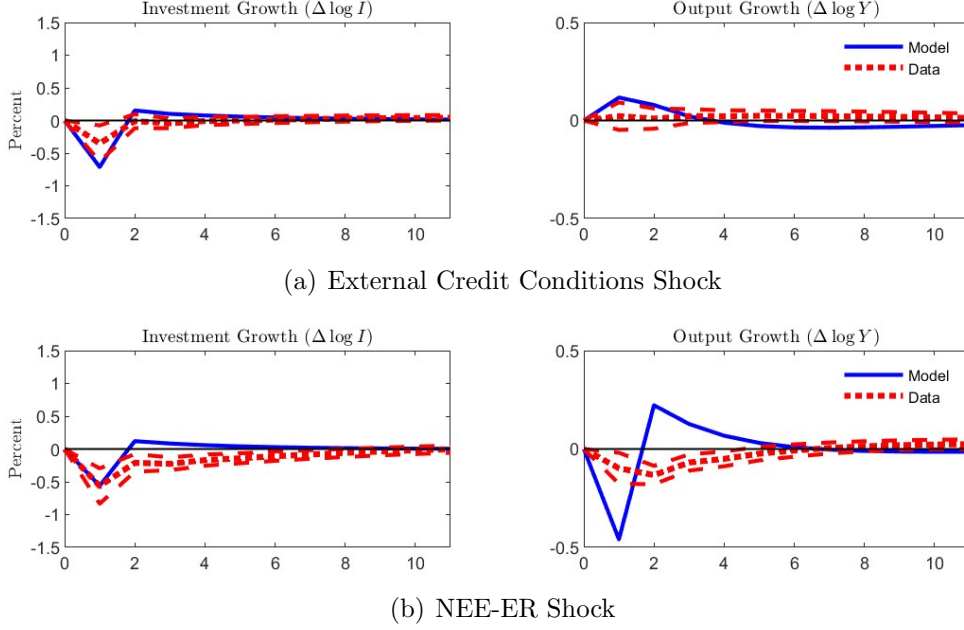


Figure 8. Model versus VAR: Economic Aggregates. This figure shows the responses of log investment ($\log I$) and log output ($\log Y$) to shocks from external credit conditions (r^w) and the NEE-ER (η). The model responses are based on the calibration in Table II. The data responses are based on the VAR specified in equations (18) and (19) using data constructed from cross-country averages. Confidence intervals are HAC-adjusted. Our sources are detailed in Appendix C and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

cates very well. According to our model, the household should borrow more from abroad in order to invest in external equities. This reallocation increases the cost of debt and forces corporations to de-leverage. Our VAR suggests that this mechanism is empirically plausible and—in the case of the NEE position—statistically significant.

In Figure 8, we conduct a similar analysis, turning our attention to macroeconomic aggregates such as investment and output growth. Our empirical analysis suggests that external shocks have a strongly contractionary impact on investment. Our model captures these patterns. We are the first to show that shocks to external equities are as relevant as shocks to external credit conditions.

In addition, our model suggests that output should decline after adverse external equity shocks but increase after external credit shocks. In the model,

an improvement in the external Sharpe ratio produces an incentive to reduce the holdings of domestic equities in favor of foreign equities. As a result, the domestic corporate sector shrinks and output declines as well. In the data, output growth likewise declines, although with a delay that is more persistent than in the model. In contrast, when external credit conditions deteriorate, the agent must offset the increase in the cost of debt by increasing labor and output. By doing so, the country reduces its external debt-to-output ratio and hence its external debt spread (equation (4)). In the model this reaction unfolds immediately, whereas in the data it picks up with a delay of about four quarters. This prediction of our model is both insightful and empirically plausible.

SOE accounting: Model vs data. Using model-simulated data, we run the same variance decomposition analysis that we perform in the data and that we presented in section C. Table IV compares the model results with the data. Since the model features only six exogenous shocks, in the data we exclude the contribution of the seventh empirical shock (white areas in Figure 1) and rescale the remaining shares to sum up to 100%, as in the model. For the sake of brevity, we focus only on four variables: NED-to-output, NEE-to-output, consumption-to-output, and output growth. We highlight several dimensions in which the model successfully replicates the data and some discrepancies for which future work will be required.

First of all, we focus on the relative importance of financial shocks versus productivity shocks. Across different horizons, financial shocks almost entirely explain the variance of the subcomponents of the NFA positions. This holds in both the model and the data, with a similar pattern for domestic corporate leverage. Financial shocks explain a lower share of the variance of output growth in the data, around 75%. Our model replicates this fact across different horizons. Consistent with the data, productivity shocks account for a moderate share of the variance of these macrofinancial variables.

Secondly, we turn our attention to the role played by external financial shocks. Across different horizons, the data suggest that external shocks are extremely important to explain fluctuations in NEE and consumption

relative to output. Simultaneously, external shocks play a minor role in explaining output growth. Our model replicates this pattern across different horizons. In the data, NED is mainly driven by internal financial shocks, not by external shocks. Over a 2-year horizon, the share of variance explained by external shocks is 25%, whereas the model predicts a value of 62%; that is, there exists a gap of 37%. Over longer horizons, however, this gap declines to 9%. Future research should focus on this mismatch. Overall, however, we find the performance of our SONOMA model satisfactory.

C. Inspecting the Mechanism

In this step, we inspect the mechanism by analyzing the various channels in our SONOMA model. To begin, we compare our SONOMA setting with its closed economy counterpart. Next, we study the role of preferences and financial frictions. We conclude by summarizing additional results reported in [Appendix G](#).

Closed SONOMA. In Table [V](#), we examine the performance of SONOMA when we remove one or more of its salient elements without altering the other elements of our benchmark calibration. For example, in the “Closed SONOMA” column we show how our results change if we analyze the closed-economy version of our model.

In the closed SONOMA, the macroeconomic moments change slightly and in the expected direction. In contrast, the asset pricing implications are very different, as the equity risk premium declines substantially. To better understand these results, in Figure [9](#) we depict key impulse responses with respect to adverse financial shocks for both our SONOMA model and its closed-economy version.

We note three key results. First, domestic credit shocks are more relevant in the closed SONOMA than in SONOMA. Not surprisingly, if internal credit shocks cannot be hedged by adjusting the external positions, their contractionary impact is stronger. Since states with tight domestic conditions feature high marginal utility, the contribution of internal shocks to the

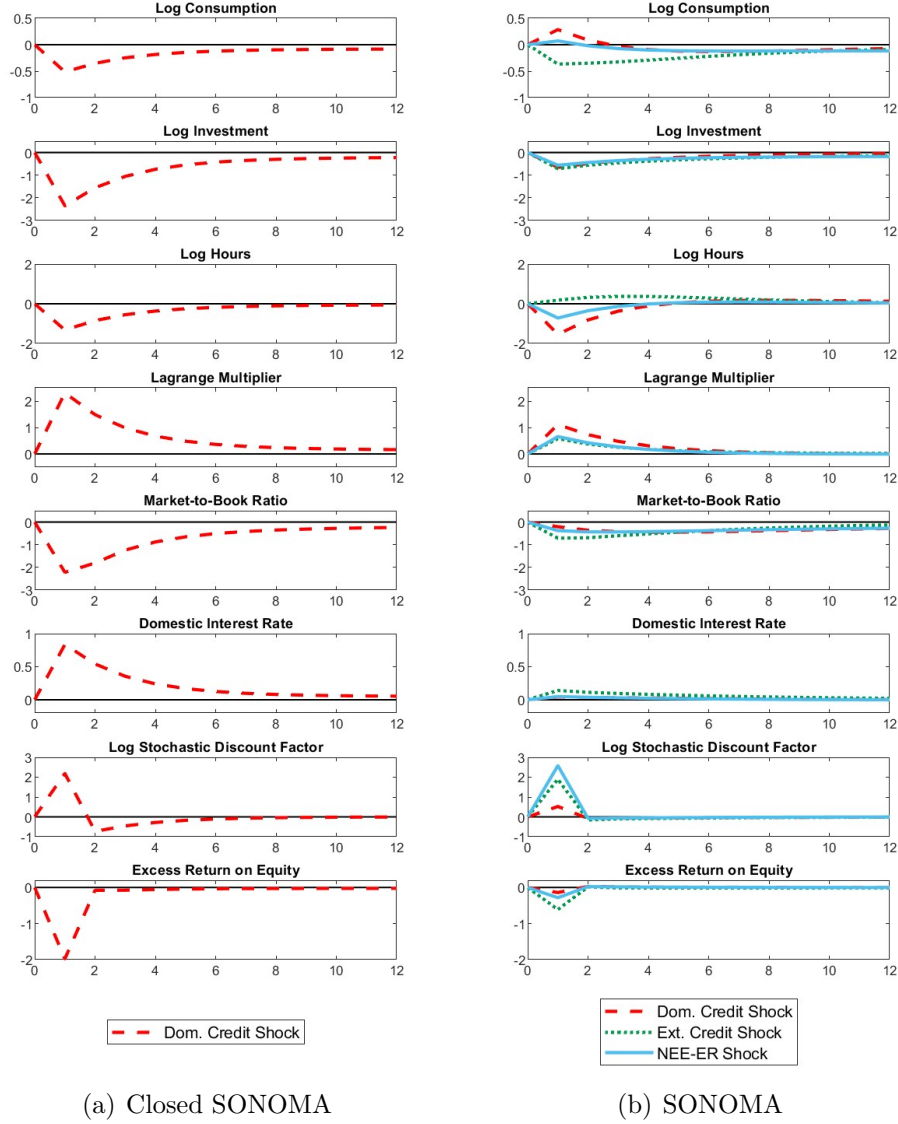


Figure 9. Impulse Responses in SONOMA vs Closed SONOMA. This figure shows percentage deviations from steady state for variables expressed in logs (percentage point deviations from steady state for variables expressed in levels) in response to a one-standard-deviation adverse shock to domestic credit conditions (ξ), external credit conditions (r^w), or the NEE-ER (η). Our benchmark calibration is reported in Table II, and Appendix A describes how we obtain the closed setting.

equity premium is strong. Under our benchmark SONOMA setting, however, shocks to either the external cost of equity or the external cost of debt are first-order and contribute to the generation of a very high risk premium

despite the adjustments of the external positions.

For completeness, we also report the responses of the same variables to productivity shocks in [Appendix G](#) (Figure [G1](#)). In all cases, the signs of the responses are in line with standard real business cycle (RBC) predictions. Moreover, the differences in responses across the closed- and open-economy specifications are consistent with prior work: for example, a negative productivity shock generates a sharper contraction in investment and a larger decline in firm values and excess returns on capital in the open economy. This result obtains because the domestic interest rate cannot adjust to cushion the shock (see, for example, Uribe and Schmitt-Grohé, [2017](#)).

The role of preferences and long-run risk. In the third column of Table [V](#), we show how our results change if we adopt time-additive log preferences as in [Jermann and Quadrini \(2012\)](#). On the macroeconomic side, we observe relevant changes: (i) investment becomes too smooth, whereas consumption becomes too volatile; and (ii) consumption and investment become highly correlated.

On the asset pricing side, we have an even stronger deterioration of the results. This should not be surprising, given that with CRRA preferences, news shocks are not priced. As a result, both long-run productivity growth changes and the long-run dynamics generated by financial shocks are no longer relevant for the determination of the risk premium.

In addition, we explore the predictions of our model when we completely turn off predictability in productivity growth. We impose $x_t = 0$ for all t by setting $\sigma_x = \rho_{rw,x} = \rho_{\eta,x} = 0$, and report our results under the column ‘No LRR’. As in the case of time-additive preferences, investment becomes too smooth and too correlated with consumption. The equity risk premium declines by two-thirds compared to SONOMA, a relevant share.

Role of financial frictions and shocks. In the rightmost two columns of Table [V](#), we assess the role of both financial frictions and shocks. For “No Dom. Financial Friction,” we retain all shocks, but we remove the equity issuance costs ($\kappa = 0$) and we replace the collateral constraint (equation [\(12\)](#))

with one that sets corporate debt equal to zero in every period ($D = 0$). In turn, these changes make the domestic credit shock irrelevant, because the firm is 100% equity financed and equity issuance is free, as in a frictionless neoclassical model. We observe a very pronounced decline in equity returns relative to the benchmark SONOMA setting. Importantly, the decline is not just mechanically due to financial leverage, as unlevered returns ($r_K - r$) fall significantly as soon as financial frictions are removed. In Table III, we report an unlevered risk premium of 3.58% for SONOMA, whereas in this case we have 3.28%. In addition, the external NFA position becomes even more volatile, in contrast to the data. These results confirm that the financial frictions studied in this manuscript are very important in all segments of the capital markets.

In the rightmost column of Table V, we retain all features of our SONOMA economy but zero out financial shocks. In this case, consumption becomes much more volatile than output, and the comovements of investment, domestic leverage, output, and labor become counterfactual. The equity risk premium increases, but that is mainly driven by the fact that in the absence of these shocks, corporations choose a higher average level of leverage. In addition, the NEE position becomes too smooth and procyclical, in contrast to the data. The NED position becomes too smooth as well, and its comovement with domestic leverage is excessively negative. The same consideration applies to the comovements between changes in the NFA position and the interest rate. These results confirm that financial shocks are relevant in explaining both equity dynamics and international positions.

In sum, consistent with the EME literature, both financial frictions and financial shocks are key elements for matching the observed behavior of investment, leverage, and international positions.

Additional results. In Appendix G.2, we document that our data analysis reveals a substantial amount of co-skewness among our external financing shocks and the domestic long-run growth conditions. Specifically, we find co-skewness between (i) the external credit shocks and long-run growth, and (ii) the NEE-ER shocks and long-run growth. After introducing a tail shock

in SONOMA that mimics the extent of our empirical downside risk, we find large negative implications for capital accumulation and welfare. In addition, we depict the impulse response of key variables in the aftermath of an adverse internal credit shock identified using the augmented VAR described in equation (20). The empirical patterns and those from simulating the model are consistent. We also study the role of a CDS shock (see [Appendix G.3](#)) and observe that the results are consistent with the findings in Bocola (2016).

V. Conclusions

This study develops a new macro-finance model for small-but-*developed* open economies (SDEs) to examine the role of financial shocks in shaping domestic economic fluctuations and corporate dynamics. We find that shocks to external debt and equity markets account for a significant portion of the medium- and long-run variance in investment, consumption, and external balances. Adverse shocks to these markets are associated with slower long-run growth, while domestic credit shocks primarily affect output growth and leverage. Our results distinguish SDEs from emerging market economies (EMEs). In SDEs, equity return shocks have a greater impact on domestic investment and corporate leverage than in EMEs, where exchange rate and credit shocks are more prominent.

Unlike studies of large developed economies, we emphasize the importance of both external credit and equity market shocks, an area previously underexplored. Building on a standard small open economy (SOE) framework, we incorporate recursive preferences, productivity growth shocks, and a decomposition of net foreign assets into debt and equity. This structure allows us to capture the dynamic trade-off between debt and equity financing in the presence of financial frictions, which influence corporate behavior and macroeconomic outcomes. Our model, SONOMA, successfully replicates key empirical patterns, including the countercyclical nature of external debt and equity positions, and the significant impact of external financial shocks on economic fluctuations.

In sum, our model demonstrates that external financial conditions affecting debt and equity markets are crucial determinants of economic dynamics in SDEs. These shocks interact with domestic financial frictions to shape both corporate leverage and macroeconomic stability.

Table I. Robustness: Relevance of Exogenous Shocks

Response	Shock	Additional Shock				
		Exchange Rate		<i>10yrYield</i>	Monetary Policy	
		<i>EUR</i>	<i>USD</i>		<i>2yrYield</i>	<i>Equity</i>
NED/Y	External Credit	−0.05	−0.03	0.00	0.00	
	NEE-ER	0.01	−0.02	0.00	0.00	
	NEE Return	0.00	−0.00	0.00	0.00	
	Domestic Credit	−0.02	0.01	−0.00	−0.00	
	LR Productivity	−0.00	−0.01	0.00	0.00	
	Additional	0.01	0.15	0.01	0.01	
NEE/Y	External Credit	0.01	0.01	−0.00	−0.00	
	NEE-ER	−0.01	−0.00	−0.00	−0.00	
	NEE Return	−0.00	0.00	0.00	0.00	
	Domestic Credit	0.00	−0.00	−0.00	−0.00	
	LR Productivity	0.01	−0.00	0.01	0.01	
	Additional	0.02	0.04	0.00	0.00	
I/Y	External Credit	0.07	0.06	−0.00	−0.00	
	NEE-ER	−0.09	−0.07	−0.01	−0.01	
	NEE Return	−0.00	−0.00	−0.00	−0.00	
	Domestic Credit	−0.00	0.01	0.00	0.00	
	LR Productivity	0.00	−0.00	−0.00	−0.00	
	Additional	0.01	0.00	0.02	0.02	
D/K	External Credit	0.03	0.02	−0.00	−0.00	
	NEE-ER	−0.03	−0.09	−0.00	0.00	
	NEE Return	0.01	−0.00	0.00	0.00	
	Domestic Credit	−0.03	0.06	0.00	0.00	
	LR Productivity	−0.03	−0.01	−0.00	−0.00	
	Additional	0.04	0.02	0.00	0.00	

Notes: This table shows the forecast error variance decomposition at the 10-year horizon of the following variables of interest: net external debt-to-output ratio (NED/Y); net external equity-to-output ratio (NEE/Y); investment-to-output ratio (I/Y); and corporate leverage (D/K). All results are based on the VAR specified in equations (18) and (19) in which we add the “Additional Shock” as the second-to-last variable. Real exchange rate fluctuations are the percent change in the local currency nominal exchange rate relative to either the EUR or USD minus the difference in inflation rates. Monetary policy shocks are computed using the market-based procedure of Cieslak and Pang (2021) applied to German bond yields or Euro STOXX50 equity returns. We use data constructed from cross-country averages. Our sources are detailed in Appendix C, and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

Table II. SONOMA Benchmark Calibration

Domestic Financial Markets		Estimate/Moment	
Average Domestic Credit Conditions	(μ_ξ)	0.24	$\mathbb{E}[D/K]$
Volatility of Domestic Credit Conditions Shock	(σ_ξ)	0.4%	0.7% (0.1%)
Persistence of Domestic Credit Conditions	(ρ_ξ)	0.97	0.96 (0.02)
Loading of Domestic Credit Conditions on r^w Shock	$(\rho_{rw,\xi})$	-0.22	-0.28 (0.79)
Loading of Domestic Credit Conditions on η Shock	$(\rho_{\eta,\xi})$	-1.00	-1.04 (0.37)
Equity Adj. Cost Coefficient	(κ)	0.146	JQ (2012)
External Credit Market			
Average World Interest Rate	(μ_{rw})	0.28%	$\mathbb{E}[r]$
Persistence of World Interest Rate	(ρ_{rw})	0.90	0.91 (0.03)
Volatility of World Interest Rate Shock	(σ_{rw})	0.13%	0.10% (0.01%)
Average External Debt Ratio	$(\overline{NEDY})$	2.40	$\mathbb{E}[NED/Y]$
Country Spread (P^{ED}) Elasticity	$(p_{ED,1})$	6	3.16 (1.78)
Country Spread (P^{ED}) Coefficient	$(p_{ED,2})$	0.008	$\sigma(P^{ED})$
Loading of r^w Shock on Long-Run Component	$(\rho_{rw,x})$	-0.04	-0.11 (0.04)
External Equity Market			
Average Net External Equities Ratio	$(\overline{NEEY})$	0.20	$\mathbb{E}[NEE/Y]$
Endog. Component of Expected Return (P^{EE}) Function Exponent	(p_{EE})	0.10	0.12 (0.02)
Persistence of Exog. Component of Expected Return (η)	(ρ_η)	0.80	0.78 (0.04)
Volatility of Exog. Component of Expected Return Shock	(σ_η)	0.2%	0.2% (0.015%)
Volatility of Return Shock	(σ_{ee})	0.10	0.15 (0.02)
Loading of r^w Shock on Return	$(\rho_{rw,ee})$	-45	-120 (81)
Loading of η Shock on Long-Run Component	$(\rho_{\eta,x})$	-0.06	-0.12 (0.02)

Notes: This table reports our benchmark quarterly calibration for the parameters that are specific to SONOMA. The rightmost column reports either a direct empirical estimate of the calibrated parameter (with HAC-adjusted standard errors in parentheses) or a specific moment that we target in our calibration procedure. “JQ (2012)” refers to parameter values obtained directly from Jermann and Quadrini (2012). All other parameters are reported in Table E1 and discussed in Appendix E.

Table III. Performance of SONOMA

Moments		Data		Model
		Est.	Range	
Macro	$\sigma(\Delta i)/\sigma(\Delta y)$	2.18	[1.76 , 2.86]	1.84
	$\sigma(\Delta c)/\sigma(\Delta y)$	0.78	[0.65 , 1.02]	1.01
	$corr(\Delta i, \Delta c)$	0.27	[0.06 , 0.57]	0.58
	$corr(\Delta i, \Delta \log(D))$	0.11	[-0.05 , 0.27]	-0.06
	$corr(\Delta y, \Delta h)$	0.28	[0.11 , 0.52]	0.53
Returns	$\mathbb{E}[r]$ (%)	0.87	[0.44 , 1.52]	1.53
	$corr(r, \Delta y)$	0.04	[-0.09 , 0.20]	0.10
	$\mathbb{E}[r_E - r]$ (%)	7.59	[5.73 , 11.28]	5.60
	$\sigma(r_E - r)$ (%)	21.43	[18.90 , 24.22]	6.51
	$\mathbb{E}[r_K - r]$ (%)	5.11	[2.85 , 7.02]	3.58
	$\sigma(r_K - r)$ (%)	14.82	[12.31 , 20.26]	4.61
	$\mathbb{E}[r - r^w]$ (%)	0.39	[0.17 , 1.39]	0.41
	$\sigma(r - r^w)$ (%)	0.25	[0.12 , 0.71]	0.55
Leverage	$\mathbb{E}[D/K]$ (%)	29.66	[16.94 , 41.30]	34.40
	$\sigma(D/K)$ (%)	5.06	[2.29 , 7.29]	6.76
	$corr(\Delta(D/K), \Delta y)$	0.34	[0.24 , 0.42]	0.10
	$\sigma(d/K)/\sigma(\Delta y)$ (%)	1.24	[1.01 , 1.89]	1.49
NFA	$\mathbb{E}[NED/Y]$ (%)	46.67	[-18.03 , 74.01]	54.71
	$\mathbb{E}[NEE/Y]$ (%)	17.65	[-21.80 , 34.56]	7.92
	$\sigma(\Delta(NED/Y))$ (%)	7.37	[4.05 , 10.04]	2.26
	$\sigma(\Delta(NEE/Y))$ (%)	9.15	[4.55 , 14.87]	5.44
	$corr(\Delta(NED/Y), \Delta y)$	-0.20	[-0.31 , -0.01]	-0.54
	$corr(\Delta(NEE/Y), \Delta y)$	-0.05	[-0.29 , 0.04]	-0.12
	$corr(\Delta(NED/Y), \Delta(NEE/Y))$	0.22	[0.16 , 0.38]	0.13
	$corr(NED/Y, D/K)$	0.17	[-0.16 , 0.87]	-0.52
	$corr(NEE/Y, D/K)$	0.51	[-0.68 , 0.85]	0.07
Current Account	$\sigma(\Delta NFA/Y)$ (%)	2.90	[1.98 , 5.10]	3.14
	$acf(\Delta NFA/Y)$	0.92	[0.65 , 0.98]	0.81
	$corr(\Delta NFA/Y, r)$	0.27	[-0.47 , 0.59]	0.24
	$corr(S/Y, I/Y)$	0.12	[-0.34 , 0.73]	-0.28

Notes: Our data sources are detailed in [Appendix C](#). The quarterly sample starts in 1995:Q1 and ends in 2018:Q4. Data moment estimates are the median of the moment values computed for each country in our sample. The data moment range shows the second-smallest and second-largest values. Model moment values are based on the calibration in [Table II](#). Both the data and the model moments are based on quarterly observations. Mean and volatility moments are annualized.

Table IV. Variance Decomposition: Model versus Data

	All Fin. Shocks		All Ext. Shocks	
	Data	Model	Data	Model
Panel A: 2 years				
NED/Y	96	89	25	62
NEE/Y	98	97	83	91
D/K	91	88	7	45
Δy	73	87	14	24
Panel B: 5 years				
NED/Y	96	79	32	57
NEE/Y	94	93	87	86
D/K	89	75	19	39
Δy	73	83	14	24
Panel C: 10 years				
NED/Y	92	76	45	54
NEE/Y	91	92	85	84
D/K	88	69	28	36
Δy	73	81	14	24

Notes: This table shows the forecast error variance decomposition over different horizons of the following variables of interest: net external debt-to-output ratio (NED/Y); net external equity-to-output ratio (NEE/Y); corporate leverage (D/K); and output growth (Δy). “All Fin. Shocks” is the sum of the FEVD values for the four financial shocks: external credit, NEE-ER, NEE return, and domestic credit. “All Ext. Shocks” is the same sum but removing the portion attributable to the domestic credit shock. All results are based on the VAR specified in equations (18) and (19) using either data constructed from cross-country averages or simulated data from our SONOMA model. The empirical shares of variance are rescaled to sum up to 100%, as in the model. Our sources are detailed in [Appendix C](#), and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

Table V. Inspecting the Mechanism

Moments		SONOMA	Closed SONOMA	No EZ	No LRR	No Dom. Financial Friction	No Financial Shocks
Macro	$\sigma(\Delta i)/\sigma(\Delta y)$	1.84	2.21	1.56	1.44	1.00	2.67
	$\sigma(\Delta c)/\sigma(\Delta y)$	1.01	0.90	1.11	1.03	0.52	1.50
	$corr(\Delta i, \Delta c)$	0.58	0.52	0.83	0.78	0.51	0.65
	$corr(\Delta i, \Delta \log(D))$	-0.06	0.17	-0.02	-0.15	–	-0.50
	$corr(\Delta y, \Delta h)$	0.53	0.33	0.51	0.55	0.90	-0.45
Returns	$\mathbb{E}[r]$ (%)	1.53	4.86	4.14	3.42	1.18	1.49
	$corr(r, \Delta y)$	0.10	-0.03	0.23	-0.04	-0.10	0.34
	$\mathbb{E}[r_E - r]$ (%)	5.60	3.00	0.26	2.08	3.28	5.75
	$\mathbb{E}[r - r^w]$ (%)	0.41	0.00	3.01	2.30	0.05	0.37
	$\sigma(r - r^w)$ (%)	0.55	0.00	1.07	0.50	0.52	0.33
Leverage	$\mathbb{E}[D/K]$ (%)	34.40	36.06	39.48	39.57	0.00	36.14
	$\sigma(D/K)$ (%)	6.76	5.88	7.29	4.93	0.00	5.28
	$corr(\Delta(D/K), \Delta y)$	0.10	0.15	0.10	0.20	–	-0.07
	$\sigma(d/K)/\sigma(\Delta y)$ (%)	1.49	1.50	1.49	1.24	0.24	1.65
NFA	$\mathbb{E}[NED/Y]$ (%)	54.71	–	59.11	58.36	54.54	55.27
	$\mathbb{E}[NEE/Y]$ (%)	7.92	–	6.27	6.98	8.28	6.65
	$\sigma(\Delta(NED/Y))$ (%)	2.26	–	1.83	2.04	3.33	0.42
	$\sigma(\Delta(NEE/Y))$ (%)	5.44	–	5.91	3.88	5.56	2.91
	$corr(\Delta(NED/Y), \Delta y)$	-0.54	–	-0.54	-0.65	-0.85	-0.18
	$corr(\Delta(NEE/Y), \Delta y)$	-0.12	–	0.06	-0.01	0.20	0.21
	$corr(\Delta(NED/Y), \Delta(NEE/Y))$	0.13	–	0.26	0.26	0.26	0.04
	$corr(NED/Y, D/K)$	-0.52	–	-0.47	-0.41	–	-0.74
Current Account	$corr(NEE/Y, D/K)$	0.07	–	0.46	0.14	–	0.45
	$\sigma(\Delta NFA/Y)$ (%)	3.14	–	2.93	2.58	3.65	2.10
	$acf(\Delta NFA/Y)$	0.81	–	0.81	0.77	0.51	0.89
	$corr(\Delta NFA/Y, r)$	0.24	–	0.12	0.38	0.25	-0.38

Notes: “Closed SONOMA” denotes the closed-economy version of our SONOMA setting (see [Appendix A](#) for more details). “No EZ” refers to the CRRA preferences case in which $\gamma = 1/\psi = 1$, as in [Jermann and Quadrini \(2012\)](#). “No LRR” refers to the case in which $\sigma_x = \rho_{rw,x} = \rho_{\eta,x} = 0$. “No Dom. Financial Friction” refers to the case in which $\kappa = 0$ and debt is equal to zero in every period ($D = 0$). In the rightmost column, we remove all financial shocks by setting their standard deviation to zero.

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Internet Appendix for
“SONOMA: a Small Open ecoNOmy for
MAcrofinance”

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ABSTRACT

This Internet Appendix provides additional derivations, data descriptions, calibration details, and empirical/quantitative results supporting the main text.

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Internet Appendix

Appendix A. Model Solution Details

In what follows, we report our derivations step by step so that our computational methods can be easily replicated. Our set of equilibrium equations is solved by high-order perturbations computed in dynare++.

Stochastic Discount Factor

The stochastic discount factor (SDF) of the household is given as

$$\widetilde{M}_{t+1} = \beta \left(\frac{\widetilde{C}_{t+1}}{\widetilde{C}_t} \right)^{-\frac{1}{\psi}} \left(\frac{U_{t+1}}{E_t \left[U_{t+1}^{1-\gamma} \right]^{\frac{1}{1-\gamma}}} \right)^{\frac{1}{\psi} - \gamma} \frac{\partial \widetilde{C}_{t+1} / \partial C_{t+1}^P}{\partial \widetilde{C}_t / \partial C_t^P}.$$

All of the variables in this expression are stationary and can be represented in Dynare++. In what follows, we use ' to denote time- $t + 1$ variables.

Solving the Household (HH) Problem

The consolidated HH problem with Lagrange multipliers (LMs) is

$$\begin{aligned} U(B, S, D, NED, A) = & \max_{\substack{C^P, H^{P,s}, \\ S', D', \\ NED', NEE'}} W \left(\widetilde{C} (C^P, 1 - H^{P,s}), \{U(S', D', NED', A')\}_{A'} \right) \\ & + \Lambda_{BC} [w_P H^{P,s} - T_{LS} + S(V^{P,ex} + d) + (1 + r_{-1})D] \\ & - \Lambda_{BC} [C^P + B' + S'V^{P,ex} + D'] \\ & + \Lambda_{BC} [NED' - (1 + r_{-1}^{ED})NED] \\ & - \Lambda_{BC} [NEE' - (1 + r^{EE})NEE], \end{aligned}$$

where we have substituted out leisure using the time constraint as $\ell = 1 - H^{P,s}$.

The FOCs are

$$\begin{aligned}
0 &= W_1 \tilde{C}_1 - \Lambda_{BC} \\
0 &= W_1 \tilde{C}_2 + \Lambda_{BC} w_P \\
0 &= \sum_{z'} W_2' U_S' - \Lambda_{BC} V^{P,ex} \\
0 &= \sum_{z'} W_2' U_D' - \Lambda_{BC} \\
0 &= \sum_{z'} W_2' U_{NED}' + \Lambda_{BC} \\
0 &= \sum_{z'} W_2' U_{NEE}' - \Lambda_{BC}.
\end{aligned}$$

The envelope conditions for S , D , and NED are

$$\begin{aligned}
U_S &= \Lambda_{BC} (V^{P,ex} + d) \\
U_D &= \Lambda_{BC} (1 + r_{-1}) \\
U_{NED} &= -\Lambda_{BC} (1 + r_{-1}^{ED}) \\
U_{NEE} &= \Lambda_{BC} (1 + r^{EE}).
\end{aligned}$$

Combine above together to get the following optimality conditions

$$\begin{aligned}
w_P &= -\frac{\tilde{C}_2}{\tilde{C}_1} \\
V^{P,ex} &= \sum_{z'} \frac{W_2' W_1' \tilde{C}_1'}{W_1 \tilde{C}_1} \left((V^{P,ex})' + d' \right) \\
1 &= \sum_{z'} \frac{W_2' W_1' \tilde{C}_1'}{W_1 \tilde{C}_1} (1 + r'^{EE}) \\
(1 + r^{ED})^{-1} &= \sum_{z'} \frac{W_2' W_1' \tilde{C}_1'}{W_1 \tilde{C}_1} \\
(1 + r)^{-1} &= \sum_{z'} \frac{W_2' W_1' \tilde{C}_1'}{W_1 \tilde{C}_1}.
\end{aligned}$$

The interest rates on domestic firm debt (r) and external debt (r^{ED}) must equal each other.

Solving the Firm Problem

The firm's consolidated problem with Lagrange multipliers is

$$\begin{aligned}
 V^P(K, D; A) = & \max_{d, H^d, I, K', D'} d + \mathbb{E} [M'V(K', D'; A')] \\
 & + \Lambda_{BC} \left[(1 - \tau_F) \left(F(K, H^{P,d}, A) - wH^d \right) - I - d - \chi(d) \right] \\
 & + \Lambda_{BC} [D' - D(1 + r_{-1}^D(1 - \tau_F))] \\
 & + \Lambda_K \left[(1 - \delta)K + I - \Phi\left(\frac{I}{K}\right)K - K' \right] \\
 & + \Lambda_{CC} \left[\xi(K' - D') - F(K, H^d, A) \right].
 \end{aligned}$$

The FOCs are

$$\begin{aligned}
 0 &= 1 - \Lambda_{BC}(1 + \chi'(d)) \\
 0 &= \Lambda_{BC}(1 - \tau_F)(F_H - w) - \Lambda_{CC}F_H \\
 0 &= -\Lambda_{BC} + \Lambda_K \left(1 - \Phi'\left(\frac{I}{K}\right) \right) \\
 0 &= \mathbb{E} [M'V'_K] - \Lambda_K + \Lambda_{CC}\xi \\
 0 &= \mathbb{E} [M'V'_D] + \Lambda_{BC} - \Lambda_{CC}\xi.
 \end{aligned}$$

The envelope conditions are

$$\begin{aligned}
 V_K &= \Lambda_{BC}(1 - \tau_F)F_K + \Lambda_K \left(1 - \delta - \frac{\partial}{\partial K} \left(\Phi\left(\frac{I}{K}\right)K \right) \right) - \Lambda_{CC}F_K \\
 V_D &= -\Lambda_{BC}(1 + r_{-1}^D(1 - \tau_F)).
 \end{aligned}$$

Combine together the FOCs and envelope conditions and then simplify to get the following optimality conditions:

$$\begin{aligned}
w_P &= \left(1 - \frac{(1 + \chi'(d)) \Lambda_{CC}}{1 - \tau_F}\right) F_H \\
(1 + \chi'(d)) \Lambda_{CC} \xi &= \frac{1}{(1 - \Phi'(\frac{I}{K}))} - \mathbb{E} \left[M' \left\{ \frac{1 + \chi'(d)}{1 + \chi'(d')} [(1 - \tau_F - (1 + \chi'(d')) \Lambda'_{CC}) F'_K] \right\} \right] \dots \\
&\quad - \mathbb{E} \left[M' \left\{ \frac{1 + \chi'(d)}{1 + \chi'(d')} \left[\frac{\left(1 - \delta - \frac{\partial}{\partial K'} \left(\Phi \left(\frac{I'}{K'} \right) K' \right) \right)}{(1 - \Phi'(\frac{I'}{K'}))} \right] \right\} \right] \\
(1 + \chi'(d)) \Lambda_{CC} \xi &= 1 - \mathbb{E} \left[M' \left\{ \frac{1 + \chi'(d)}{1 + \chi'(d')} (1 + r(1 - \tau_F)) \right\} \right]
\end{aligned}$$

These equations can be rewritten and expanded as

$$w_P = \left(1 - \frac{\tilde{\Lambda}_{CC}}{1 - \tau_F}\right) F_H \quad (\text{A1})$$

$$1 = \mathbb{E} [M' R'_K] \quad (\text{A2})$$

$$R'_K = \frac{q'_d}{q_d} \left[\frac{\left((1 - \tau_F) - \tilde{\Lambda}'_{CC} \right) F'_K + q'_K \left(1 - \delta - \frac{\partial}{\partial K'} \left(\Phi \left(\frac{I'}{K'} \right) K' \right) \right)}{q_K - \tilde{\Lambda}_{CC} \xi} \right] \quad (\text{A3})$$

$$\tilde{\Lambda}_{CC} = \frac{1 - \mathbb{E} \left[M' \left\{ \frac{q'_d}{q_d} (1 + r(1 - \tau_F)) \right\} \right]}{\xi} \quad (\text{A4})$$

$$q_K = \frac{1}{1 - \Phi'(\frac{I}{K})} \quad (\text{A5})$$

$$q_d = \frac{1}{1 + \chi'(d)}, \quad (\text{A6})$$

where we have defined

$$\begin{aligned}
\tilde{\Lambda}_{CC} &\equiv (1 + \chi'(d)) \Lambda_{CC} \\
&\equiv \frac{\Lambda_{CC}}{q_d}.
\end{aligned}$$

From this expression, the value of the exact LM is recovered as

$$\Lambda_{CC} = \frac{\tilde{\Lambda}_{CC}}{(1 + \chi'(d))}$$

or

$$\Lambda_{CC} = q_d \tilde{\Lambda}_{CC},$$

where $\tilde{\Lambda}_{CC}$ is pinned down by the model equation above. If the collateral constraint binds in equilibrium then $\Lambda_{CC} > 0$ and

$$\xi (K' - D') = F \left(K, H^{P,d}, A \right).$$

This constraint can be rearranged to define D' as

$$D' = K' - \frac{F \left(K, H^{P,d}, A \right)}{\xi}.$$

From this expression, we can see that $\xi > 0$ must be set sufficiently high in order for $D' > 0$ given a set of values for K' and $F \left(K, H^{P,d}, A \right)$.

If $\Lambda_{CC} \leq 0$, the collateral constraint may not be binding and D is determined by the following equation:

$$\begin{aligned} 1 &= \mathbb{E} \left[M' \left\{ \frac{1 + \chi' (d)}{1 + \chi' (d')} (1 + r (1 - \tau_F)) \right\} \right] \\ &= (1 + r (1 - \tau_F)) \mathbb{E} \left[M' \left\{ \frac{1 + \chi' (d)}{1 + \chi' (d')} \right\} \right]. \end{aligned}$$

If $\chi' (d) = \chi' (d') = 0$ (e.g., $\kappa = 0$), then the above optimality simplifies to

$$\Lambda_{CC} \xi = 1 - \mathbb{E} [M'] (1 + r (1 - \tau_F)).$$

From the HH optimality conditions, we know that

$$\mathbb{E} [M'] = \frac{1}{1 + r},$$

hence if $\tau_F > 0$ we obtain

$$\mathbb{E} [M'] (1 + r (1 - \tau_F)) < 1,$$

meaning that $\Lambda_{CC} > 0$ with certainty and the collateral constraint binds in equilibrium. We solve the model assuming that the collateral constraint binds and check ex-post that Λ_{CC} is positive in our simulations.

Closed Economy Setting

Solving the closed-economy version of our model requires us to modify only a few equations. We replace the following equations

$$\begin{aligned}r_t^{ED} &= r_{t-1}^w + P_t^{ED} \\r_t^{EE} &= \eta_{t-1} + P_t^{EE} + \sigma_{ee}\epsilon_{ee,t} \\1 &= E_t [M_{t+1}r_{t+1}^{EE}] ,\end{aligned}$$

with equations that impose

$$NED_t = NEE_t = r_t^{EE} = 0 \quad \forall t.$$

Appendix B. Convex Distress Cost

This section shows the results from using a convexified cost instead of the collateral constraint in benchmark calibrations. As we can see in Table B1, our results are very similar regardless of whether we employ convex distress costs or assume a binding constraint.

In what follows, we describe in detail how the model equations are modified to include a convex distress cost instead of the binding constraint. We no longer include the equation for the collateral constraint, rather we convexify the penalty of violating the constraint. Specifically, the firm's condensed problem is

$$V^P(K, D; A) = \max_{d, H^P, d, I, K', D'} d + \mathbb{E} [M'V(K', D'; A')]$$

subject to

$$(1 - \tau_F) \left(F(K, H^d, A) - wH^d \right) - I = d + \chi(d) + D(1 + r_{-1}^D(1 - \tau_F)) - D' + CC$$

$$K' = (1 - \delta)K + I - \Phi\left(\frac{I}{K}\right)K$$

where the constraint $\xi_t(K' - D') \geq F(K, H^d, A)$ is now captured by the convexified cost function CC . Following Croce et al. (2012), we use the following functional form

$$CC = A_{t-1} \times cc_1 \times e^{-cc_2[\xi_t(K' - D') - F(K, H^d, A)]}$$

and set the parameter cc_1 to be close to zero ($cc_1 = 0.001$), whereas the parameter cc_2 is chosen to be large ($cc_2 = 1000$).

The firm's problem is

$$V^P(K, D; A) = \max_{d, H^P, d, I, K', D'} d + \mathbb{E} [M'V(K', D'; A')]$$

$$+ \Lambda_{BC} \left[(1 - \tau_F) \left(F(K, H^d, A) - wH^d \right) \right]$$

$$+ \Lambda_{BC} \left[-I - d - \chi(d) + D' - D(1 + r_{-1}^D(1 - \tau_F)) - CC \right]$$

$$+ \Lambda_K \left[(1 - \delta)K + I - \Phi\left(\frac{I}{K}\right)K - K' \right],$$

where Λ denotes Lagrangian multipliers. The associated first order conditions are:

$$\begin{aligned}
0 &= 1 - \Lambda_{BC} (1 + \chi' (d)) \\
0 &= \Lambda_{BC} \left[(1 - \tau_F) (F_H - w) - \frac{\partial CC}{\partial H^d} \right] \\
0 &= -\Lambda_{BC} + \Lambda_K \left(1 - \Phi' \left(\frac{I}{K} \right) \right) \\
0 &= \mathbb{E} [M' V'_K] - \Lambda_K - \Lambda_{BC} \frac{\partial CC}{\partial K'} \\
0 &= \mathbb{E} [M' V'_D] + \Lambda_{BC} \left(1 - \frac{\partial CC}{\partial D'} \right).
\end{aligned}$$

The envelope conditions are

$$\begin{aligned}
V_K &= \Lambda_{BC} \left[(1 - \tau_F) F_K - \frac{\partial CC}{\partial K} \right] + \Lambda_K \left(1 - \delta - \frac{\partial}{\partial K} \left(\Phi \left(\frac{I}{K} \right) K \right) \right) \\
V_D &= -\Lambda_{BC} (1 + r_{-1}^D (1 - \tau_F)).
\end{aligned}$$

Combine together the FOCs and envelope conditions to get the following optimality conditions

$$\begin{aligned}
\Lambda_{BC} &= \frac{1}{1 + \chi' (d)} \\
(1 - \tau_F) (F_H - w_P) &= \frac{\partial CC}{\partial H^{P,d}} \\
\Lambda_{BC} &= \Lambda_K \left(1 - \Phi' \left(\frac{I}{K} \right) \right) \\
\Lambda_K + \Lambda_{BC} \frac{\partial CC}{\partial K'} &= \mathbb{E} \left[M' \left\{ \Lambda'_{BC} \left[(1 - \tau_F) F'_K - \left(\frac{\partial CC}{\partial K} \right)' \right] \right\} \right] \\
&= + \mathbb{E} \left[M' \left\{ \Lambda'_K \left(1 - \delta - \frac{\partial}{\partial K'} \left(\Phi \left(\frac{I'}{K'} \right) K' \right) \right) \right\} \right] \\
\Lambda_{BC} \left(1 - \frac{\partial CC}{\partial D'} \right) &= \mathbb{E} \left[M' \left(\Lambda'_{BC} (1 + r (1 - \tau_F)) \right) \right].
\end{aligned}$$

After some simplifications:

$$\begin{aligned}
(1 - \tau_F) (F_H - w_P) &= \frac{\partial CC}{\partial H^{P,d}} \\
\left(\frac{1}{1 + \chi'(d)} \right) \left(\frac{1}{(1 - \Phi'(\frac{I}{K}))} + \frac{\partial CC}{\partial K'} \right) &= \mathbb{E} \left[M' \left\{ \frac{1}{1 + \chi'(d')} \left[(1 - \tau_F) F'_K - \left(\frac{\partial CC}{\partial K} \right)' \right] \right\} \right] \\
&= + \mathbb{E} \left[M' \left\{ \frac{\left(1 - \delta - \frac{\partial}{\partial K'} \left(\Phi \left(\frac{I'}{K'} \right) K' \right) \right)}{(1 + \chi'(d')) (1 - \Phi'(\frac{I'}{K'}))} \right\} \right] \\
1 - \frac{\partial CC}{\partial D'} &= (1 + r(1 - \tau_F)) \mathbb{E} \left[M' \left(\frac{1 + \chi'(d)}{1 + \chi'(d')} \right) \right].
\end{aligned}$$

These equations can be rewritten and expanded as

$$\begin{aligned}
\frac{\partial CC}{\partial H^d} &= (1 - \tau_F) (F_H - w) \\
1 &= \mathbb{E} [M' R'_K] \\
R'_K &= \left(\frac{q'_d}{q_d} \right) \left(\frac{\left[(1 - \tau_F) F'_K - \left(\frac{\partial CC}{\partial K} \right)' \right] + q'_K \left(1 - \delta - \frac{\partial}{\partial K'} \left(\Phi \left(\frac{I'}{K'} \right) K' \right) \right)}{(q_K + \frac{\partial CC}{\partial K'})} \right) \\
\frac{\partial CC}{\partial D'} &= 1 - \mathbb{E} \left[M' \left(\frac{q'_d}{q_d} \right) (1 + r(1 - \tau_F)) \right] \\
q_K &= \frac{1}{1 - \Phi'(\frac{I}{K})} \\
q_d &= \frac{1}{1 + \chi'(d)}.
\end{aligned}$$

Using the functional form

$$CC = A_{t-1} \times cc_1 \times e^{-cc_2 [\xi_t(K' - D') - F(K, H^d, A)]}$$

we have the following expressions for the derivatives

$$\begin{aligned}\frac{\partial CC}{\partial H^{P,d}} &= CC \times cc_2 \times F_H \\ \left(\frac{\partial CC}{\partial K}\right)' &= CC' \times cc_2 \times F'_K \\ \frac{\partial CC}{\partial K'} &= CC \times (-cc_2) \times \xi \\ \frac{\partial CC}{\partial D'} &= CC \times cc_2 \times \xi\end{aligned}$$

Hence $CC \times cc_2$ replaces the multiplier $(\tilde{\Lambda}_{CC})$ in front of the borrowing constraint in our benchmark model.

Table B1: Model with Convex Distress Costs

Moments		Binding Constraint	Convexified Constraint
Macro	$\sigma(\Delta i)/\sigma(\Delta y)$	1.84	1.80
	$\sigma(\Delta c)/\sigma(\Delta y)$	1.01	0.97
	$corr(\Delta i, \Delta c)$	0.58	0.57
	$corr(\Delta i, \Delta \log(D))$	-0.06	-0.06
	$corr(\Delta y, \Delta h)$	0.53	0.52
Returns	$\mathbb{E}[r]$ (%)	1.53	1.50
	$corr(r, \Delta y)$	0.10	0.08
	$\mathbb{E}[r_E - r]$ (%)	5.60	5.56
	$\sigma(r_E - r)$ (%)	6.51	6.47
	$\mathbb{E}[r_K - r]$ (%)	3.58	3.58
	$\sigma(r_K - r)$ (%)	4.61	4.62
	$\mathbb{E}[r - r^w]$ (%)	0.41	0.38
	$\sigma(r - r^w)$ (%)	0.55	0.63
Leverage	$\mathbb{E}[D/K]$ (%)	34.40	33.67
	$\sigma(D/K)$ (%)	6.76	7.01
	$corr(\Delta(D/K), \Delta y)$	0.10	0.19
	$\sigma(d/K)/\sigma(\Delta y)$ (%)	1.49	1.49
NFA	$\mathbb{E}[NED/Y]$ (%)	54.71	54.94
	$\mathbb{E}[NEE/Y]$ (%)	7.92	7.83
	$\sigma(\Delta(NED/Y))$ (%)	2.26	2.21
	$\sigma(\Delta(NEE/Y))$ (%)	5.44	5.37
	$corr(\Delta(NED/Y), \Delta y)$	-0.54	-0.54
	$corr(\Delta(NEE/Y), \Delta y)$	-0.12	-0.06
	$corr(\Delta(NED/Y), \Delta(NEE/Y))$	0.13	0.15
	$corr(NED/Y, D/K)$	-0.52	-0.53
	$corr(NEE/Y, D/K)$	0.07	0.12
Current Acc.	$\sigma(\Delta NFA/Y)$ (%)	3.14	3.10
	$acf(\Delta NFA/Y)$	0.81	0.81
	$corr(\Delta NFA/Y, r)$	0.24	0.27

Notes: The calibration for our benchmark open economy setting SONOMA is reported in Table II.

Appendix C. Data Sources and Availability

In the table below, we summarize our main sources by data type.

Table C1: Summary of Data Sources

Panel A: List of Sources	
Name	Acronym
Bank of International Settlements	BIS
International Monetary Fund	IMF
Organisation for Economic Co-Operation and Development	OECD
Penn World Table	PWT
Ken French Data Library	KF
Bloomberg	BLOOM
Panel B: Data Sources	
Variable	Source
National aggregates (GDP, C, I)	OECD
Depreciation	PWT
Labor hours	OECD, PWT
Private sector debt	BIS
Net external debt	IMF
Domestic interest rates	OECD
Public equity data	KF, BLOOM
Inflation	IMF
Exchange rates	IMF

Notes: This table summarizes our main data sources.

National Aggregates. Quarterly gross domestic product, investment, and consumption data are from the OECD. The data series are pulled from the statistics tool (<https://stats.oecd.org/>) as the full time series are considered estimates and therefore not posted on the main OECD website. Other comprehensive databases such as the IMF's *International Financial Statistics* database only include the recent period (e.g., from the mid 1990s for most European countries). Peru is not in the OECD database and only has data from 2007 in the IMF database. Given these limitations, data for Peru are sourced from Haver Analytics.

Depreciation. Annual average depreciation rates are from the PWT.

Labor Income Share. Annual labor income share values are from the PWT.

Labor. Quarterly and annual hours worked are computed from OECD labor data.

We compute quarterly (annual) hours worked as the product of quarterly (annual) number of persons employed by the average annual number of hours worked. For Peru, which is not in the OECD database, we interpolate the annual employment time series from the PWT.

Private Sector Debt. Quarterly credit to non-financial corporations from all lenders is available from the BIS. For Peru, which is not in the BIS database, we interpolate the annual time series for nonfinancial corporate debt (loans and debt securities) reported in the IMF’s Global Debt Database (GDD).

Net External Debt. Quarterly and annual asset and liability positions in U.S. dollars are available from the IMF’s *Balance of Payments and International Investment Position Statistics* database. We follow the same method as Lane and Milesi-Ferretti (2007) to compute net external debt from the reported asset and liability positions. While Lane and Milesi-Ferretti (2007) do make available their *External Wealth of Nations* dataset, they only provide annual data.

Domestic Interest Rates. Quarterly long-term interest rates are available from the OECD with a two exceptions. Mexico’s rate data are from the Banco de Mexico. Peru’s rate data are from Bloomberg.

Public Equity Data. For SDEs except Portugal, monthly equity returns with dividends (r^{cum}) and without dividends (r^{ex}) are available from the Ken French Data Library. The raw returns are measured in U.S. dollars. We compute quarterly price-dividend ratios in three steps. First, we compute monthly price (p) and dividend (d) series using the recursion $p_t = (1+r_t^{ex})p_{t-1}$ and $d_t = (1+r_t^{cum})p_{t-1} - p_t$ after initializing $p_0 = 1$. Second, we compute monthly price-dividend ratios as $PD_t = p_t / \sum_{\tau=t-11}^t d_\tau$. Third and finally, we compute quarterly price-divided ratios as the average of the monthly values in a given quarter. For Portugal and the EMEs, which are not available from the Ken French Data Library, monthly returns and the quarterly price-dividend series are from Bloomberg. Specifically we use the MSCI Portugal Index (*MXPT Index*), Ibovespa Brasil Sao Paulo Stock Exchange Index (*IBOV Index*), MSCI Chile Index (*MXCL Index*), MSCI Colombia Index (*MXCO Index*) S&P/BMV IPC Index (*MEXBOL Index*), and S&P/BVL Peru General Index (*SPBLPGPT Index*).

Inflation. Quarterly consumer price indices for all items are available from the IMF’s *International Financial Statistics* database. We use these series in two ways. First, we construct measures of quarterly inflation in order to convert nominal returns into real returns. Second, we deflate nominal quantities using the price

index in order to compute real growth rates.

Exchange Rates. Quarterly exchange rates are available from the IMF’s *International Financial Statistics* database. We use these data to compute gross domestic product figures in U.S. dollars in order to compute net external debt ratios. The external debt data are only provided in U.S. dollars.

World Price-Dividend Ratio. We use the “Major Markets” international index provided by the Ken French Data Library to construct the P/D ratio. This index is composed from all the country-level indexes in the library. The raw returns are measured in U.S. dollars. We follow the same method to construct quarterly price-dividend ratios as described in Public Equity Data for individual countries. The Major Markets index is also used to represent world equity returns in our country-level volatility-adjusted spread (*VAS*) measures.

European Corporate Bond Yields. We use daily yields from the ICE BofA Euro Corporate Bond Indexes for AAA- and BBB-rated issuers. Each index is constructed from euro-denominated corporate bonds with remaining maturities of five to seven years. Daily values are aggregated to the quarterly frequency using arithmetic averages

Sovereign CDS Spreads. We use daily 5-year sovereign credit default swap (CDS) spreads from Markit. The data are reported at the country level and correspond to USD-denominated contracts referencing each country’s sovereign debt. Daily values are aggregated to the quarterly frequency using arithmetic averages

Data Availability by Country

The countries listed in this section comprise the SDEs and EMEs included in our empirical analysis. Table C2 reports the starting year for each constructed country-level measurement, while Table C3 reports the availability of the underlying raw data series used to construct these variables.

Table C2: Data Availability for Country-Level Measurements				
Country	External Financial	Internal Credit	TFP Growth	Price-Dividend Ratio
	r^w and η	ξ	Δa	PD
SDEs				
Austria	2005	1995	1995	1987
Belgium	1987	1980	1970	1975
Denmark	1992	1994	1970	1989
Finland	1988	1970	1960	1988
Italy	1991	1960	1995	1975
Netherlands	2003	1990	1975	1975
Portugal	1997	1979	1970	1995
Spain	1982	1980	1977	1975
Sweden	1987	1980	1963	1975
Switzerland	1984	1999	1998	1975
EMEs				
Brazil	2002	1995	1995	1998
Chile	2004	2002	1995	1995
Colombia	2005	2005	2005	1995
Mexico	2002	1994	1990	2002
Peru	2006	2001	1992	2002

Notes: This table reports the starting year for each constructed series. See section A for a description of the construction of these measurements.

Table C3: Data Availability by Country										
Country	National	Depreciation	Labor		Private	Net External		Domestic	Public	Inflation
	Aggregates				Debt	Positions	Interest	Rate		
	Quarterly	Annual	Quarterly	Annual	Quarterly	Quarterly	Annual	Quarterly	Monthly	Quarterly
SDEs										
Austria	1960	1950	1995	1950	1995	2005	2005	1990	1987	1958
Belgium	1960	1950	1983	1950	1980	2005	1981	1955	1975	1956
Denmark	1960	1950	1995	1950	1994	2005	1991	1987	1989	1958
Finland	1960	1950	1990	1950	1970	1975	1975	1988	1988	1956
Italy	1960	1950	1995	1950	1950	1999	1972	1991	1975	1956
Netherlands	1960	1950	1995	1950	1990	2003	2003	1959	1975	1958
Portugal	1960	1950	1995	1950	1979	1999	1996	1993	NA	1956
Spain	1960	1950	1995	1950	1980	1999	1981	1980	1975	1955
Sweden	1960	1950	1993	1950	1980	1999	1986	1987	1975	1956
Switzerland	1960	1950	2010	1950	1999	1999	1983	1955	1975	1956
EMEs										
Brazil	1995	1950	2002	1950	1994	2001	2001	1995	1990	1981
Chile	1995	1951	1986	1951	2002	2003	1997	2004	1988	1970
Colombia	2005	1950	NA	1950	1996	2000	1996	2003	1993	1970
Mexico	1960	1950	2005	1950	1994	2009	2001	2001	1994	1958
Peru	1992	1950	NA	1950	2001	1993	1993	2006	1990	1958

Notes: This table reports the starting year for each raw data series.

Appendix D. Additional Empirical Results

In Table D1, we report the results obtained when extracting the long-run component.

Table D1: Extracting Long-Run Productivity Growth (x)

Log Price-Dividend Ratio	0.003*** (5.12)
Inflation	0.164*** (5.23)
Risk-free Rate	0.012 (0.16)
R ²	0.067
N	942

Notes: This table shows the coefficient estimates from the following panel regression:

$$\Delta a_t^{j,dm} = \beta' X_{t-1}^{j,dm} + \epsilon_{a,t}^j$$

where the dependent variable is the productivity growth rate. All variables are de-meaned at the country level and therefore we omit the constant term. Standard errors are heteroskedasticity-consistent. t -statistics are in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

In Table D2, we report our country-level results for the estimation of the external interest rate spread (P^{ED}) coefficients.

In Figure D1, we depict the fundamental exogenous processes used in our empirical analysis. In each panel, we depict a cross-country average of the process of interest.

Table D2: Extracting External Credit Conditions (r^w)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Panel	Finland	Italy	Portugal	Spain	Sweden	Switzerland
Instrumented NED/Y	0.0026*** (0.0007)	0.0015 (0.0009)	0.0037*** (0.0008)	0.0052*** (0.0012)	0.0026*** (0.0006)	0.0036** (0.0018)	0.0017*** (0.0005)
Constant	0.0008** (0.0004)	0.0009 (0.0006)	0.0019*** (0.0005)	0.0029** (0.0012)	0.0007 (0.0007)	0.0019*** (0.0006)	-0.0008 (0.0006)
R ²	0.242	0.102	0.464	0.594	0.439	0.028	0.327
N	877	96	96	88	96	96	96

Notes: This table shows the coefficient estimates from the following panel regression:

$$\ln(1 + P_t^{ED,j}) = \ln(p_{ED,2}^j) + p_{ED,1}^j (NED_t^j / Y_t^j - \overline{NEDY}^j) + \varepsilon_t^j$$

where the dependent variable is the log of one plus the difference between the real interest rate of country j and the real German rate. The independent variable is the the demeaned net external debt ratio for country j instrumented with its lagged value. See section A for additional details. Standard errors are HAC-adjusted. t -statistics are in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

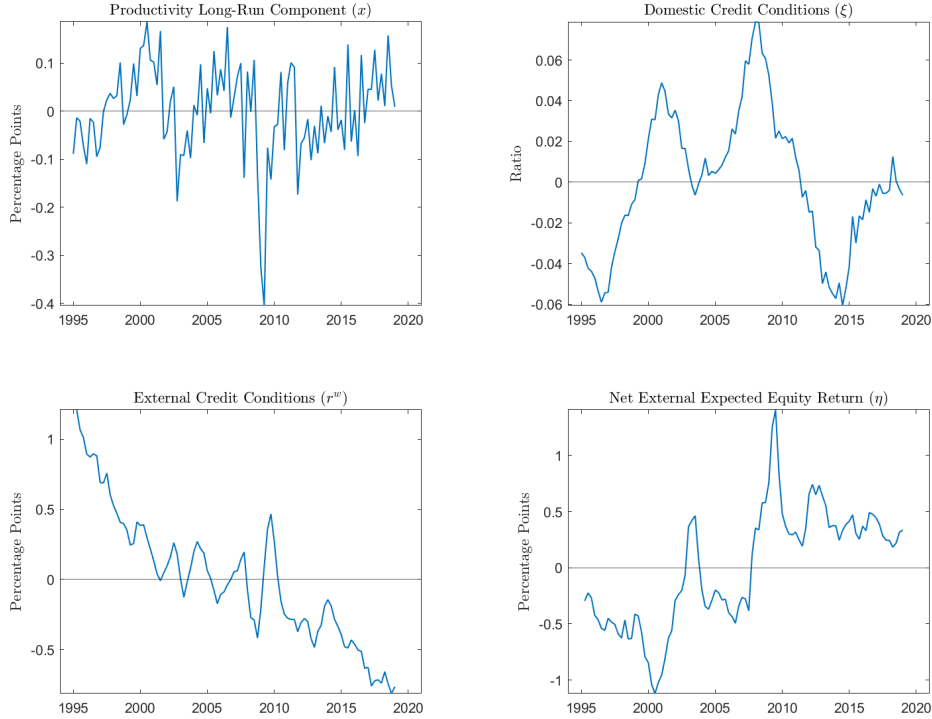


Fig. D1. Common Estimated Processes. This figure shows our estimated long-run component (x), domestic credit conditions (ξ), external cost of debt (r^w) and external expected return on net equity (η). All results are based on a cross sectional average across the countries in our sample. All series are demeaned.

As an additional robustness check, we verify that our main IRF results are not an artifact of the specific Cholesky ordering assumed in equation (19b). We re-estimate the VAR under all 120 possible orderings of our five exogenous variables and, for each, record whether three key sign predictions continue to hold at 2 and 3 quarters after the shock: (i) a negative response of x to the NEE-ER shock, (ii) a negative response of x to the external credit shock, and (iii) a negative response of ξ to the NEE-ER shock. Table D3 reports the fraction of orderings for which each response has the correct sign, and, separately, the correct sign combined with statistical significance (evaluated at two standard errors).

Table D3: Robustness to Cholesky Ordering

Response	Correct Sign		Correct Sign & Significant	
	2Q	3Q	2Q	3Q
x to NEE-ER shock	100%	100%	100%	100%
x to external credit shock	100%	100%	50%	17%
ξ to NEE-ER shock	100%	100%	75%	100%

Notes: This table reports, for all 120 possible Cholesky orderings of the five exogenous variables in equation (19b), the fraction of orderings for which each impulse response continues to match the directional response in Figure 3, and separately, the predicted sign combined with statistical significance (evaluated at two standard errors), at 2 and 3 quarters after the shock.

Appendix D.1. SDEs vs EMEs

In EMEs, estimating the long-run component x and the friction P^{ED} with statistical power is challenging. Hence, when comparing EMEs and SDEs in our VAR we use Δa and r^{ED} , as opposed to x and r^w . Figure D2 is constructed as our benchmark Figure 1, but it obtained using Δa and r^{ED} . As shown in Figure D3, the quantitative impact of these substitutions is minimal, with only small differences between the two sets of values. Thus, the VAR results obtained for EMEs are comparable to our benchmark results for SDEs.

The precise FEVD share values used in the top panel of Figure 2 for the SDEs and EMEs can be seen in the bottom panels of Figures D4 and D5, respectively.

In Figure D4, the bottom panels show the relative shares of variance explained by each variable, expressed as a proportion of the total variance explained in the top panels. We emphasize these relative shares because they provide a more meaningful basis for comparison with results for emerging economies, as they account for differences in unexplained variance across country groups. This approach allows

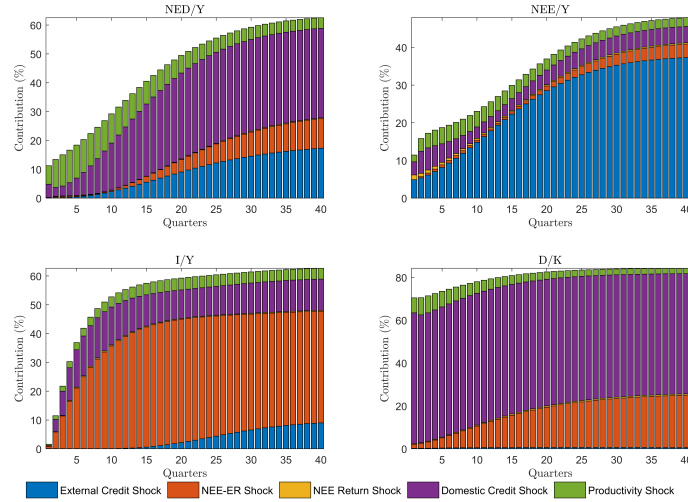


Fig. D2. Relevance of Exogenous Shocks for SDEs Using r^{ED} and Δa . This figure shows the forecast error variance decomposition over a 40-quarter horizon for a set of variables. Relative to the results in Figure 1, there are two differences in the underlying VAR specification. The first variable is the raw the real interest rate instead of the exogenous component of the real interest rate. The fifth variable is the productivity growth instead of the long-run component of productivity growth. Otherwise, see the notes for Figure 1.

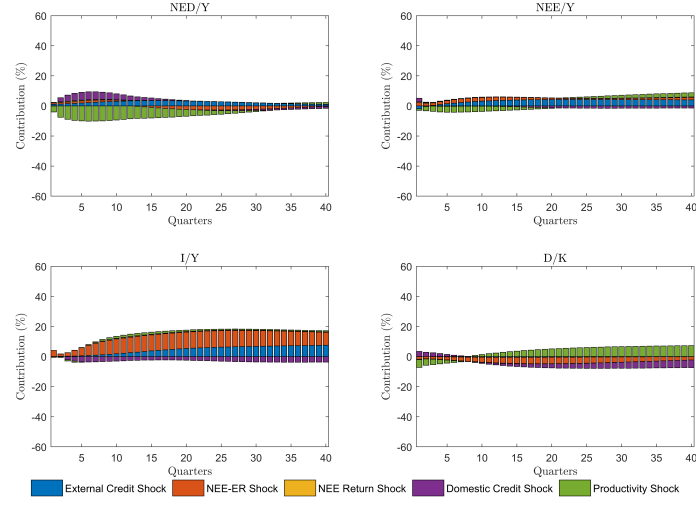
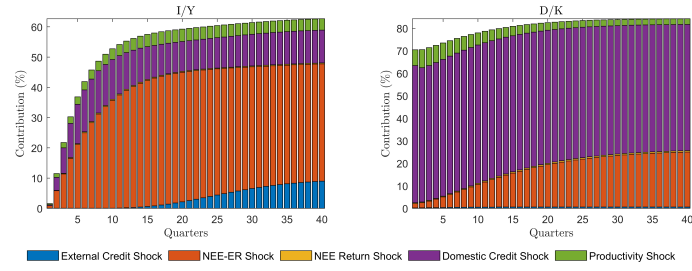
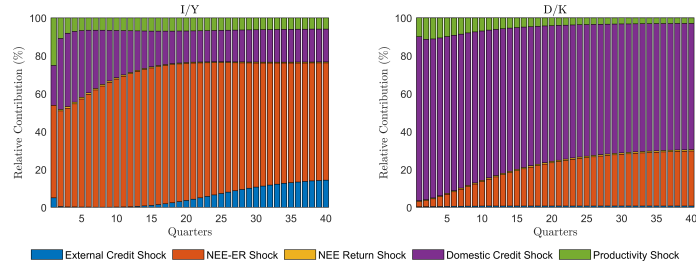


Fig. D3. Difference in Relevance of Exogenous Shocks for SDEs When Using r^{ED} and Δa . This figure reports the differences in the values reported in Figures 1 and Figures D2.

for a fairer comparison of explanatory power across the two settings.

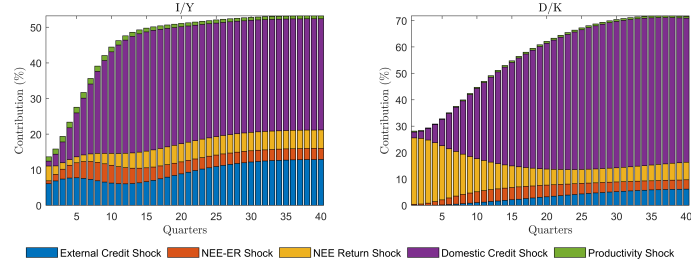


(a) FEVD Values

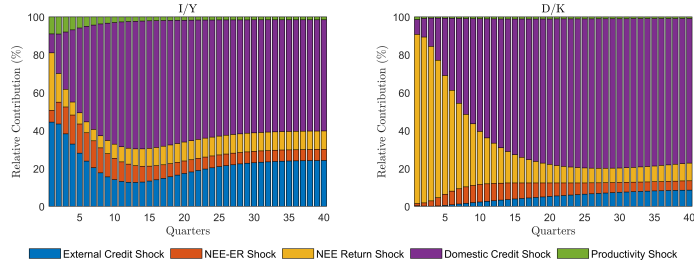


(b) FEVD Shares

Fig. D4. Relevance of Exogenous Shocks for Domestic Investment and Leverage: SDEs. This figure shows the forecast error variance decomposition over a 40-quarter horizon for a set of variables. Relative to the results in Figure D2, the underlying VAR specification is the same and therefore the same notes apply. The top panel replicates the bottom row of Figure D2. The bottom panel shows the relative shares of variance explained by each variable, expressed as a proportion of the total variance explained in the top panel.



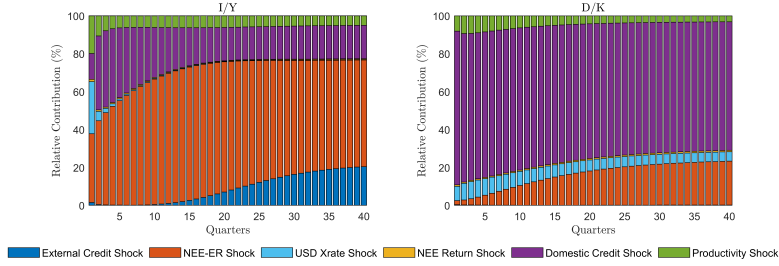
(a) FEVD Values



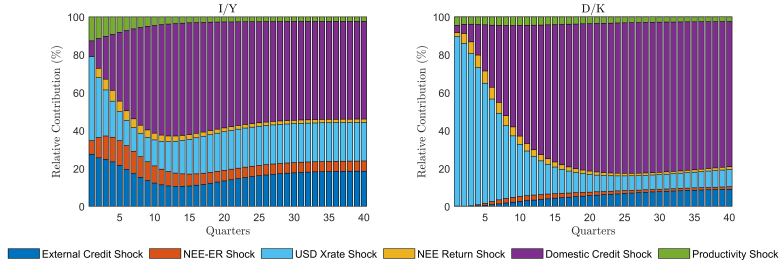
(b) FEVD Shares

Fig. D5. Relevance of Exogenous Shocks for Domestic Investment and Leverage: EMEs. This figure shows the forecast error variance decomposition over a 40-quarter horizon for a set of variables for the following set of EMEs: Brazil, Chile, Colombia, Mexico, and Peru. Relative to the results in Figure D4, there only difference is that the underlying series are measured as the equal-weighted average of the aforementioned set of EMEs rather than the set of SDEs studied in section III. Otherwise, see the notes for Figure D4.

We present the results for our set of EMEs in Figure D5, in which the bottom panels are directly comparable to those for SDEs as shown in Figure D4. The numbers reported in the bottom panels are used in computing the differences reported in Figure 2.



(a) SDEs



(b) EMEs

Fig. D6. Relevance of Exogenous Shocks Including Exchange Rates for Domestic Investment and Leverage. This figure shows the forecast error variance decomposition over a 40-quarter horizon for a set of variables. The values reported are the relative shares of variance explained by each variable, expressed as a proportion of the total variance explained among the given variables. Relative to the results in the bottom panels of Figures D4 and D5, there is one difference in the underlying VAR specification. We add the percent change in the real exchange rates of the domestic currency relative to the U.S. dollar as the third variable in the VAR ordering.

In Figure D6, we replicate the results presented in the bottom panels of Figures D4 and D5 using an augmented VAR specification. Specifically, we add the percent change in the real exchange rates of the domestic currency relative to the U.S. dollar as the third variable in the VAR ordering. We do so to account for the potential influence of exchange rate fluctuations on domestic investment and leverage.

Figure D7 documents that domestic investment is more sensitive to unexpected external equity shocks in EMEs than in SDEs. Figure D8 has the same structure of Figure 2, but with a focus on NED- and NEE-to-GDP ratios. We note two important results: (i) unexpected equity return shocks are relatively more important in EMEs than in SDEs, and (ii) external credit shocks explain a larger portion of short-run NED variance in EMEs than in SDEs, consistent with Uribe and Yue

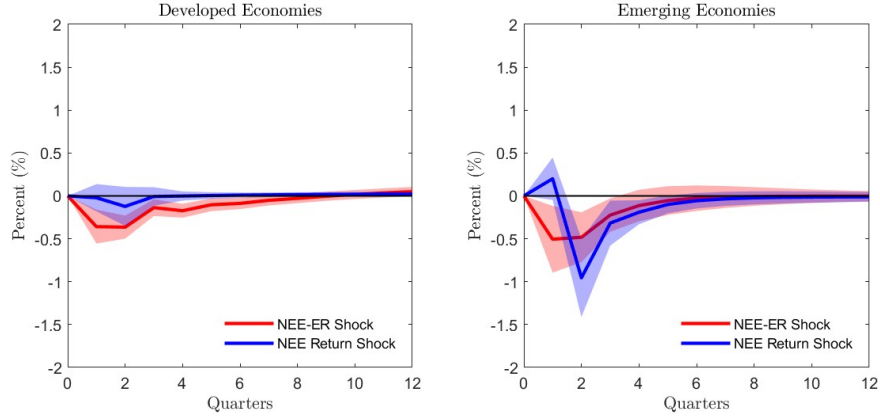


Fig. D7. Empirical Impulse Response of Domestic Investment to External Equity Return Shocks. This figure shows the response of the net external equity-to-output ratio ($\Delta \log I$) to shocks to the expected return (NEE-ER, η) or unexpected return ($r^E E$) of external equities. All results are based on the VAR specified in equations (18) and (19) using data constructed from cross-country averages. Confidence intervals are HAC-adjusted. Our sources are detailed in [Appendix C](#).

(2006).

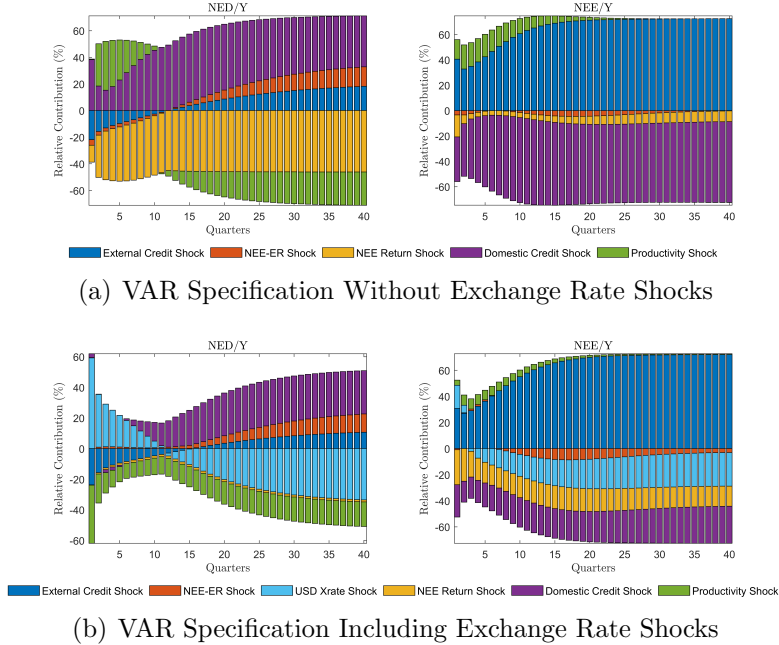


Fig. D8. Difference in Relevance of Exog. Shocks: SDEs Minus EMEs (II). This figure repeats the analysis in Figure 2 but for the NED- and NEE-to-GDP ratios.

Appendix D.2. Country-Level Results

In Figures D9 and D10, we depict impulse responses computed using country-level data, as opposed to cross country averages.

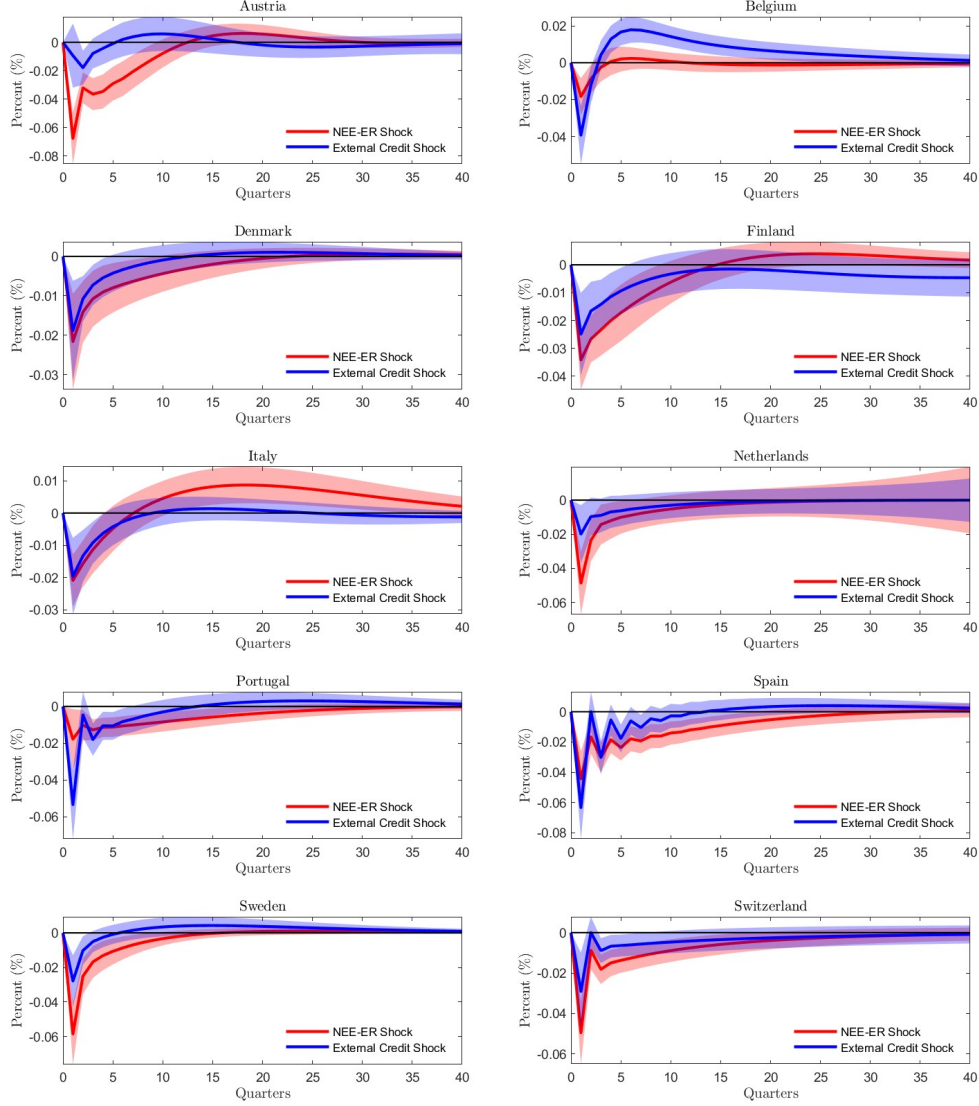


Fig. D9. Impulse Responses in Data VAR by Country: Productivity Long-Run Component (x). This figure shows the responses of the long-run component of productivity (x) to adverse shocks from external credit (r^w) and equity (NEE-ER, η) markets. All results are based on the VAR specified in equations (18)-(19) using individual country-level data. Confidence intervals are HAC-adjusted. Our sources are detailed in Appendix C and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

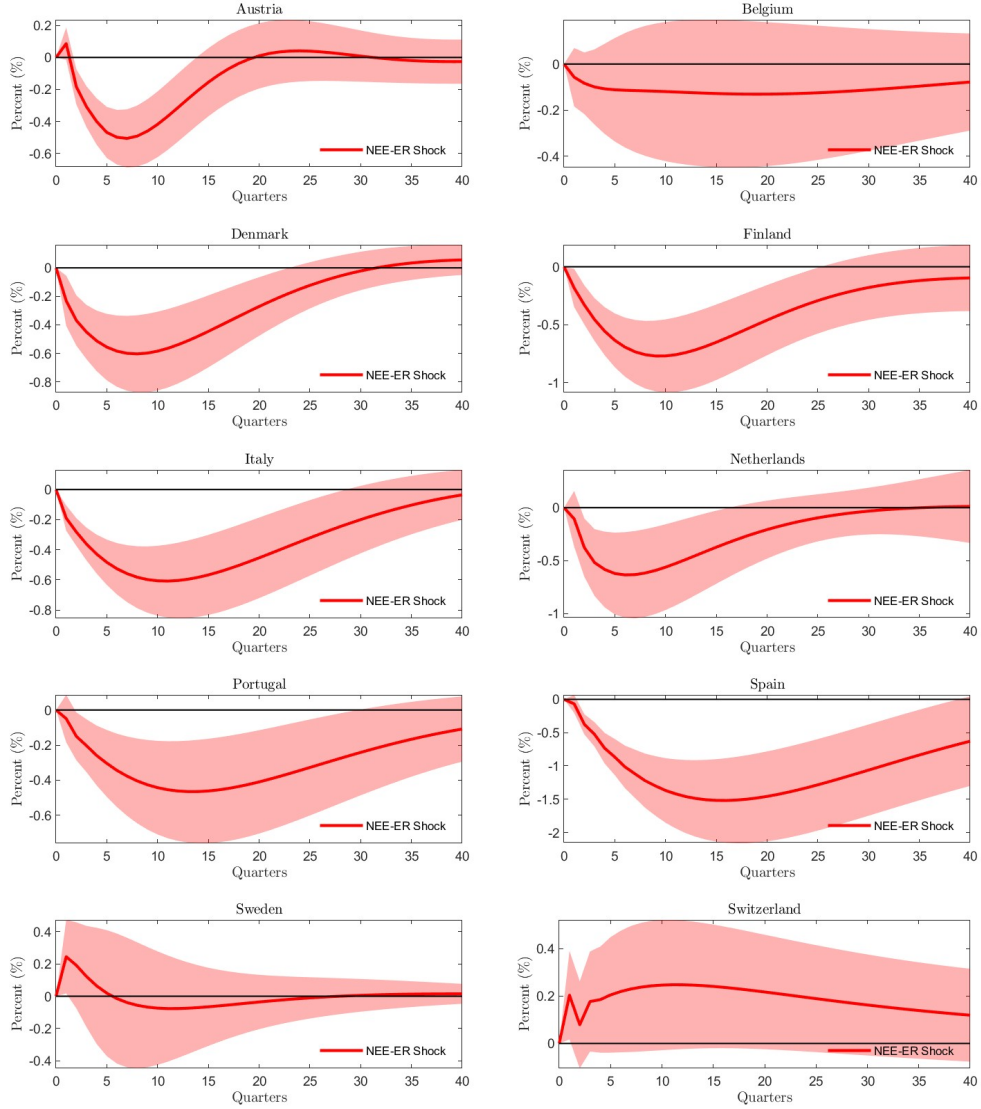


Fig. D10. Impulse Responses in Data VAR by Country: Domestic Credit Conditions (ξ). This figure shows the responses of domestic credit conditions (ξ) to adverse shocks from external equity (NEE-ER, η) markets. All results are based on the VAR specified in equations (18)-(19) using individual country-level data. Confidence intervals are HAC-adjusted. Our sources are detailed in [Appendix C](#) and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

Figure D11 shows the joint distribution of country-level shocks recovered from our country-level VARs.



Fig. D11. VAR Residual Scatter Plots. This figure shows the VAR residuals (i.e., innovations) from the external credit conditions (r^w) and NEE-ER (η) equations against the long-run productivity news (ϵ_x). All results are based on the VAR specified in equations (18)-(19) using country-level data. Our sources are detailed in [Appendix C](#) and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

Appendix D.3. Panel VAR Results

As an additional robustness exercise, we re-estimate our empirical VAR using the underlying country-level panel rather than the cross-country averages employed in the benchmark specification. Following the panel VAR methodology of Uribe and Yue (2006), we estimate the following first-order fixed-effects panel VAR equation-by-equation:

$$Y_{i,t} = A_1 Y_{i,t-1} + \alpha_i + \varepsilon_{i,t}, \quad (\text{D7})$$

where $Y_{i,t} = (r_{i,t}^w, \eta_{i,t}, r_{i,t}^e, \xi_{i,t}, x_{i,t})'$ denotes the vector of endogenous variables for country i in quarter t , α_i denotes country fixed effects, and $\varepsilon_{i,t}$ is the vector of reduced-form innovations. Consistent with the benchmark specification, we estimate a first-order panel VAR, maintain the same recursive ordering of variables, and identify structural shocks using a Cholesky decomposition of the estimated residual covariance matrix.

Figure D12 reports the resulting orthogonalized impulse responses for the long-run component of productivity (x) and domestic credit conditions (ξ). Confidence intervals are computed using a numerical implementation of the delta method following the asymptotic inference procedure described in Uribe and Yue (2006). Specifically, we numerically differentiate each orthogonalized impulse response with respect to the estimated VAR coefficients using finite differences and combine these derivatives with the estimated covariance matrix of the coefficient estimates to obtain asymptotic standard errors.

The panel VAR produces impulse responses that are qualitatively very similar to those obtained from the benchmark VAR based on cross-country averages. In particular, adverse NEE-ER shocks generate persistent declines in both the long-run component of productivity and domestic credit conditions, while external credit shocks produce qualitatively similar responses, although generally with smaller magnitudes. The timing, persistence, and overall shape of the impulse responses are also largely unchanged. Taken together, these results indicate that our benchmark findings are not driven by the aggregation of country-level data into cross-country averages.

Although the panel VAR and benchmark VAR produce qualitatively similar conclusions, their estimated magnitudes need not coincide exactly because the two estimators exploit different sources of variation. The benchmark VAR is estimated on cross-country averages and therefore primarily reflects variation common across

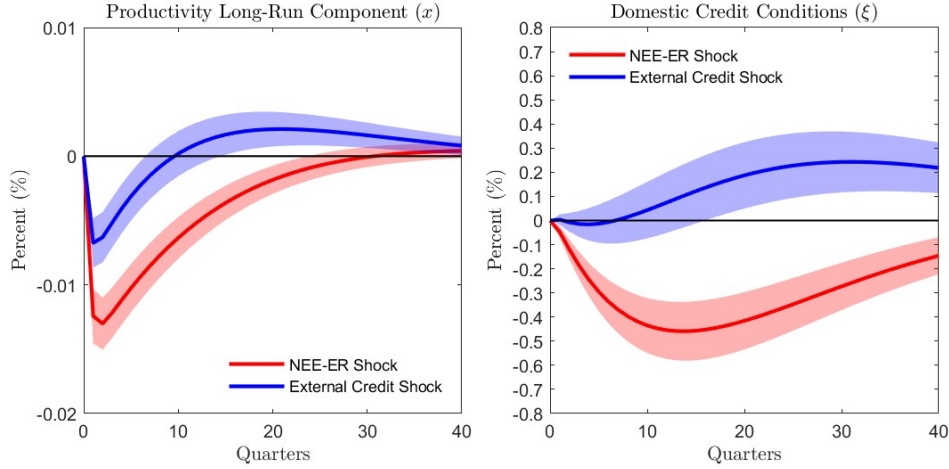


Fig. D12. Panel VAR Impulse Responses to External Financial Shocks. This figure shows the responses of the long-run component of productivity (x) and domestic credit conditions (ξ) to adverse shocks from external credit (r^w) and equity (NEE-ER, η) markets. Relative to Figure 3, all results are based on a first-order fixed-effects panel VAR estimated on the underlying country-level quarterly panel, following the empirical approach of Uribe and Yue (2006). Confidence intervals are computed using a numerical implementation of the delta method that follows the asymptotic inference procedure described in Uribe and Yue (2006). Our sources are detailed in Appendix C, and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

countries. By contrast, the panel VAR exploits both within-country and cross-country variation through country fixed effects. The close agreement across the two approaches therefore suggests that the qualitative conclusions of the benchmark VAR are not an artifact of aggregating country-level data into cross-country averages.

Using the same panel VAR specification, we next replicate the forecast error variance decomposition presented in Figure 1. Figure D13 reports the resulting forecast error variance decompositions. The panel VAR yields qualitative conclusions that are very similar to those obtained from the benchmark VAR estimated on cross-country averages. In particular, domestic credit shocks continue to account for the largest share of fluctuations in net external debt and corporate leverage, external credit shocks remain the dominant determinant of net external equity fluctuations, and NEE-ER shocks continue to explain the largest share of investment fluctuations.

Although the quantitative contributions of the exogenous shocks are generally smaller than those reported in Figure 1, this difference is expected because the

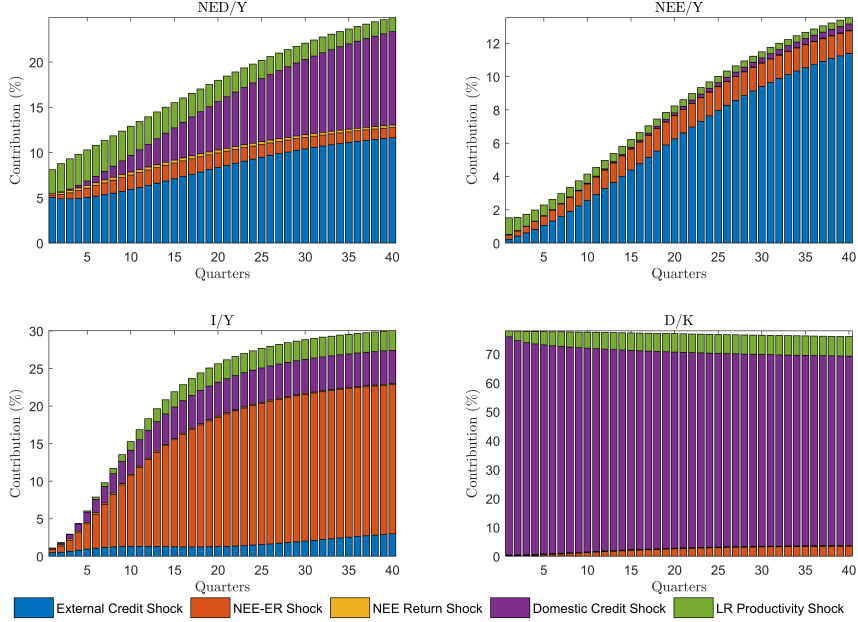


Fig. D13. Relevance of Exogenous Shocks using Panel VAR. This figure shows the forecast error variance decomposition over a 40-quarter horizon for the following variables of interest: net external debt-to-output ratio (NED/Y); net external equity-to-output ratio (NEE/Y); investment-to-output ratio (I/Y); and corporate leverage, i.e., debt-to-assets ratio (D/K). Relative to Figure 1, all results are based on a first-order fixed-effects panel VAR estimated on the underlying country-level quarterly panel, following the empirical approach of Uribe and Yue (2006). Our sources are detailed in Appendix C, and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

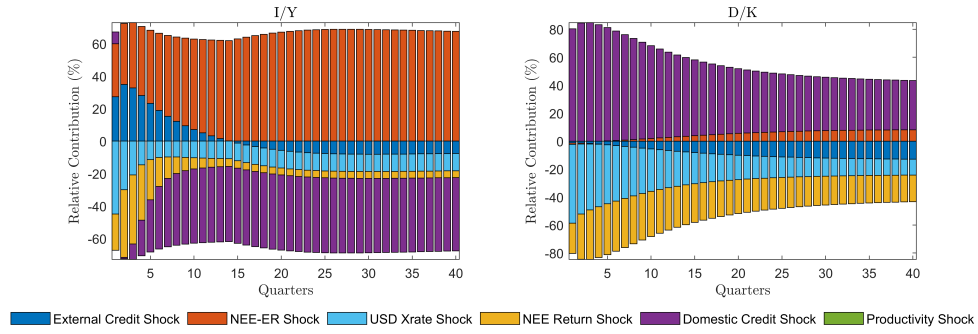
panel VAR exploits both cross-country and within-country variation, thereby attributing a larger fraction of forecast error variance to country-specific innovations. Nevertheless, the qualitative ranking of the exogenous shocks remains largely unchanged. These results are consistent with the comparison of impulse responses discussed above and further confirm that our principal conclusions. Namely, the relative importance of external and domestic financial shocks are robust to estimating the VAR directly on the underlying country-level panel rather than on aggregated time-series data.

Finally, we replicate the SDE-versus-EME comparison reported in Figure 2 using the corresponding panel VAR specifications. Figure D14 reports the resulting differences in forecast error variance shares between the two groups of countries. The panel VAR yields conclusions that are qualitatively very similar

to those reported in the benchmark analysis. Across both VAR specifications, NEE-ER shocks remain relatively more important for investment fluctuations in SDEs, whereas domestic credit, productivity, and unexpected equity return shocks continue to account for a relatively larger share of investment and leverage fluctuations in EMEs. Likewise, when exchange-rate shocks are incorporated into the VAR, exchange-rate innovations remain substantially more important in EMEs than in SDEs. Although the estimated magnitudes differ somewhat from those in Figure 2, the qualitative ranking of the shocks and the principal differences between SDEs and EMEs remain unchanged. These findings provide additional evidence that our conclusions are not an artifact of estimating the benchmark VAR on cross-country averages.



(a) VAR Specification Without Exchange Rate Shocks



(b) VAR Specification Including Exchange Rate Shocks

Fig. D14. Difference in Relevance of Exogenous Shocks using Panel VAR: SDEs Minus EMEs. This figure replicates Figure 2 using first-order fixed-effects panel VARs estimated on the underlying country-level quarterly panels, following the empirical approach of Uribe and Yue (2006). The top panel reports the benchmark specification, while the bottom panel additionally incorporates exchange-rate shocks relative to the U.S. dollar. In all panels, the underlying values for each set of countries are the relative shares of forecast error variance explained by each exogenous shock, expressed as a proportion of the total variance explained by the reported exogenous shocks. A positive (negative) bar indicates that a given shock accounts for a relatively larger share of forecast error variance in SDEs (EMEs).

Appendix D.4. Alternative Construction of $\hat{x}$

Our benchmark empirical analysis extracts the long-run component of productivity, $\hat{x}$, using a set of forecasting variables that includes country-level price-dividend ratios. Expected returns in the external equity market, η , are instead constructed using the world price-dividend ratio. Although these two price-dividend ratios are distinct, a potential concern is that the use of price-dividend ratios in both constructions may contribute to the estimated relation between external equity-market shocks and long-run productivity. In this subsection, we examine this possibility by reconstructing $\hat{x}$ without using the domestic price-dividend ratio. We then re-estimate our VARs while leaving the construction of expected external equity returns and all remaining aspects of the empirical procedure unchanged.

Figure D15 compares the impulse response of the long-run component of productivity to an adverse NEE-ER shock under the benchmark and alternative constructions of $\hat{x}$. Excluding the domestic price-dividend ratio attenuates the estimated response. Consistent with the impulse responses shown in the figure, the estimated η -to- x coefficient in the five-variable VAR declines in magnitude from -0.12 (standard error 0.02) under our benchmark specification to -0.04 (standard error 0.01) under the alternative specification. Thus, excluding the domestic price-dividend ratio materially attenuates the estimated η -to- x relation, reducing its magnitude by approximately two-thirds, although the relation remains negative and statistically significant. The greater magnitude under our benchmark construction is consistent with the domestic price-dividend ratio containing additional information about future productivity growth.

Motivated by this result, we next ask whether the attenuation in the productivity response materially alters our principal empirical conclusions regarding the relative importance of international shocks. Figure D16 repeats the forecast error variance decomposition from our benchmark analysis after reconstructing $\hat{x}$ without using the domestic price-dividend ratio and re-estimating the complete six-variable VAR. All other aspects of the empirical procedure, including the recursive ordering, identification strategy, sample period, and estimation methodology, are unchanged.

As Figure D16 shows, the resulting forecast error variance decompositions are very similar to their benchmark counterparts. Across all four endogenous variables, the same shocks continue to account for the largest shares of medium- and long-

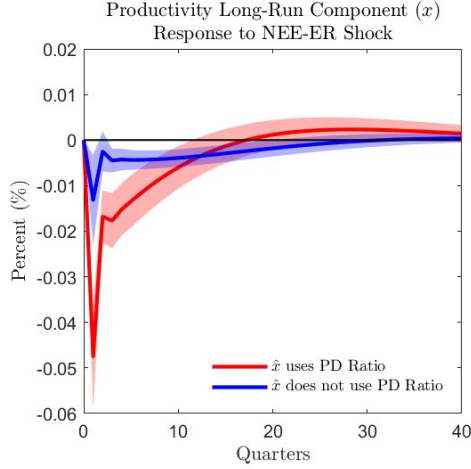


Fig. D15. Empirical Impulse Response of Long-Run Productivity (x) to NEE-ER Shock: Role of Price-Dividend Ratio. This figure shows the responses of the long-run component of productivity (x) to an adverse shock from the external equity (NEE-ER, η) market depending on whether we use the domestic price-dividend ratio when extracting ($\hat{x}$). All results are based on the VAR specified in equations (18) and (19b) using data constructed from cross-country averages. Confidence intervals are HAC-adjusted. Our sources are detailed in Appendix C, and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

run forecast error variance. Domestic credit shocks remain the dominant source of fluctuations in net external debt and corporate leverage, while external credit shocks remain the primary determinant of net external equity dynamics. Most importantly, expected equity-return (NEE-ER) shocks continue to account for the majority of medium- and long-horizon investment fluctuations. Thus, although excluding the domestic price-dividend ratio attenuates the estimated response of long-run productivity itself, it does not materially alter our empirical conclusions regarding the relative importance of international shocks.

The attenuation also has a direct implication for the calibration of $\rho_{\eta,x}$, the channel through which external equity shocks affect long-run growth in the model. In our benchmark calibration, we set $\rho_{\eta,x} = -0.06$, a value that is already conservative relative to the benchmark empirical estimate of -0.12 . The alternative construction of $\hat{x}$ instead yields an empirical estimate of -0.04 . To assess the quantitative implications of this attenuation, we therefore consider an alternative calibration in which we reduce $\rho_{\eta,x}$ from -0.06 to -0.04 , while holding all other parameters at their benchmark values. This exercise isolates the quantitative effect of weakening the specific η -to- x transmission channel identified above.

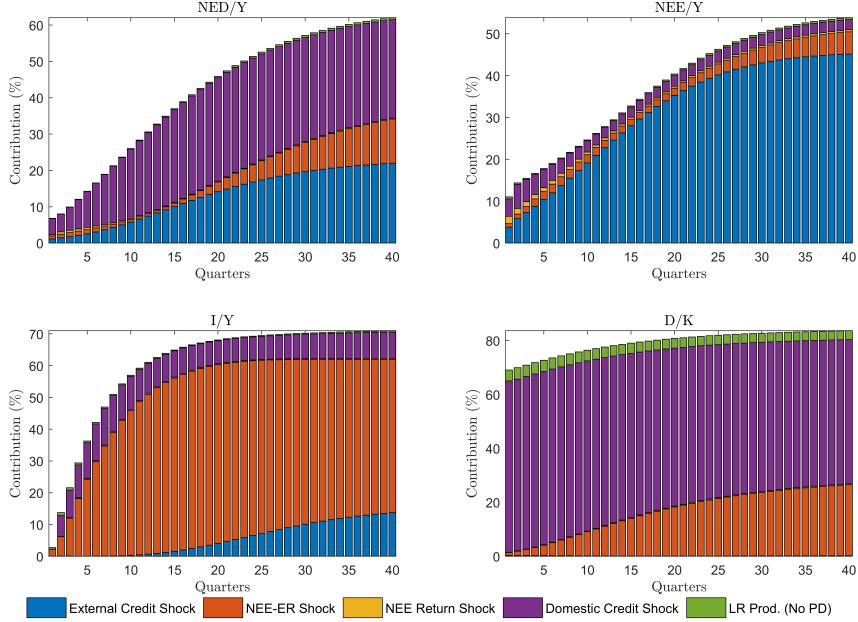


Fig. D16. Relevance of Exogenous Shocks Using an Alternative Measure of Long-Run Productivity. This figure repeats the forecast error variance decomposition reported in Figure 1 after reconstructing the long-run productivity component, $\hat{x}$, without using the domestic price-dividend ratio. Relative to Figure 1, all remaining aspects of the empirical procedure, including the VAR specification, recursive ordering, identification strategy, sample period, and estimation methodology, are unchanged.

Table D4 compares the model moments under the benchmark and alternative calibrations. Reducing $\rho_{\eta,x}$ from -0.06 to -0.04 produces only modest changes in the model moments. In particular, the model continues to generate essentially the same macroeconomic volatilities and correlations, return moments, leverage dynamics, international positions, and current-account dynamics as under the benchmark calibration.

Finally, Figure D17 examines the dynamic implications of the alternative calibration. To provide an internally consistent comparison, the data responses in this figure are obtained from the VARs estimated using the alternative measure of long-run productivity that excludes the domestic price-dividend ratio, while the model responses use the alternative calibration $\rho_{\eta,x} = -0.04$. The figure reports the responses of fundamentals, external positions and leverage, and economic aggregates to an adverse NEE-ER shock.

The recalibrated model continues to reproduce the corresponding empirical dy-

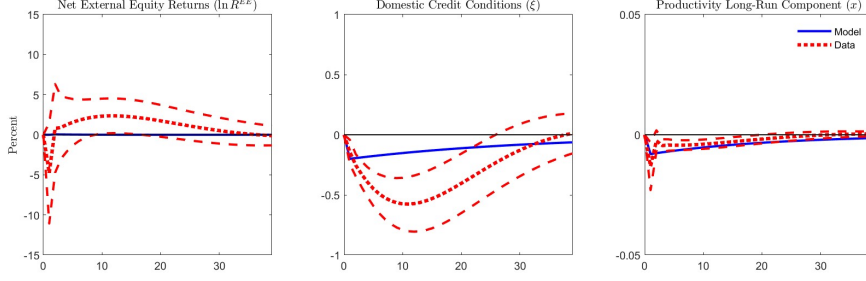
Table D4: Model Moments under the Alternative Calibration

Moments		Benchmark ($\rho_{\eta,x} = -0.06$)	Modified ($\rho_{\eta,x} = -0.04$)
Macro	$\sigma(\Delta i)/\sigma(\Delta y)$	1.84	1.84
	$\sigma(\Delta c)/\sigma(\Delta y)$	1.01	1.01
	$corr(\Delta i, \Delta c)$	0.58	0.58
	$corr(\Delta i, \Delta \log(D))$	-0.06	-0.06
	$corr(\Delta y, \Delta h)$	0.53	0.53
Returns	$\mathbb{E}[r]$ (%)	1.53	1.54
	$corr(r, \Delta y)$	0.10	0.10
	$\mathbb{E}[r_E - r]$ (%)	5.60	5.60
	$\sigma(r_E - r)$ (%)	6.51	6.51
	$\mathbb{E}[r_K - r]$ (%)	3.58	3.58
	$\sigma(r_K - r)$ (%)	4.61	4.61
	$\mathbb{E}[r - r^w]$ (%)	0.41	0.42
	$\sigma(r - r^w)$ (%)	0.55	0.55
Leverage	$\mathbb{E}[D/K]$ (%)	34.40	34.38
	$\sigma(D/K)$ (%)	6.76	6.76
	$corr(\Delta(D/K), \Delta y)$	0.10	0.10
	$\sigma(d/K)/\sigma(\Delta y)$ (%)	1.49	1.49
NFA	$\mathbb{E}[NED/Y]$ (%)	54.71	54.74
	$\mathbb{E}[NEE/Y]$ (%)	7.92	7.91
	$\sigma(\Delta(NED/Y))$ (%)	2.26	2.26
	$\sigma(\Delta(NEE/Y))$ (%)	5.44	5.43
	$corr(\Delta(NED/Y), \Delta y)$	-0.54	-0.54
	$corr(\Delta(NEE/Y), \Delta y)$	-0.12	-0.12
	$corr(\Delta(NED/Y), \Delta(NEE/Y))$	0.13	0.14
	$corr(NED/Y, D/K)$	-0.52	-0.52
	$corr(NEE/Y, D/K)$	0.07	0.08
Current Acc.	$\sigma(\Delta NFA/Y)$ (%)	3.14	3.13
	$acf(\Delta NFA/Y)$	0.81	0.81
	$corr(\Delta NFA/Y, r)$	0.24	0.23

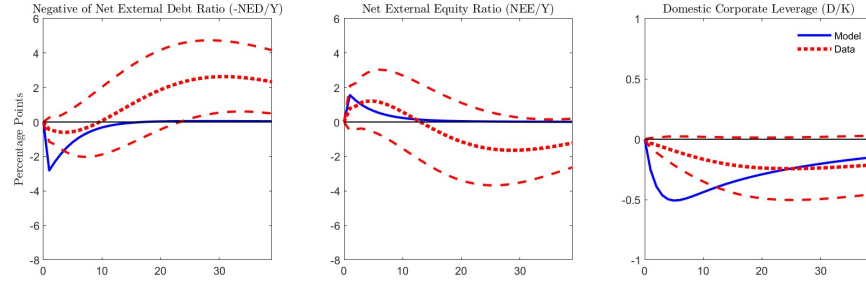
Notes: The calibration for our benchmark open economy setting SONOMA is reported in Table II.

namics well. In particular, the weaker response of long-run productivity is reflected in the model, while the responses of external positions, leverage, investment, and output remain close to those obtained under our benchmark specification. Taken together, these results show that excluding the domestic price-dividend ratio materially attenuates the estimated empirical link between expected external equity returns and long-run productivity. Accounting for this attenuation through a cor-

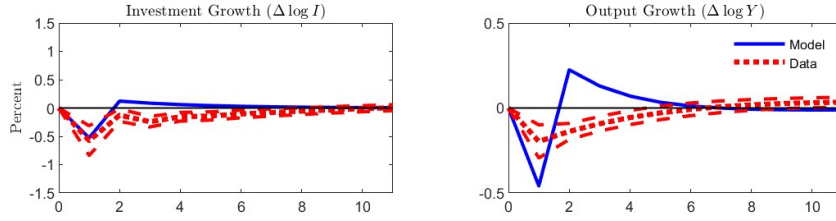
responding reduction in $\rho_{\eta,x}$ weakens this transmission channel in the model, but produces only modest changes in the model's quantitative implications and leaves our main conclusions unchanged.



(a) Fundamental Shocks



(b) External Positions and Leverage



(c) Economic Aggregates

Fig. D17. Recalibrated Model versus VAR Using the Alternative Measure of Long-Run Productivity. The top panel shows the responses of net external equity returns ($\ln R^{EE}$), domestic credit conditions (ξ), and the long-run component of productivity (x); the middle panel shows the responses of the negative of the net external debt ratio ($-NED/Y$), the net external equity ratio (NEE/Y), and the domestic book leverage ratio (D/K); and the bottom panel shows the responses of log investment ($\log I$) and log output ($\log Y$). All responses are to shocks from the NEE-ER (η). The model responses use the benchmark calibration in Table II, except that $\rho_{\eta,x}$ is reduced from -0.06 to -0.04 . The data responses are based on VARs estimated using data constructed from cross-country averages, as specified in equations (18) and (19b) for the top panel and equations (18)–(19) for the middle and bottom panels. Confidence intervals are HAC-adjusted. Our sources are detailed in Appendix C, and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

Appendix E. Calibration (II)

In this section, we discuss the calibration of canonical parameters that are common to many production-based asset pricing models. Focusing on preferences, we choose the coefficient of relative risk aversion γ , the subjective discount factor β , and the intertemporal elasticity of substitution ψ in the spirit of the long-run risk literature (see, among others, Bansal et al. 2012 and Croce 2014). Specifically, the subjective discount factor β , is set to match the average country-specific interest rate, r_t . The IES is calibrated to be greater than one and enables us to match the volatility of consumption to that of output. The relative risk aversion is set to 10 and enables us to produce a high average equity excess return, $E[r^{ex}]$.

Since we use a CES aggregator to construct the consumption bundle $\tilde{C}_t$ in equation (A), we need to calibrate two parameters: the consumption weight φ , and the elasticity of substitution coefficient f . Our choices of parameter are comparable to other studies in the international macroeconomics literature such as, for example, Zetlin-Jones and Shourideh (2017). In addition, we need to calibrate the smoothing parameter for the standards of living process. We choose a moderate value, $\theta = 0.50$.

The countries in our cross section are all developed and share production characteristics that are similar to those in other developed economies, such as the United States. As a result, we choose values for capital share of output α and capital depreciation rate δ that are close to values used in previous studies by Kaltenbrunner and Lochstoer (2010), Jermann and Quadrini (2012), and Croce (2014). We set the coefficient of the elasticity of adjustment cost ϕ_2 to 1.5 in order to match the volatility of investment growth relative to that of output growth. Our value is smaller than $\phi_2 = 7$ used by Croce (2014), but it is larger than 0.7 used by Kaltenbrunner and Lochstoer (2010). The corporate tax rate changes across countries, but $\tau_F = 0.35$ is a good representative value, given our sample. In Jermann and Quadrini (2012) economy, the value of τ_F determines whether the enforcement constraint is binding. In our model, the enforcement constraint is not always binding in simulations. However, this does not affect the results. As a robustness check, we carry out an exercise where instead of a collateral constraint, the economy faces a convex distress cost (which imposes a steep convex penalty). As shown in Appendix B, the output of this exercise are very close to the benchmark model that features collateral constraints.

Table E1: SONOMA Benchmark Calibration

Preferences			Estimate/Moment
Relative Risk Aversion	(γ)	10	$\mathbb{E}[r^{ex}]$
Intertemporal Elasticity of Substitution	(ψ)	2	$\sigma(\Delta c)/\sigma(\Delta y)$
Subjective Discount Rate	(β)	0.99	$\mathbb{E}[r^{ED}]$
Consumption-Leisure Aggregator			
Consumption Coefficient	(φ)	0.35	$\mathbb{E}[C/Y]$
Elasticity of Substitution	(f)	0.3	$corr(\Delta y, \Delta h)$
Production			
Capital Share	(α)	0.36	JQ(2012)
Capital Depreciation Rate	(δ)	0.10/4	JQ(2012)
Capital Adjustment Cost Elasticity	(ϕ_2)	1.5	$\sigma(\Delta i)/\sigma(\Delta y)$
Corporate Tax Rate	(τ_F)	0.35	JQ(2012)
Productivity Growth Rate			
Average	(μ_a)	0.020/4	$\mathbb{E}[\Delta y]$
Volatility of Short-Run Shock	(σ_a)	0.044/2	0.041/2 (0.003/2)
Persistence of Long-Run Component	(ρ_x)	0.95	0.93 (0.02)
Volatility of Long-Run Shock	(σ_x)	0.10 σ_a	0.08 (0.01)

Notes: This table reports our benchmark quarterly calibration for a subset of canonical parameters. The other parameter values are reported in Table II. The last column reports either a direct empirical estimate of the calibrated parameter or a specific moment that we target in our calibration procedure. JQ(2012) refers to parameter values that we get directly from Jermann and Quadrini (2012). Our smoothing parameter θ for the process determining the standards of living process, SL , is equal to 0.50.

We next turn to the productivity growth process. Its average growth, μ_a , is set to match the average growth rate in the sample. We jointly estimate the volatility of short-run shocks (σ_a), the persistence of the productivity long-run component (ρ_x); and the relative volatility of the long-run shocks (σ_x/σ_a) from an unbalanced panel of countries in our sample.² The two correlations between the long-run component and external financial condition shocks, $\rho_{rw,x}$ and $\rho_{\eta,x}$, are based on our VAR estimation.

²We use an unbalanced panel since the balanced panel starts in 1995, and is thus too short for estimating this parameter satisfactory. In addition, in order to minimize short-sample noise, the persistence parameter is estimated using a 4-quarter moving average of the long-run component.

Appendix F. Replication of Jermann and Quadrini

In this section, we first show that we are able to replicate the results in Jermann and Quadrini (2012). Next, we show how these results change when modifying the parameter values to those in SONOMA.

Table F1 shows the calibration that we use to replicate the benchmark model in Jermann and Quadrini (2012). These parameter values are the same as those in their paper with the following exceptions: we set $(\sigma_a, \mu_\xi, \sigma_\xi) = (0.0045 \times 3/2, 0.35, 0.005)$ instead of $(\sigma_a, \mu_\xi, \sigma_\xi) = (0.0045, 0.1634, 0.0098)$. We choose these values in order to match (i) their steady-state quarterly debt-to-output ratio of 3.36; and (ii) the impulse response functions seen in Figure 6 of Jermann and Quadrini (2012).

Table F1: Calibration to Replicate Jermann and Quadrini (2012)

		Replication	JQ (w. CAC)	Stronger CAC, Prod. LRR
Preferences				
Relative Risk Aversion	(γ)	1.1	1.1	1.1
Intertemporal Elasticity of Substitution	(ψ)	0.91	0.91	0.91
Subjective Discount Rate	(β)	0.98	0.98	0.99
Consumption-Leisure Aggregator				
Consumption Coefficient	$(\tilde{w}_1)$	1.00	1.00	0.35
Leisure Coefficient	$(\tilde{w}_2)$	1.88	1.88	0.65
Elasticity of Substitution	(f)	1.0	1.0	1.0
Production				
Capital Share	(α)	0.36	0.36	0.36
Capital Depreciation Rate	(δ)	0.10/4	0.10/4	0.10/4
Capital Adjustment Cost Elasticity	(ϕ_2)	1000	5	1.5
Corporate Tax Rate	(τ_F)	0.35	0.35	0.35
Productivity Growth Rate				
Average	(μ_a)	0.000/4	0.000/4	0.020/4
Volatility of Short-Run Shock	(σ_a)	0.014/2	0.014/2	0.044/2
Persistence of Long-Run Component	(ρ_x)	0.95	0.95	0.95
Volatility of Long-Run Shock	(σ_x)	$0.00\sigma_a$	$0.00\sigma_a$	$0.10\sigma_a$
Domestic Financial Markets				
Average Domestic Credit Conditions	(μ_ξ)	0.35	0.35	0.35
Persistence of Domestic Credit Conditions	(ρ_ξ)	0.97	0.97	0.97
Volatility of Domestic Credit Conditions Shock	(σ_ξ)	0.010/2	0.010/2	0.010/2
Equity Adj. Cost Coefficient	(κ)	0.146	0.146	0.146

Notes: Parameter values to replicate the benchmark model in Jermann and Quadrini (2012).

The main reason of this change is that their production function, $F(A_t, K_t, H_t) = A_t K_t^\theta H_t^{1-\theta}$, is not consistent with balanced growth when A is non-stationary. By using $F(A_{t-1}, K_t, H_t) = K_t^\theta (A_{t-1} H_t^{1-\theta})$ as production function, balanced growth is preserved and we simply need to modify slightly the shock sizes and average level of ξ to quantitatively match their output.

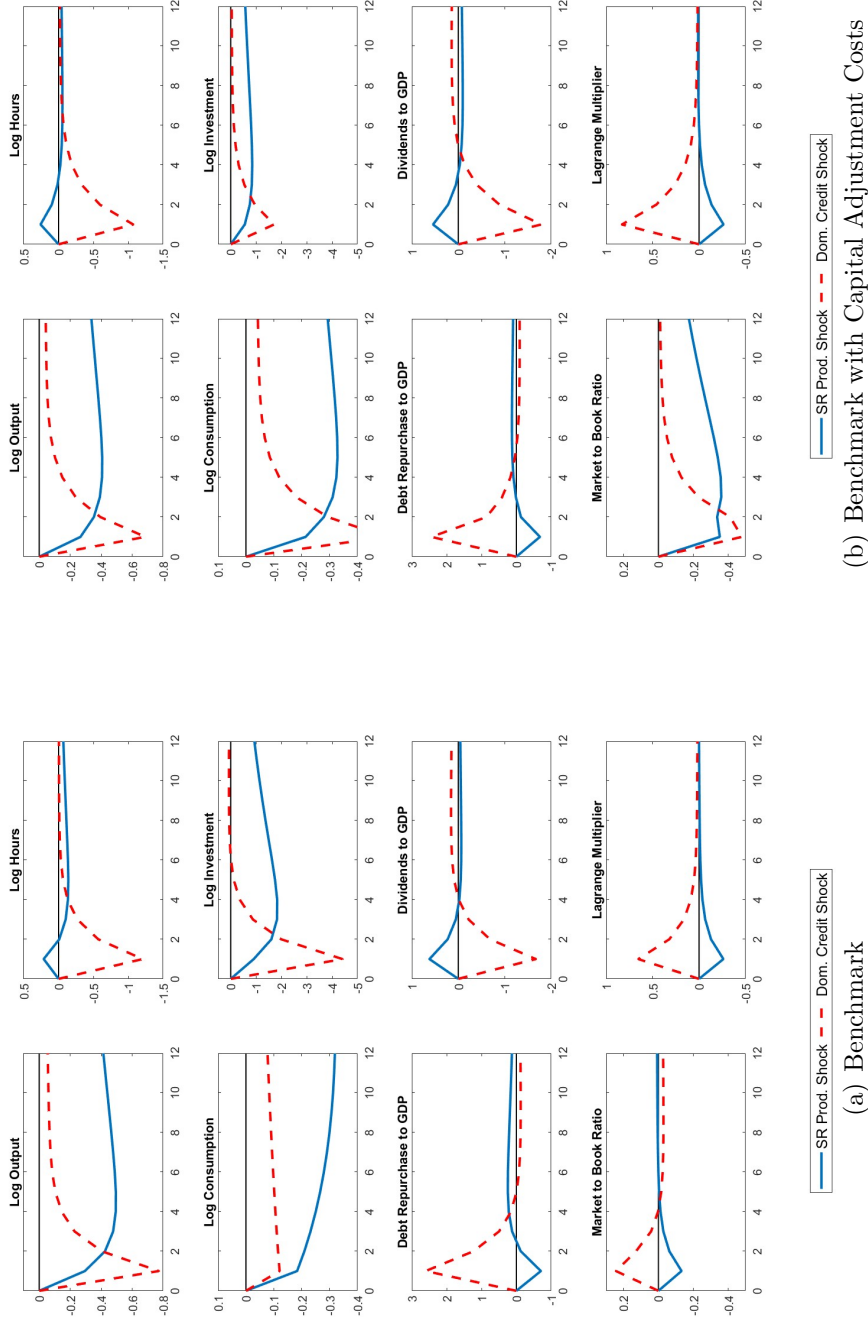


Fig. F1. Replication of Impulse Responses. This figure shows percentage deviations from steady state for variables expressed in logs and percentage point deviations from steady state for variables expressed in levels. “Benchmark” refers to the benchmark calibration in Jermann and Quadrini (2012) (see Table F1). To add capital adjustment costs, we set the capital adjustment costs parameter as $\phi_2 = 5$. All responses refer to the realization of adverse shocks to productivity and credit conditions.

Table F2: From Closed Replication to Closed SONOMA

Moments		JQ (w. CAC)	Stronger CAC, Prod. LRR	Closed SONOMA
Macro	$\sigma(\Delta i)/\sigma(\Delta y)$	2.46	1.11	2.21
	$\sigma(\Delta c)/\sigma(\Delta y)$	0.73	0.98	0.90
	$corr(\Delta i, \Delta c)$	1.00	0.99	0.52
	$corr(\Delta i, \Delta \log(D))$	0.51	-0.26	0.17
	$corr(\Delta y, \Delta h)$	0.85	0.14	0.33
Returns	$\mathbb{E}[r]$ (%)	7.12	7.05	4.86
	$corr(r, \Delta y)$	-0.34	0.27	-0.03
	$\mathbb{E}[r_E - r]$ (%)	0.08	0.28	3.00
	$\sigma(r_E - r)$ (%)	1.13	5.27	7.80
	$\mathbb{E}[r_K - r]$ (%)	0.03	0.13	2.11
	$\sigma(r_K - r)$ (%)	0.45	2.40	4.77
	$\mathbb{E}[r - r^w]$ (%)	0.00	0.00	0.00
	$\sigma(r - r^w)$ (%)	0.00	0.00	0.00
Leverage	$\mathbb{E}[D/K]$ (%)	53.85	54.56	36.06
	$\sigma(D/K)$ (%)	2.27	4.21	5.88
	$corr(\Delta(D/K), \Delta y)$	0.49	0.42	0.15
	$\sigma(d/K)/\sigma(\Delta y)$ (%)	1.30	1.34	1.50

Notes: “JQ (w. CAC)” is our Jermann and Quadrini (2012) replication calibration (Table F1) with capital adjustment costs set to $\phi_2 = 5$. “Stronger CAC, Prod. LRR” means that we (i) further tighten capital adjustment costs to the level in SONOMA; (ii) increase the short-run shock volatility to level in SONOMA; and (iii) add long-run risk. The Closed SONOMA calibration is the same as the benchmark open economy calibration except that the open economy parameters are superfluous.

In Figure F1, we depict impulse responses that confirm our replication effort. Panel (a) (left) can be compared against Figure 6 of Jermann and Quadrini (2012), and panel (b) (right) can be compared against their Figure 7. In Table F2, we show how key simulated moments change as we move towards the Closed SONOMA calibration. By comparing the moments across columns, we see three key outcomes that differ in Closed SONOMA.

First, the correlation between investment growth and consumption growth becomes smaller than one. Second, average excess returns (i.e., the equity premium) increase significantly as well as the volatility of excess returns. This result was to be expected given that Closed SONOMA features EZ preferences, higher risk aversion, and an intertemporal elasticity of substitution greater than one. Third,

we see the volatility of the firm debt to capital ratio increases, meaning that the firm is more actively changing its debt position.

Appendix G. Additional Results

Appendix G.1. Impulse Response Functions

In this section, we show impulse response functions of potential interest to the reader.

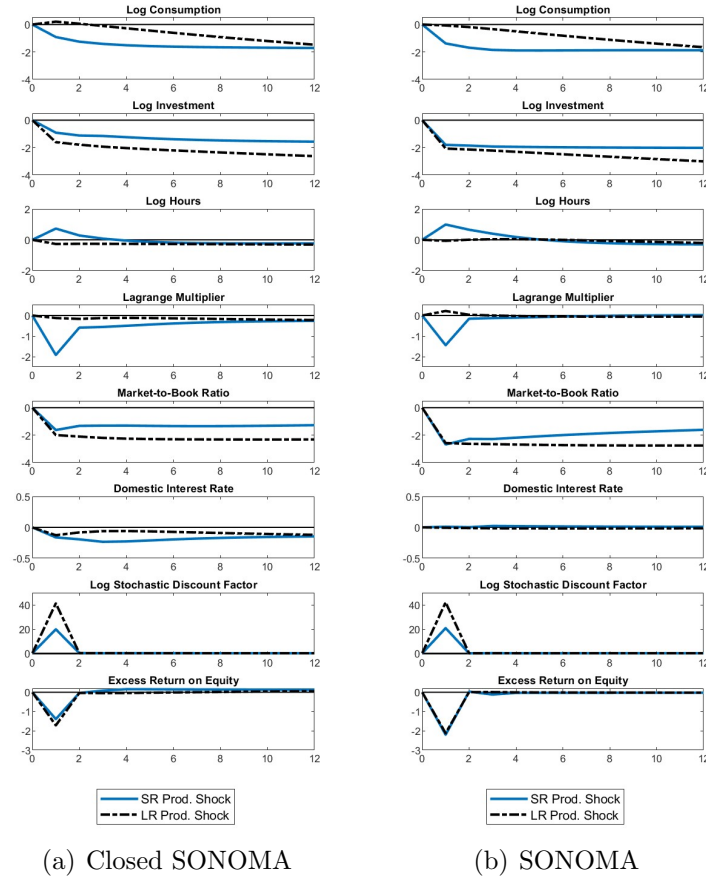


Fig. G1. Impulse Responses to Productivity Shocks in SONOMA vs Closed SONOMA. This figure shows percentage deviations from steady state for variables expressed in logs (percentage point deviations from steady state for variables expressed in levels) in response to a one-standard-deviation adverse shock to long-run productivity growth (x), or short-run productivity growth (Δa). Our benchmark calibration is reported in Table II and Appendix A describes how we obtain the closed setting.

Figure G1 reports key impulse responses with respect to adverse productivity shocks for both our SONOMA model and its closed-economy version.

Appendix G.2. Co-skewness

A visual inspection of the residuals estimated using country-level VARs reveals a notable clustering of extreme adverse external financial shocks and downside long-run growth (see Figure D11, Appendix D). Equivalently, the data suggest that there may be an adverse co-skewness between these variables.

In what follows, we formally assess whether downside long-run growth shocks are realized simultaneously with either extreme positive (i.e., adverse) NEE-ER or external credit condition shocks. To test this hypothesis, we compute Harvey and Siddique (2000) co-skewness measures,

$$CoSk(\epsilon_z, \epsilon_y) = \frac{\mathbb{E}[\epsilon_{z,t+1} \cdot \epsilon_{y,t+1}^2]}{\sqrt{\mathbb{E}[\epsilon_{z,t+1}^2] \cdot \mathbb{E}[\epsilon_{y,t+1}^2]}}.$$

Specifically, we first recover the residuals from estimating the VAR in equations (18)–(19) with country-level data. Next, we estimate both $CoSk(\epsilon_{r^w}, \epsilon_x)$ and $CoSk(\epsilon_\eta, \epsilon_x)$ using a generalized method of moments (GMM) estimator.³ Since adverse external shocks take positive values, a positive co-skewness refers to bad states of the world.

Our estimated co-skewness values for $CoSk(\epsilon_{r^w}, \epsilon_x)$ and $CoSk(\epsilon_\eta, \epsilon_x)$ are positive and statistically significant (Table G1). Hence we cannot reject the existence of a common adverse tail shock in both external credit conditions and long-run productivity growth, and between NEE-ER and long-run productivity growth. To the best of our knowledge, our study is the first one documenting this empirical fact.

³For each co-skewness coefficient, $CoSk$, we use the following set of over-identifying moment conditions:

$$\mathbb{E}[\epsilon_{z,t}^j \cdot (\epsilon_{y,t}^j)^2 - CoSk] = 0 \quad j = 1, \dots, 10,$$

where all residuals have been standardized first. For robustness, we also perform this estimation using an unbalanced panel regression. The resulting estimates are both quantitatively and qualitatively similar to those obtained from our two-step GMM procedure (Table G1).

Table G1: Co-Skewness with Long-Run Growth Shocks

External Variable	Moment	Estimate	Confidence Interval
External Credit Conditions	$CoSk(\epsilon_{rw}, \epsilon_x)$	0.245	[0.187 , 0.303]
NEE-ER	$CoSk(\epsilon_\eta, \epsilon_x)$	0.240	[0.129 , 0.351]

Notes: This table reports estimates of the co-skewness ($CoSk$) between country-level innovations to the long-run component of productivity (ϵ_x^j) and either shocks to the external credit conditions (ϵ_{rw}) or the NEE-ER, i.e., the external expected equity returns (ϵ_η). Numbers in square brackets represent the 10th and 90th percentile values based on Newey and West (1987) HAC-adjusted standard errors. All results are based on the VAR specified in equation (18)-(19) using country-level data. Our sources are detailed in [Appendix C](#) and our quarterly sample starts in 1995:Q1 and ends in 2018:Q4.

Since we find no significant skewness in domestic credit shocks and no other statistically relevant results on co-skewness across other shocks, in our model we treat all other processes as subject to symmetric innovations.

Tail shocks in SONOMA. Given our empirical evidence, we augment SONOMA with tail shocks by including a common jump shock, J_t , in the NEE-ER, the external credit conditions process, and the long-run component:

$$r_t^w = (1 - \rho_{rw})(\mu_{rw} - j_0) + \rho_{rw}r_{t-1}^w + \sigma_{rw}\epsilon_{rw,t} + (1 - \rho_{rw})J_t \quad (G8)$$

$$\eta_t = (1 - \rho_\eta)(\mu_{rw} + s_{rw} - j_0) + \rho_\eta\eta_{t-1} + \sigma_\eta\epsilon_{\eta,t} + (1 - \rho_\eta)J_t \quad (G9)$$

$$x_t = (1 - \rho_x)j_0\rho_x x_{t-1} + \sigma_x\epsilon_{x,t+1} + \rho_{rw,x}\sigma_{rw}\epsilon_{rw,t+1} + \rho_{\eta,x}\sigma_\eta\epsilon_{\eta,t+1} - (1 - \rho_x)\epsilon_{x,t} \quad (G10)$$

where, $J_t = j_0 e^{j_1 \cdot \epsilon_{j,t}} > 0$, $\epsilon_{j,t} \sim N(0, 1)$, and $j_1 = 1.9$ and $j_0 = 2.e^{-5}$ so that this process is on average close to zero but it also feature infrequent large realizations.

We report simulation results from this model in [Table G2](#). With the inclusion of an adverse tail shock, SONOMA generates positive co-skewness values that are within the empirical confidence intervals presented in [Table G1](#).

Compared to our benchmark SONOMA, this model setting produces a higher equity premium (by 22 basis points) and a 1% decline in the average level of capital. Given this lower level of capital accumulation, the representative agent suffers a welfare loss of about 12% of life-time consumption. These results suggest that adverse tail shocks may have strong detrimental effects on both the real economy

Table G2: SONOMA with Adverse Jump Shocks

Moments		Without Jump Shock	With Jump Shock
Macro	$\sigma(\Delta i)/\sigma(\Delta y)$	1.84	1.84
	$\sigma(\Delta c)/\sigma(\Delta y)$	1.01	1.00
	$corr(\Delta i, \Delta c)$	0.58	0.58
	$corr(\Delta i, \Delta \log(D))$	-0.06	-0.08
	$corr(\Delta y, \Delta h)$	0.53	0.53
Returns	$\mathbb{E}[r]$ (%)	1.53	1.45
	$corr(r, \Delta y)$	0.10	0.07
	$\mathbb{E}[r_E - r]$ (%)	5.60	5.82
	$\sigma(r_E - r)$ (%)	6.51	6.53
	$\mathbb{E}[r_K - r]$ (%)	3.58	3.70
	$\sigma(r_K - r)$ (%)	4.61	4.58
	$\mathbb{E}[r - r^w]$ (%)	0.41	0.33
Leverage	$\sigma(r - r^w)$ (%)	0.55	0.40
	$\mathbb{E}[D/K]$ (%)	34.40	36.42
	$\sigma(D/K)$ (%)	6.76	6.48
	$corr(\Delta(D/K), \Delta y)$	0.10	0.11
NFA	$\sigma(d/K)/\sigma(\Delta y)$ (%)	1.49	1.42
	$\mathbb{E}[NED/Y]$ (%)	54.71	53.60
	$\mathbb{E}[NEE/Y]$ (%)	7.92	7.99
	$\sigma(\Delta(NED/Y))$ (%)	2.26	2.40
	$\sigma(\Delta(NEE/Y))$ (%)	5.44	6.24
	$corr(\Delta(NED/Y), \Delta y)$	-0.54	-0.54
	$corr(\Delta(NEE/Y), \Delta y)$	-0.12	-0.18
	$corr(\Delta(NED/Y), \Delta(NEE/Y))$	0.13	0.10
	$corr(NED/Y, D/K)$	-0.52	-0.55
	$corr(NEE/Y, D/K)$	0.07	-0.05
Current Acc.	$\sigma(\Delta NFA/Y)$ (%)	3.14	3.17
	$acf(\Delta NFA/Y)$	0.81	0.80
	$corr(\Delta NFA/Y, r)$	0.24	0.32
Co-skewness	$CoSk(\epsilon_{rw}, \epsilon_x)$	-0.01	0.35
	$CoSk(\epsilon_\eta, \epsilon_x)$	0.01	0.30
Welfare	$E[\log(U/A)]$	-1.51	-1.63
	$E[\log(K/A)]$	1.72	1.72

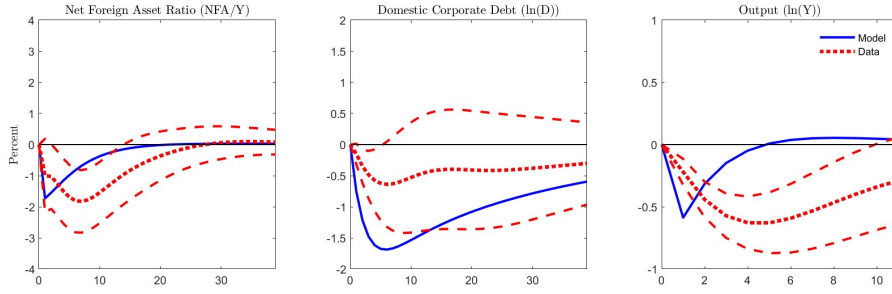
Notes: The calibration for SONOMA (i.e., “Without Jump Shock”) is reported in Table II. The common adverse jump shock is modeled as in equations (G8)–(G10) and the corresponding parameter values ($j_0 = 2.e^{-5}$ and $j_1 = 1.9$) are chosen to match the observed co-skewness values in the data (Table G1).

and welfare.

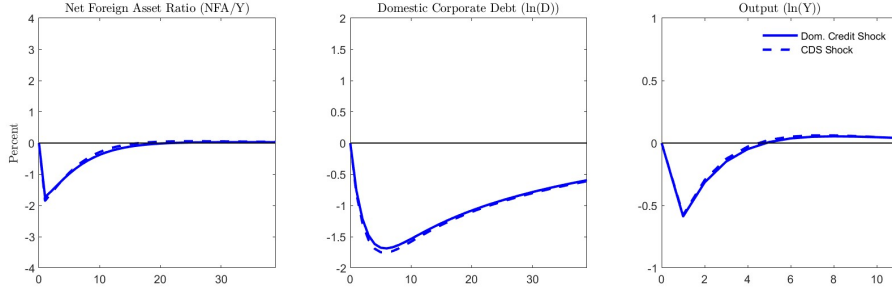
Appendix G.3. Credit Spread and CDS Shocks

In the top portion of figure G2, we show the response of the NFA-to-GDP ratio, domestic corporate leverage, and GDP to a negative shock to internal credit conditions coming from a credit spread shock, as identified by the VAR specified in equation (20). The patterns resulting from simulating the model are consistent with the empirical ones.

In the bottom portion of the same figure, we also look at the role of a CDS shock. In the model, a CDS shock corresponds to both an adverse credit spread



(a) Simulating a Credit Spread Shock



(b) CDS Shock versus Credit Spread Shock

Fig. G2. Credit Spread and CDS Shocks. The top portion of this figure shows the response of NFA, domestic corporate leverage and cumulative GDP growth in response to a credit spread shock, that is, a shock to internal credit conditions identified by the augmented VAR specified in equations (18) and (20). The entries for the data are obtained from the same VAR. In the bottom portion of this figure, we compare the model responses across a credit spread and a CDS shock. A CDS shock differs from a credit spread shock in that it features also an increase in the external cost of borrowing, consistent with the results in Figure 4.

shock and a simultaneous increase in the external cost of borrowing. As mentioned in main text, in our data, CDS shocks play a marginal role. As a result, the increase in the external cost of borrowing is small and the model impulse responses are very similar to those obtained while simulating a negative shock to internal credit conditions. Overall, our model results are consistent with the findings in [Bocola \(2016\)](#).

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